



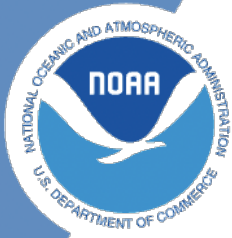
NOAA Technical Memorandum NMFS-XXX-##

Analyses to support the Dolphinfish Management Strategy Evaluation in the U.S. Atlantic

Marcel Kuipers

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National Marine Fisheries Service
Northwest Fisheries Science Center



**NOAA
FISHERIES**

Analyses to support the Dolphinfish Management Strategy Evaluation in the U.S. Atlantic

Mandy Karnauskas¹

1. NOAA Fisheries, Southeast Fisheries Science Center

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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 NOAA Fisheries recreational landings data

Here we will partition recreational landings data from NOAA Fisheries' Marine Recreational Information Program (MRIP), a standardized, peer-reviewed suite of recreational fishing surveys used to produce catch and effort estimates for quota monitoring and stock assessment in the United States. MRIP is a standardized survey suite that employs a complex statistical sampling design stratified across time, regions of the coast, and different fishing modes, which allows the intercept data to be scaled up by effort estimates to calculate total landings.

2.1 Data access and upload

Data are downloaded directly from the Miami server using the files provided to SERO for ACL monitoring. The current location is: M://SFD/SECM-SFD/ACL/2025_Aug19_-MRIP_FES/. The landings are in FES MRIP and the units are in pounds whole weight.

```
# clear workspace
rm(list = ls())

# read in data file
load("data/mrip_fes_rec81_25wv2_23June25.RData") # most recent file on SFD server
head(dat)
```

	REC_ACL_MRIP_FES_ID	YEAR	MONTH	WAVE	SUB_REG	NEW_ST	NEW_STA	NEW_MODE	NEW_MODEN
1	73407425	1989	NA	4	6	9	NC	4	Priv
2	73407426	1989	NA	4	6	9	NC	4	Priv
3	73407427	1989	NA	4	6	9	NC	4	Priv
4	73407428	1989	NA	4	6	9	NC	4	Priv
5	73407429	1989	NA	4	6	9	NC	4	Priv
6	73407430	1989	NA	4	6	9	NC	4	Priv

	AREA_X	NEW_AREA	NEW_AREAN	FL_REG	NC_REG	DS	NEW_SCI
1	5	5	Inshore	NA	S	MRIP Centropristis	striata
2	5	5	Inshore	NA	N	MRIP Centropristis	striata

2 NOAA Fisheries recreational landings data

3	2	2	Ocean>3mi	NA	N MRIP	Centropristis striata
4	2	2	Ocean>3mi	NA	S MRIP	Centropristis striata
5	1	1	Ocean<=3mi	NA	N MRIP	Centropristis striata
6	1	1	Ocean<=3mi	NA	S MRIP	Centropristis striata
	NEW_COM	SP_CODE	SPECIES_ITIS	SPECIES	AB1	B2
1	black sea bass	8835020301	167687		34181.4033	121019.1368
2	black sea bass	8835020301	167687		795.8612	4445.0527
3	black sea bass	8835020301	167687		549.5390	2919.4840
4	black sea bass	8835020301	167687		71823.2390	24548.2535
5	black sea bass	8835020301	167687		415.7642	87.5928
6	black sea bass	8835020301	167687		48728.5632	11859.6428
	A	B1	LBSEST_SECWWT	LBSEST_SECSOURCE	SAMPLE_SIZE_USED	
1	31761.4755	2419.9278	11457.8425	srysmwa	41	
2	584.4155	211.4457	266.7782	srysmwa	41	
3	549.5390	0.0000	484.4319	srysmwa	119	
4	33343.1844	38480.0546	63313.9276	srysmwa	119	
5	415.7642	0.0000	294.0623	srysmwa	71	
6	40585.0152	8143.5480	34464.8156	srysmwa	71	
	AVG_LEN	AVG_WGT	SRYSMWA_AVG_LEN	SRYSMWA_AVG_WGT	SRYSMWA_SAMPLE_SIZE	
1	203.4572	0.3352069	203.4572	0.3352069	41	
2	203.4572	0.3352069	203.4572	0.3352069	41	
3	296.2590	0.8815243	296.2590	0.8815243	119	
4	296.2590	0.8815243	296.2590	0.8815243	119	
5	261.9949	0.7072816	261.9949	0.7072816	71	
6	261.9949	0.7072816	261.9949	0.7072816	71	
	SRYSMWA_FLAGGED	SRYSMW_AVG_LEN	SRYSMW_AVG_WGT	SRYSMW_SAMPLE_SIZE		
1	NA	269.2563	0.7310037	231		
2	NA	269.2563	0.7310037	231		
3	NA	269.2563	0.7310037	231		
4	NA	269.2563	0.7310037	231		
5	NA	269.2563	0.7310037	231		
6	NA	269.2563	0.7310037	231		
	SRYSMW_FLAGGED	SRYSMW_AVG_LEN	SRYSMW_AVG_WGT	SRYSMW_SAMPLE_SIZE	SRYSMW_FLAGGED	
1	NA	259.7694	0.6948228	556	NA	
2	NA	259.7694	0.6948228	556	NA	
3	NA	259.7694	0.6948228	556	NA	
4	NA	259.7694	0.6948228	556	NA	
5	NA	259.7694	0.6948228	556	NA	
6	NA	259.7694	0.6948228	556	NA	
	SRYS_AVG_LEN	SRYS_AVG_WGT	SRYS_SAMPLE_SIZE	SRYS_FLAGGED	SRYS_AVG_LEN	
1	270.955	0.8099641	982	NA	269.1276	

2 NOAA Fisheries recreational landings data

2	270.955	0.8099641		982	NA	269.1276		
3	270.955	0.8099641		982	NA	269.1276		
4	270.955	0.8099641		982	NA	269.1276		
5	270.955	0.8099641		982	NA	269.1276		
6	270.955	0.8099641		982	NA	269.1276		
	SRY_AVG_WGT	SRY_SAMPLE_SIZE	SRY_FLAGGED	SR_AVG_LEN	SR_AVG_WGT	SR_SAMPLE_SIZE		
1	0.8085211	1703	NA	303.6871	1.050403	46055		
2	0.8085211	1703	NA	303.6871	1.050403	46055		
3	0.8085211	1703	NA	303.6871	1.050403	46055		
4	0.8085211	1703	NA	303.6871	1.050403	46055		
5	0.8085211	1703	NA	303.6871	1.050403	46055		
6	0.8085211	1703	NA	303.6871	1.050403	46055		
	SR_FLAGGED	S_AVG_LEN	S_AVG_WGT	S_SAMPLE_SIZE	S_FLAGGED	VAR_B2	VAR_AB1	
1	NA	344.4373	1.364082	305089	NA	4072492794	310099094	
2	NA	344.4373	1.364082	305089	NA	2866289	258947	
3	NA	344.4373	1.364082	305089	NA	0	0	
4	NA	344.4373	1.364082	305089	NA	105731413	534702898	
5	NA	344.4373	1.364082	305089	NA	0	0	
6	NA	344.4373	1.364082	305089	NA	22265768	286196169	
	SA_LABEL	GOM_LABEL	REC_ACL	MIGRA_GRP	AGG_MODEN	JURISDICTION	PS_REG	
1	Snapper	Grouper	1		Private	State	6	
2	Snapper	Grouper	1		Private	State	5	
3	Snapper	Grouper	1		Private	Federal	5	
4	Snapper	Grouper	1		Private	Federal	6	
5	Snapper	Grouper	1		Private	State	5	
6	Snapper	Grouper	1		Private	State	6	
	COUNCILREG	GFMC_FMP	SAFMC_FMU	LBSEST_SECGWT	AGGR_AREA	AREA	FIRST_MONTH	
1	6		Y	9710.0360		NA	NA	
2	5		Y	226.0832		NA	NA	
3	5		Y	410.5355		NA	NA	
4	6		Y	53655.8708		NA	NA	
5	5		Y	249.2054		NA	NA	
6	6		Y	29207.4709		NA	NA	
	LAST_MONTH	ALT_FLAG	CHTS_CL	CHTS_H	CHTS_RL	CHTS_WAB1C	CHTS_WAB1H	MODE_FX
1	NA	0	0	0	0	0	0	7
2	NA	0	0	0	0	0	0	7
3	NA	0	0	0	0	0	0	7
4	NA	0	0	0	0	0	0	7
5	NA	0	0	0	0	0	0	7
6	NA	0	0	0	0	0	0	7
	NEW_SEASN	SEASON						

2 NOAA Fisheries recreational landings data

```
1      NA
2      NA
3      NA
4      NA
5      NA
6      NA
```

```
#apply(dat[2:18], 2, table, useNA = "always")
```

```
# confirm nomenclature for dolphin and subset by species -----
table(dat$NEW_COM[grepl("dolphin", dat$NEW_COM)])
```

```
dolphin
9377
```

```
table(dat$NEW_SCI[grepl("dolphin", dat$NEW_COM)])
```

```
Coryphaena hippurus
9377
```

```
d <- dat[which(dat$NEW_COM == "dolphin"), ]
#apply(d[2:18], 2, table, useNA = "always")
d <- d[which(d$YEAR < 2025), ] # take out 2025 because incomplete year
```

2.2 Separate Gulf and Atlantic recreational landings

Now we filter the data set to separate out the Gulf versus Atlantic landings.

```
# codes for WFL regions: 1 (NW FL Panhandle- Escambia to Dixie co.), 2 (SW FL Peninsula- Le
table(d$FL_REG, useNA = "always")
```

```
1    2    3    4    5 <NA>
387 110 568 1022 312 6943
```

2 NOAA Fisheries recreational landings data

```
# subregional codes: 6 is Atlantic and 7 is Gulf
table(d$FL_REG, d$SUB_REG, useNA = "always")
```

	4	5	6	7	<NA>
1	0	0	0	387	0
2	0	0	0	110	0
3	0	0	0	568	0
4	0	0	1022	0	0
5	0	0	312	0	0
<NA>	96	585	3836	2426	0

```
unique(d$NEW_STA)
```

```
[1] "NC"      "FLE"      "FLW"      "SC"      "GA"      "MS"      "AL"      "LA"
[9] "TX"      "FLE/GA"   "RI"       "VA"      "DE"      "MD"      "NJ"      "CT"
[17] "MA"      "NY"       "LA/MS"    "AL/FLW"
```

```
lis <- c("TX", "LA", "LA/MS", "MS", "AL", "AL/FLW") # filter out the Gulf states
d1 <- d[-which(d$NEW_STA %in% lis), ]
unique(d1$NEW_STA)
```

```
[1] "NC"      "FLE"      "FLW"      "SC"      "GA"      "FLE/GA"   "RI"      "VA"
[9] "DE"      "MD"      "NJ"      "CT"      "MA"      "NY"
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
1	0	0	0	0	387	0	0	0	0	0	0	0	0	0
2	0	0	0	0	110	0	0	0	0	0	0	0	0	0
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0
<NA>	14	114	0	1794	174	54	20	125	1344	108	73	62	644	165

<NA>

2 NOAA Fisheries recreational landings data

```
1      0
2      0
3      0
4      0
5      0
<NA>   0
```

```
dim(d1)
```

```
[1] 7090   83
```

```
d1 <- d1[-which(d1$FL_REG == 1 | d1$FL_REG == 2), ] # filter out FL regions 1 and 2 but r
dim(d1)
```

```
[1] 6593   83
```

```
d1 <- d1[-which(is.na(d1$FL_REG) & d1$NEW_STA == "FLW"), ] # filter out FLW with unknown F
dim(d1)
```

```
[1] 6419   83
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always") # retained are all Atlantic states plus reg
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0
<NA>	14	114	0	1794	0	54	20	125	1344	108	73	62	644	165

```
<NA>
3      0
4      0
5      0
<NA>   0
```

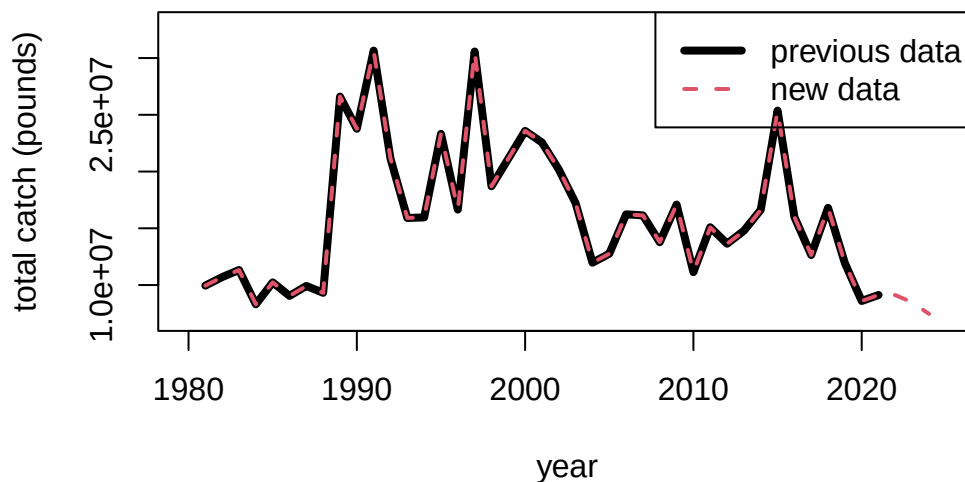
2 NOAA Fisheries recreational landings data

Now we can summarize the total dolphin landings in weight for the U.S. Atlantic, and compare to previous values sent for quota monitoring. We see that the values are almost identical.

```
# summarize by year and compare -----  
  
tot1 <- tapply(d1$LBSEST_SECWWT, d1$YEAR, sum, na.rm = T)  
head(tot1)
```

```
      1981      1982      1983      1984      1985      1986  
9952644 10701659 11325575  8303073 10236339  9047438
```

```
old <- read.csv("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", skip = 6)  
plot(old$Year, old$ATL, type = "l", xlab = "year", ylab = "total catch (pounds)", lwd = 4,  
      xlim = c(1980, 2025), ylim = c(7*10^6, 3.3*10^7))  
lines(as.numeric(names(tot1)), tot1, lwd = 2, col = 2, lty = 2)  
legend("topright", c("previous data", "new data"), col = c(1, 2), lwd = c(4, 2), lty = c(1,
```



2.3 Partition catches by area and sector

Now we will partition out the total catches according to the dolphin MSE areas. The areas are as follows:

- FLK: Florida Keys to Indian River County County (FL)
- NCFL: Brevard (FL) to Southern NC (South of Hatteras)
- NNC: Northern NC (North of Hatteras to NC/VA border)
- VBM: Virginia to Maine

```
# separate out 4 regions -----

d1$region <- "VBM"

# select for FLK
# FL_REG codes: 4: SE FL- Miami-Dade to Indian River County.; 5: NE FL- Brevard to Nassau C
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0
<NA>	14	114	0	1794	0	54	20	125	1344	108	73	62	644	165

	<NA>
3	0
4	0
5	0
<NA>	0

```
d1$region[which(d1$FL_REG == 3 | d1$FL_REG == 4)] <- "FLK"

# select for NCFL
d1$region[which(d1$FL_REG == 5)] <- "NCFL"
lis <- c("SC", "GA", "FLE/GA") # NCFL states
d1$region[which(d1$NEW_STA %in% lis)] <- "NCFL"
# NC_REG codes: N: North of Cape Hatteras; S: South of Cape Hatteras
table(d1$NC_REG, d1$NEW_STA, useNA = "always")
```

2 NOAA Fisheries recreational landings data

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	14	114	1334	1794	568	54	20	125	586	108	73	62	644	165
N	0	0	0	0	0	0	0	0	420	0	0	0	0	0
S	0	0	0	0	0	0	0	0	338	0	0	0	0	0
<NA>	0	0	0	0	0	0	0	0	0	0	0	0	0	0

<NA>	0
N	0
S	0
<NA>	0

```
d1$region[which(d1$NC == "S")] <- "NCFL"

# select for NNC
d1$region[which(d1$NC == "N")] <- "NNC"

table(d1$FL_REG, d1$NEW_STA, d1$region)
```

, , = FLK

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = NCFL

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0

, , = NNC

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
--	----	----	-----	--------	-----	----	----	----	----	----	----	----	----	----

2 NOAA Fisheries recreational landings data

3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = VBM

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0

```
table(d1$NC_REG, d1$NEW_STA, d1$region) # check classification
```

, , = FLK

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	0	0	1022	0	568	0	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = NCFL

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	0	0	312	1794	0	54	0	0	0	0	0	0	644	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	338	0	0	0	0	0

, , = NNC

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0	420	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = VBM

2 NOAA Fisheries recreational landings data

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	14	114	0	0	0	0	20	125	586	108	73	62	0	165
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Now we will separate out the catches by recreational sector and plot the totals by region and sector.

```
# separate out by for hire vs. private rec -----
d1$sector <- "ForHire"

d1$sector[which(d1$NEW_MODEN == "Shore")] <- "Rec"
d1$sector[which(d1$NEW_MODEN == "Priv")] <- "Rec"

table(d1$sector, d1$NEW_MODEN, useNA = "always") # check classification
```

	Cbt	Cbt/Hbt	Hbt	Priv	Shore	<NA>
ForHire	1616	119	2912	0	0	0
Rec	0	0	0	1750	22	0
<NA>	0	0	0	0	0	0

```
tot <- tapply(d1$LBSEST_SECWWT, list(d1$YEAR, d1$region, d1$sector), sum, na.rm = T)

head(tot)
```

, , ForHire

	FLK	NCFL	NNC	VBM
1981	354639.5	74702.50	256755.24	1400.247
1982	347458.2	92271.42	27578.44	2749.224
1983	1020005.8	41784.04	88690.53	5379.177
1984	1021086.8	74658.81	NA	4203.048
1985	717523.0	172989.65	58514.13	8880.857
1986	1562354.0	63555.89	817192.26	61658.193

, , Rec

2 NOAA Fisheries recreational landings data

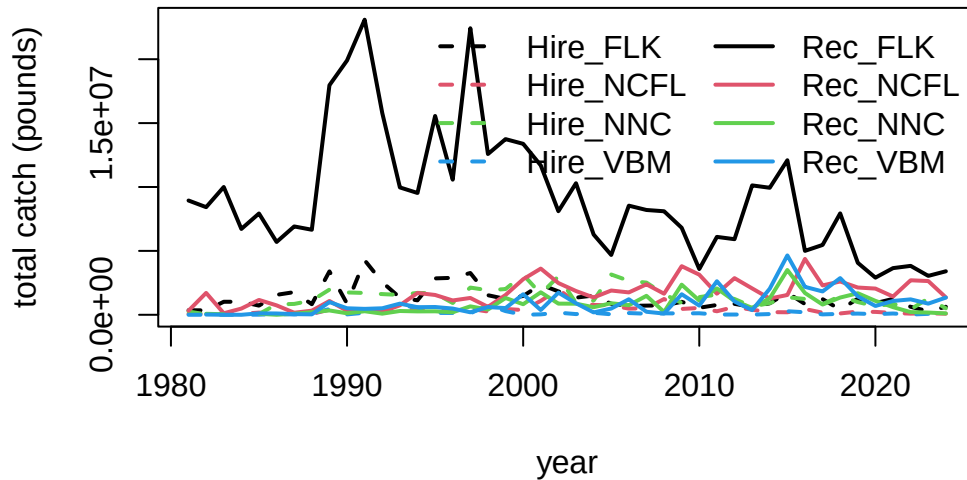
	FLK	NCFL	NNC	VBM
1981	8942641	322505.7	NA	NA
1982	8422673	1711834.7	78367.161	18727.290
1983	10000662	119370.4	44226.032	5456.237
1984	6717044	486079.7	NA	0.000
1985	7925301	1162708.4	77477.096	112944.274
1986	5699311	736872.0	2549.586	103945.134

```
fin <- data.frame(cbind(tot[, , 1], tot[, , 2]))
names(fin) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_NNC")
head(fin)
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981	354639.5	74702.50	256755.24	1400.247	8942641	322505.7	NA
1982	347458.2	92271.42	27578.44	2749.224	8422673	1711834.7	78367.161
1983	1020005.8	41784.04	88690.53	5379.177	10000662	119370.4	44226.032
1984	1021086.8	74658.81	NA	4203.048	6717044	486079.7	NA
1985	717523.0	172989.65	58514.13	8880.857	7925301	1162708.4	77477.096
1986	1562354.0	63555.89	817192.26	61658.193	5699311	736872.0	2549.586
	Rec_VBM						
1981	NA						
1982	18727.290						
1983	5456.237						
1984	0.000						
1985	112944.274						
1986	103945.134						

```
matplot(rownames(fin), fin, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), type = "l", las = 1,
        xlab = "year", ylab = "total catch (pounds)")
legend("topright", colnames(fin), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2)
```

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```
# check that totals match
cor(rowSums(fin, na.rm = T), tot1)
```

```
[1] 1
```

```
mean(rowSums(fin, na.rm = T) - tot1) # very tiny differences
```

```
[1] 4.233284e-11
```

Here are the catches (in pounds) for years 1981 to 2024.

```
fin
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981	354639.5	74702.50	256755.24	1400.247	8942641	322505.7	NA
1982	347458.2	92271.42	27578.44	2749.224	8422673	1711834.7	78367.161
1983	1020005.8	41784.04	88690.53	5379.177	10000662	119370.4	44226.032
1984	1021086.8	74658.81	NA	4203.048	6717044	486079.7	NA
1985	717523.0	172989.65	58514.13	8880.857	7925301	1162708.4	77477.096

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1986	1562354.0	63555.89	817192.26	61658.193	5699311	736872.0	2549.586
1987	1774558.7	61243.58	848994.25	24967.877	6913585	161627.1	79540.365
1988	826480.6	62392.87	1114627.05	94676.212	6655923	321023.1	202768.950
1989	3398703.5	405140.01	1970776.23	451477.768	17965047	1079951.5	339371.567
1990	868572.6	277783.17	1749965.55	31079.442	19900865	312333.3	118539.960
1991	4268773.7	225552.64	1706751.50	159172.855	23093044	457667.1	277389.646
1992	2548407.4	189514.34	1619685.06	161149.703	15788853	245395.8	89160.860
1993	1343093.6	391254.33	1555850.09	741955.369	9967356	742741.8	296399.199
1994	1131831.7	437642.72	1713981.98	573551.716	9532763	1708942.1	264907.193
1995	2848691.7	667475.30	1673435.48	145798.587	15558489	1550136.0	271442.073
1996	2892436.3	309604.42	923128.81	141976.431	10574454	1121937.8	196579.537
1997	3271271.6	466891.67	2129471.46	110115.485	22423256	1316502.7	660257.843
1998	1532460.0	252620.73	1921834.08	779364.855	12578694	619824.1	412304.292
1999	1238462.0	457880.10	2014752.29	288625.130	13745135	1548217.4	1314805.374
2000	1471481.3	369553.33	3139555.98	3016.589	13379647	2784424.2	826980.337
2001	2361751.6	1090421.00	1638707.54	34335.241	11690170	3614370.3	1760721.144
2002	1828918.7	1993778.99	3005172.92	168406.810	8105262	2483946.6	871930.660
2003	1342339.1	958997.12	928894.01	55690.419	10294918	1860435.4	868200.254
2004	1509660.5	771820.65	1116373.11	155510.796	6281794	1354303.7	593257.716
2005	858090.6	734896.20	3153480.89	38020.553	4681210	1898945.9	904360.210
2006	700089.5	483337.12	2656240.22	136715.477	8540432	1751853.5	749541.413
2007	718651.9	506776.30	2518430.49	84908.300	8200488	2383417.1	1483630.590
2008	736213.2	1235299.71	1620865.87	113592.877	8100550	1648662.2	223565.185
2009	1016583.6	458562.82	928419.86	106262.915	6809447	3805127.7	2359177.817
2010	552080.9	534671.06	1384434.93	120050.410	3579591	3139199.6	1110479.730
2011	768597.7	275546.65	1577019.96	17844.729	6097050	1651096.4	2063315.642
2012	856261.4	615430.34	1236410.76	21887.318	5916144	2867599.1	1082607.063
2013	614150.1	394570.11	640465.60	16064.763	10124799	2064256.6	540676.961
2014	897508.4	210198.76	916695.96	56760.758	9940020	1272477.3	1289704.416
2015	1729825.9	196326.97	1374908.91	277867.986	12093928	1548428.5	3505194.743
2016	841662.5	454053.51	1236565.76	195837.854	5002273	4372622.6	1704661.669
2017	1227827.8	161188.31	835080.86	31185.061	5464893	2297930.0	801324.151
2018	401703.0	105328.49	1465087.58	72211.785	7935729	2606091.5	1347724.203
2019	1261827.1	250624.40	998102.45	93718.699	4064867	2132483.1	1687239.863
2020	910431.4	225493.79	667501.79	51621.152	2906913	2052222.0	1089245.171
2021	1244842.2	146893.15	871760.94	100997.979	3652596	1428508.3	596672.480
2022	627020.6	90504.71	427953.34	18514.133	3828113	2701619.8	211394.537
2023	274532.5	118528.80	1281254.55	72728.020	3045801	2639445.1	175189.224
2024	629327.2	85572.19	506558.07	18752.239	3404227	1358519.1	106307.278
	Rec_VBM						
1981	NA						

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1982	18727.290
1983	5456.237
1984	0.000
1985	112944.274
1986	103945.134
1987	52139.772
1988	33566.163
1989	996945.390
1990	509424.981
1991	464786.733
1992	509297.261
1993	871948.601
1994	594467.414
1995	609302.619
1996	486106.187
1997	197871.963
1998	606769.298
1999	523623.688
2000	1608478.920
2001	372623.273
2002	1732356.036
2003	904780.558
2004	186538.567
2005	489236.451
2006	1214497.040
2007	236769.123
2008	96818.461
2009	1607919.004
2010	717410.858
2011	2626410.052
2012	1041694.282
2013	364492.286
2014	2040329.550
2015	4645796.249
2016	2186126.672
2017	1826822.849
2018	2871124.410
2019	1440420.940
2020	689099.471
2021	1083268.811
2022	1206278.324

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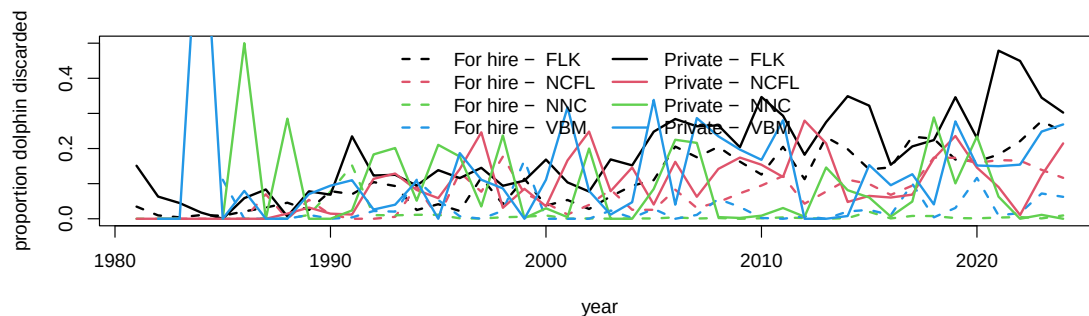
2023 872490.886

2024 1346606.842

2.4 Looking at extent of discarding

Discard estimates (known as B2 estimates) are provided in numbers by the MRIP survey but are not available in the Southeast Region Headboat Survey. We can look at the overall numbers of dolphinfish caught versus reported released from the MRIP survey to get an understanding of the extent of discarding.

```
d1$perdis <- d1$B2 / (d1$AB1 + d1$B2) # percent discarded is B2 (fish released alive) / to  
  
# calculate mean discard rate by year, region, sector -----  
dis <- tapply(d1$perdis, list(d1$YEAR, d1$region, d1$sector), mean, na.rm = T)  
disr <- tapply(d1$perdis, list(d1$region, d1$sector), mean, na.rm = T)  
  
matplot(rownames(dis), cbind(dis[, , 1], dis[, , 2]), lty = c(rep(2, 4), rep(1, 4)), col =  
        type = "l", lwd = 2, xlab = "year", ylab = "proportion dolphin discarded", ylim = c  
legend("top", c(paste("For hire - ", colnames(dis)), paste("Private - ", colnames(dis))),  
        lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2, ncol = 2, bty = "n")
```



```
round(disr * 100, 2) # discarding rates by region and sector
```

	ForHire	Rec
FLK	10.79	18.76
NCFL	9.44	11.23

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NNC	0.94	9.62
VBM	2.94	12.69

According to the MRIP survey and the reported numbers, discarding generally increases over time, excluding some highly anomalous estimates from the early years which are driven by low sample sizes. Discarding is generally higher in the private sector compared to the for-hire sector, and is highest in the Florida Keys region.

2.5 Split the catches by season

The dolphin MSE operates on a seasonal time step and thus the catches and discards must be parsed apart by individual season and year. This is not straightforward because MRIP estimates are reported in two-month waves (Jan - Feb, Mar - Apr, May - Jun, etc.), and the seasons are 3-month periods (Dec - Feb, Mar - May, Jun - Aug, Sep - Nov). For the two waves that have to be split into seasons (e.g., December is lumped with January and February; June is lumped with July and August), we split the wave 50/50%, i.e, we make the assumption that half of the catch in that wave occurs in one season and the other half occurs in the next season.

```
# define seasons -----  
table(d1$WAVE, useNA = "always")
```

0	1	2	3	4	5	6	<NA>
5	407	718	1230	1439	1095	581	944

```
d1$YRWAVE <- paste0(d1$YEAR, "_", d1$WAVE)  
  
totw <- tapply(d1$LBSEST_SECWWT, list(d1$YRWAVE, d1$region, d1$sector), sum, na.rm = T)  
  
f <- data.frame(cbind(totw[, , 1], totw[, , 2]))  
names(f) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_NNC")  
head(f)
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981_2	NA	NA	NA	NA	2793285.5	270968.46	NA
1981_3	300873.764	NA	254049.420	NA	3915698.3	NA	NA

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1981_4	50017.460	NA	2705.821	NA	905621.2	51537.26	NA
1981_5	3748.301	NA	NA	NA	479834.0	NA	NA
1981_6	NA	NA	NA	NA	848201.5	NA	NA
1981_NA	NA	74702.5	NA	1400.247	NA	NA	NA

	Rec_VBM
1981_2	NA
1981_3	NA
1981_4	NA
1981_5	NA
1981_6	NA
1981_NA	NA

```
f$YEAR <- as.numeric(substr(rownames(f), 1, 4))
f$WAVE <- as.numeric(substr(rownames(f), 6, 8))
head(f) # table with catches separated by wave and year
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981_2	NA	NA	NA	NA	2793285.5	270968.46	NA
1981_3	300873.764	NA	254049.420	NA	3915698.3	NA	NA
1981_4	50017.460	NA	2705.821	NA	905621.2	51537.26	NA
1981_5	3748.301	NA	NA	NA	479834.0	NA	NA
1981_6	NA	NA	NA	NA	848201.5	NA	NA
1981_NA	NA	74702.5	NA	1400.247	NA	NA	NA

	Rec_VBM	YEAR	WAVE
1981_2	NA	1981	2
1981_3	NA	1981	3
1981_4	NA	1981	4
1981_5	NA	1981	5
1981_6	NA	1981	6
1981_NA	NA	1981	NA

```
f[is.na(f)] <- 0

f <- f[which(f$YEAR >= 1985), ]
f <- f[which(f$YEAR <= 2024), ]

table(f$YEAR)
```

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1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
7	7	7	7	7	7	7	7	7	7	7	7	6	6	6	6
2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
6	6	6	6	6	6	6	6	6	6	6	6	7	7	7	6
2017	2018	2019	2020	2021	2022	2023	2024								
6	6	6	6	6	6	6	6								

```
table(f$WAVE)
```

```

0  1  2  3  4  5  6
15 40 40 40 40 40 40

```

```

# deal with NAs in many locations -----
# calculate percentages of catch by wave, for each region and fleet
avw <- tapply(d1$LBSEST_SECWWT, list(d1$WAVE, d1$region, d1$sector), sum, na.rm = T)
avw <- avw[-1, , ]
avw <- data.frame(cbind(avw[, , 1], avw[, , 2]))
names(avw) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Re
avwp <- apply(avw, 2, function (x) x / sum(x, na.rm = T))
#colSums(avwp)

# go through each column, find NAs and fill in waves by percentages
pre_colsum <- colSums(f, na.rm = T) # column sums to check later

lis <- which(f$WAVE == 0) # list of rows with NA wave data

for (i in 1:8) { # loop through each column
  for(j in lis) { # loop through list of NA waves
    if (f[j, i] > 0) { # if there are NA wave catches, split by known proportion from other
      miscatch <- avwp[, i] * f[j, i] # missing catch is known proportion * NA wave catch
      f[which(f$YEAR == f$YEAR[j] & f$WAVE != 0), i] <- f[which(f$YEAR == f$YEAR[j] & f$WAVE
      f[j, i] <- 0 # set to zero after catch is attributed to waves
    }
  }
}
}
f[lis, 1:8] # should be all zeros now

```

```
Hire_FLK Hire_NCFL Hire_NNC Hire_VBM Rec_FLK Rec_NCFL Rec_NNC Rec_VBM
```

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1985_NA	0	0	0	0	0	0	0	0
1986_NA	0	0	0	0	0	0	0	0
1987_NA	0	0	0	0	0	0	0	0
1988_NA	0	0	0	0	0	0	0	0
1989_NA	0	0	0	0	0	0	0	0
1990_NA	0	0	0	0	0	0	0	0
1991_NA	0	0	0	0	0	0	0	0
1992_NA	0	0	0	0	0	0	0	0
1993_NA	0	0	0	0	0	0	0	0
1994_NA	0	0	0	0	0	0	0	0
1995_NA	0	0	0	0	0	0	0	0
1996_NA	0	0	0	0	0	0	0	0
2013_0	0	0	0	0	0	0	0	0
2014_0	0	0	0	0	0	0	0	0
2015_0	0	0	0	0	0	0	0	0

```
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8] # check that catch was parsed correctly - column sums s
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

```
f <- f[which(f$WAVE != 0), ] # remove NA waves which now contain no catch

# split waves 3 and 6 50/50% to parse into seasons -----
lis <- which(f$WAVE == 3 | f$WAVE == 6) # list of 3 and 6 waves for all years

for (i in lis) {
  # go through list of waves
  temp <- f[i, ] / 2 # split catch in half
  temp[9] <- f$YEAR[i] # reformat year and wave data columns
  temp[10] <- f$WAVE[i]
  temp <- rbind(temp, temp) # create two half-waves and relabel with new wave reference n
  temp[1, 10] <- temp[1, 10] - 0.5
  temp[2, 10] <- temp[2, 10] + 0.5
  f <- rbind(f, temp) # concatenate with main data table
}

f <- f[-lis, ] # remove waves that have been split in half
```

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```
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8] # check that catch was parsed correctly - column sums s
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

```
f <- f[order(f$YEAR, f$WAVE), ] # reorder table in chronological order

f$YEAR[which(f$WAVE == 6.5)] <- f$YEAR[which(f$WAVE == 6.5)] + 1 # add year to December wa
f$WAVE[which(f$WAVE == 6.5)] <- 0.5 # change December wave to small number so it gets summ

table(f$WAVE)
```

0.5	1	2	2.5	3.5	4	5	5.5
40	40	40	40	40	40	40	40

```
f$seas <- cut(f$WAVE, breaks = c(0, 1.5, 3, 4.5, 6))
#table(f$seas)
f$seas1 <- NA
f$seas1[which(f$seas == "(0,1.5)")] <- "DJF" # Dec Jan Feb is 0.5 and 1
f$seas1[which(f$seas == "(1.5,3)")] <- "MAM" # Mar Apr May is 2 and 2.5
f$seas1[which(f$seas == "(3,4.5)")] <- "JJA" # Jun Jul Aug is 3.5 and 4
f$seas1[which(f$seas == "(4.5,6)")] <- "SON" # Sep Oct Nov is 5 and 5.5

temp <- tapply(f[, 1], list(f$seas1, f$YEAR), sum, na.rm = T)
temp1 <- matrix(temp)
for (i in 2:8) {
  temp2 <- matrix(tapply(f[, i], list(f$seas1, f$YEAR), sum, na.rm = T))
  temp1 <- cbind(temp1, temp2)
}

findat <- data.frame(sort(rep(as.numeric(colnames(temp)), 4)),
                     rep(rownames(temp), ncol(temp)), temp1)
names(findat) <- c("year", "season", names(f)[1:8])

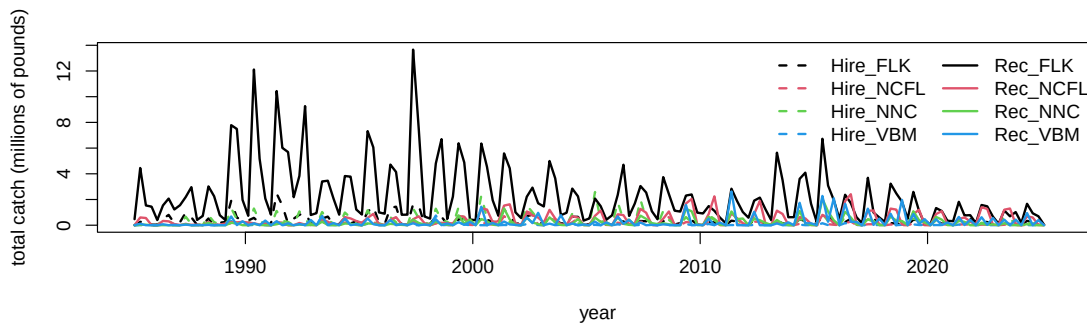
colSums(f[, 1:8], na.rm = T) == colSums(findat[, 3:10], na.rm = T) # check that total catc
```

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```
Hire_FLK Hire_NCFL Hire_NNC Hire_VBM Rec_FLK Rec_NCFL Rec_NNC Rec_VBM
      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

We plot the results by region and sector, over the time series of year and season. We can clearly see the seasonality in the catch by region.

```
findat$yrseas <- findat$year + as.numeric(as.factor(findat$season))/4-0.125
matplot(findat$yrseas, findat[, 3:10]/10^6, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2),
        xlab = "year", ylab = "total catch (millions of pounds)")
legend("topright", colnames(findat[, 3:10]), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2))
```



```
# check that totals match
cbind(colSums(findat[, 3:10], na.rm = T), colSums(fin[-c(1:5), ], na.rm = T))
```

	[,1]	[,2]
Hire_FLK	55604999	54887476
Hire_NCFL	16909916	16736926
Hire_NNC	57944903	57886388
Hire_VBM	5836953	5828072
Rec_FLK	371462933	363537632
Rec_NCFL	70803866	69641157
Rec_NNC	32556596	32479119
Rec_VBM	40081529	39968585

```
# the values should be close but slightly >1 because the last data set includes December 19
colSums(findat[, 3:10], na.rm = T) / colSums(fin[-c(1:5), ], na.rm = T)
```

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```
Hire_FLK Hire_NCFL Hire_NNC Hire_VBM Rec_FLK Rec_NCFL Rec_NNC Rec_VBM
1.013073 1.010336 1.001011 1.001524 1.021800 1.016696 1.002385 1.002826
```

```
#write.csv(findat, file = "data/outputs/MRIP_catch_by_season_4_areas.csv", row.names = FALSE)
```

Here we will format the data for input to the operating model

```
tab3 <- findat

flt <- c(rep("Hire", nrow(tab3)*4), rep("Rec", nrow(tab3)*4))

yrmon <- as.numeric(tab3$yrseas)
yrs <- floor(yrmon)
qrt <- (yrmon - floor(yrmon)) * 4 + 0.5
area <- c(rep("FLK", nrow(tab3)), rep("NCFL", nrow(tab3)),
          rep("NNC", nrow(tab3)), rep("VBM", nrow(tab3)))
area <- rep(area, 2)

findat1 <- data.frame(yrs, qrt, flt, area, matrix(as.matrix(tab3[,3:10])))
findat1
```

	yrs	qrt	flt	area	matrix.as.matrix.tab3...3.10...
1	1985	1	Hire	FLK	4.462090e+03
2	1985	2	Hire	FLK	2.534874e+05
3	1985	3	Hire	FLK	3.808114e+05
4	1985	4	Hire	FLK	4.146999e+04
5	1986	1	Hire	FLK	4.054541e+04
6	1986	2	Hire	FLK	5.011846e+05
7	1986	3	Hire	FLK	8.024703e+05
8	1986	4	Hire	FLK	1.511596e+05
9	1987	1	Hire	FLK	2.414829e+05
10	1987	2	Hire	FLK	9.787720e+05
11	1987	3	Hire	FLK	6.087522e+05
12	1987	4	Hire	FLK	2.491889e+04
13	1988	1	Hire	FLK	5.112349e+04
14	1988	2	Hire	FLK	5.236336e+05
15	1988	3	Hire	FLK	2.796967e+04
16	1988	4	Hire	FLK	1.960574e+05
17	1989	1	Hire	FLK	2.524921e+05

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18	1989	2 Hire	FLK	1.898385e+06
19	1989	3 Hire	FLK	5.918365e+05
20	1989	4 Hire	FLK	3.962056e+05
21	1990	1 Hire	FLK	3.673921e+05
22	1990	2 Hire	FLK	5.735168e+05
23	1990	3 Hire	FLK	1.672854e+05
24	1990	4 Hire	FLK	5.481602e+04
25	1991	1 Hire	FLK	1.200051e+05
26	1991	2 Hire	FLK	2.463709e+06
27	1991	3 Hire	FLK	1.530429e+06
28	1991	4 Hire	FLK	1.329583e+05
29	1992	1 Hire	FLK	5.113129e+05
30	1992	2 Hire	FLK	8.083241e+05
31	1992	3 Hire	FLK	9.688738e+05
32	1992	4 Hire	FLK	1.566697e+05
33	1993	1 Hire	FLK	2.333242e+05
34	1993	2 Hire	FLK	5.197939e+05
35	1993	3 Hire	FLK	6.692056e+05
36	1993	4 Hire	FLK	3.734484e+04
37	1994	1 Hire	FLK	8.981145e+04
38	1994	2 Hire	FLK	5.934543e+05
39	1994	3 Hire	FLK	1.406221e+05
40	1994	4 Hire	FLK	2.984649e+05
41	1995	1 Hire	FLK	1.390055e+05
42	1995	2 Hire	FLK	1.039095e+06
43	1995	3 Hire	FLK	1.461029e+06
44	1995	4 Hire	FLK	2.027951e+05
45	1996	1 Hire	FLK	7.773956e+04
46	1996	2 Hire	FLK	1.352282e+06
47	1996	3 Hire	FLK	1.439914e+06
48	1996	4 Hire	FLK	4.336462e+04
49	1997	1 Hire	FLK	3.874707e+04
50	1997	2 Hire	FLK	1.629481e+06
51	1997	3 Hire	FLK	1.514418e+06
52	1997	4 Hire	FLK	8.428625e+04
53	1998	1 Hire	FLK	9.909438e+04
54	1998	2 Hire	FLK	6.713363e+05
55	1998	3 Hire	FLK	6.563503e+05
56	1998	4 Hire	FLK	9.008397e+04
57	1999	1 Hire	FLK	1.387479e+05
58	1999	2 Hire	FLK	4.497292e+05

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59	1999	3	Hire	FLK	5.063335e+05
60	1999	4	Hire	FLK	1.268359e+05
61	2000	1	Hire	FLK	8.315590e+04
62	2000	2	Hire	FLK	7.265192e+05
63	2000	3	Hire	FLK	4.308598e+05
64	2000	4	Hire	FLK	2.502685e+05
65	2001	1	Hire	FLK	2.460745e+05
66	2001	2	Hire	FLK	1.233226e+06
67	2001	3	Hire	FLK	7.170208e+05
68	2001	4	Hire	FLK	1.353495e+05
69	2002	1	Hire	FLK	1.250031e+05
70	2002	2	Hire	FLK	7.042666e+05
71	2002	3	Hire	FLK	8.214989e+05
72	2002	4	Hire	FLK	2.144944e+05
73	2003	1	Hire	FLK	1.173219e+05
74	2003	2	Hire	FLK	6.037893e+05
75	2003	3	Hire	FLK	4.592291e+05
76	2003	4	Hire	FLK	1.306186e+05
77	2004	1	Hire	FLK	1.665554e+05
78	2004	2	Hire	FLK	6.537249e+05
79	2004	3	Hire	FLK	6.539262e+05
80	2004	4	Hire	FLK	6.991629e+04
81	2005	1	Hire	FLK	8.617154e+04
82	2005	2	Hire	FLK	3.823569e+05
83	2005	3	Hire	FLK	3.634808e+05
84	2005	4	Hire	FLK	4.281886e+04
85	2006	1	Hire	FLK	6.246935e+04
86	2006	2	Hire	FLK	2.945357e+05
87	2006	3	Hire	FLK	2.947750e+05
88	2006	4	Hire	FLK	4.741635e+04
89	2007	1	Hire	FLK	3.433560e+04
90	2007	2	Hire	FLK	3.291462e+05
91	2007	3	Hire	FLK	3.047614e+05
92	2007	4	Hire	FLK	4.854292e+04
93	2008	1	Hire	FLK	8.213770e+04
94	2008	2	Hire	FLK	3.145215e+05
95	2008	3	Hire	FLK	2.804606e+05
96	2008	4	Hire	FLK	4.715174e+04
97	2009	1	Hire	FLK	1.731352e+05
98	2009	2	Hire	FLK	4.032198e+05
99	2009	3	Hire	FLK	3.926185e+05

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100	2009	4	Hire	FLK	5.327003e+04
101	2010	1	Hire	FLK	3.615959e+04
102	2010	2	Hire	FLK	2.687809e+05
103	2010	3	Hire	FLK	2.193667e+05
104	2010	4	Hire	FLK	3.321194e+04
105	2011	1	Hire	FLK	2.245187e+04
106	2011	2	Hire	FLK	3.516103e+05
107	2011	3	Hire	FLK	2.843143e+05
108	2011	4	Hire	FLK	8.316655e+04
109	2012	1	Hire	FLK	8.049080e+04
110	2012	2	Hire	FLK	3.419584e+05
111	2012	3	Hire	FLK	3.499724e+05
112	2012	4	Hire	FLK	9.702987e+04
113	2013	1	Hire	FLK	7.075256e+04
114	2013	2	Hire	FLK	3.346052e+05
115	2013	3	Hire	FLK	1.512397e+05
116	2013	4	Hire	FLK	5.990593e+04
117	2014	1	Hire	FLK	3.864672e+04
118	2014	2	Hire	FLK	4.783451e+05
119	2014	3	Hire	FLK	3.063555e+05
120	2014	4	Hire	FLK	7.951533e+04
121	2015	1	Hire	FLK	1.126962e+05
122	2015	2	Hire	FLK	6.722192e+05
123	2015	3	Hire	FLK	7.334898e+05
124	2015	4	Hire	FLK	1.694061e+05
125	2016	1	Hire	FLK	1.344365e+05
126	2016	2	Hire	FLK	2.417873e+05
127	2016	3	Hire	FLK	5.033407e+05
128	2016	4	Hire	FLK	1.743354e+04
129	2017	1	Hire	FLK	1.170229e+04
130	2017	2	Hire	FLK	6.117980e+05
131	2017	3	Hire	FLK	5.660774e+05
132	2017	4	Hire	FLK	3.588338e+04
133	2018	1	Hire	FLK	2.329070e+04
134	2018	2	Hire	FLK	1.653072e+05
135	2018	3	Hire	FLK	1.361174e+05
136	2018	4	Hire	FLK	8.056149e+04
137	2019	1	Hire	FLK	2.832672e+04
138	2019	2	Hire	FLK	4.269701e+05
139	2019	3	Hire	FLK	3.745493e+05
140	2019	4	Hire	FLK	4.160198e+05

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141	2020	1	Hire	FLK	2.420522e+04
142	2020	2	Hire	FLK	5.096188e+05
143	2020	3	Hire	FLK	7.868668e+04
144	2020	4	Hire	FLK	3.096802e+05
145	2021	1	Hire	FLK	2.795122e+04
146	2021	2	Hire	FLK	5.011873e+05
147	2021	3	Hire	FLK	2.016812e+05
148	2021	4	Hire	FLK	5.132135e+05
149	2022	1	Hire	FLK	2.639086e+04
150	2022	2	Hire	FLK	3.009903e+05
151	2022	3	Hire	FLK	2.459965e+05
152	2022	4	Hire	FLK	5.880012e+04
153	2023	1	Hire	FLK	1.499705e+04
154	2023	2	Hire	FLK	1.198942e+05
155	2023	3	Hire	FLK	9.200433e+04
156	2023	4	Hire	FLK	5.271621e+04
157	2024	1	Hire	FLK	3.809446e+03
158	2024	2	Hire	FLK	3.415764e+05
159	2024	3	Hire	FLK	2.234813e+05
160	2024	4	Hire	FLK	4.784226e+04
161	2025	1	Hire	FLK	1.563611e+04
162	2025	2	Hire	FLK	NA
163	2025	3	Hire	FLK	NA
164	2025	4	Hire	FLK	NA
165	1985	1	Hire	NCFL	1.294748e+02
166	1985	2	Hire	NCFL	9.054349e+04
167	1985	3	Hire	NCFL	8.062661e+04
168	1985	4	Hire	NCFL	1.555487e+03
169	1986	1	Hire	NCFL	3.360800e+02
170	1986	2	Hire	NCFL	2.800349e+04
171	1986	3	Hire	NCFL	2.370312e+04
172	1986	4	Hire	NCFL	1.143834e+04
173	1987	1	Hire	NCFL	4.103876e+02
174	1987	2	Hire	NCFL	3.073598e+04
175	1987	3	Hire	NCFL	2.499202e+04
176	1987	4	Hire	NCFL	5.105764e+03
177	1988	1	Hire	NCFL	3.474931e+02
178	1988	2	Hire	NCFL	3.493479e+04
179	1988	3	Hire	NCFL	1.686078e+04
180	1988	4	Hire	NCFL	1.018500e+04
181	1989	1	Hire	NCFL	4.729552e+02

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182	1989	2	Hire	NCFL	1.780131e+05
183	1989	3	Hire	NCFL	2.109711e+05
184	1989	4	Hire	NCFL	1.556368e+04
185	1990	1	Hire	NCFL	7.824121e+02
186	1990	2	Hire	NCFL	1.476583e+05
187	1990	3	Hire	NCFL	1.154803e+05
188	1990	4	Hire	NCFL	1.385021e+04
189	1991	1	Hire	NCFL	6.480516e+02
190	1991	2	Hire	NCFL	1.118934e+05
191	1991	3	Hire	NCFL	9.679768e+04
192	1991	4	Hire	NCFL	1.054628e+04
193	1992	1	Hire	NCFL	6.204025e+03
194	1992	2	Hire	NCFL	1.125717e+05
195	1992	3	Hire	NCFL	6.841681e+04
196	1992	4	Hire	NCFL	7.686783e+03
197	1993	1	Hire	NCFL	8.603992e+02
198	1993	2	Hire	NCFL	2.689442e+05
199	1993	3	Hire	NCFL	1.201582e+05
200	1993	4	Hire	NCFL	1.839589e+03
201	1994	1	Hire	NCFL	2.400880e+02
202	1994	2	Hire	NCFL	2.077107e+05
203	1994	3	Hire	NCFL	2.113411e+05
204	1994	4	Hire	NCFL	1.834842e+04
205	1995	1	Hire	NCFL	3.394897e+02
206	1995	2	Hire	NCFL	3.896585e+05
207	1995	3	Hire	NCFL	2.726127e+05
208	1995	4	Hire	NCFL	4.841136e+03
209	1996	1	Hire	NCFL	8.213613e+03
210	1996	2	Hire	NCFL	1.778849e+05
211	1996	3	Hire	NCFL	1.015929e+05
212	1996	4	Hire	NCFL	2.140903e+04
213	1997	1	Hire	NCFL	3.872523e+03
214	1997	2	Hire	NCFL	2.720876e+05
215	1997	3	Hire	NCFL	1.493999e+05
216	1997	4	Hire	NCFL	3.987688e+04
217	1998	1	Hire	NCFL	3.176531e+03
218	1998	2	Hire	NCFL	1.367443e+05
219	1998	3	Hire	NCFL	1.057200e+05
220	1998	4	Hire	NCFL	6.394769e+03
221	1999	1	Hire	NCFL	1.617787e+04
222	1999	2	Hire	NCFL	2.315459e+05

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223	1999	3	Hire	NCFL	1.855573e+05
224	1999	4	Hire	NCFL	2.427385e+04
225	2000	1	Hire	NCFL	9.786403e+03
226	2000	2	Hire	NCFL	2.209559e+05
227	2000	3	Hire	NCFL	1.004109e+05
228	2000	4	Hire	NCFL	3.691695e+04
229	2001	1	Hire	NCFL	1.444289e+04
230	2001	2	Hire	NCFL	7.099938e+05
231	2001	3	Hire	NCFL	3.323576e+05
232	2001	4	Hire	NCFL	3.669715e+04
233	2002	1	Hire	NCFL	3.452582e+03
234	2002	2	Hire	NCFL	1.067933e+06
235	2002	3	Hire	NCFL	8.876937e+05
236	2002	4	Hire	NCFL	3.528256e+04
237	2003	1	Hire	NCFL	3.840700e+03
238	2003	2	Hire	NCFL	5.318615e+05
239	2003	3	Hire	NCFL	4.048370e+05
240	2003	4	Hire	NCFL	1.538532e+04
241	2004	1	Hire	NCFL	4.478344e+03
242	2004	2	Hire	NCFL	2.297842e+05
243	2004	3	Hire	NCFL	5.213514e+05
244	2004	4	Hire	NCFL	1.644082e+04
245	2005	1	Hire	NCFL	6.818692e+03
246	2005	2	Hire	NCFL	4.295189e+05
247	2005	3	Hire	NCFL	2.923724e+05
248	2005	4	Hire	NCFL	7.924777e+03
249	2006	1	Hire	NCFL	2.665228e+03
250	2006	2	Hire	NCFL	1.996467e+05
251	2006	3	Hire	NCFL	1.077716e+05
252	2006	4	Hire	NCFL	1.744758e+05
253	2007	1	Hire	NCFL	2.214669e+03
254	2007	2	Hire	NCFL	2.668275e+05
255	2007	3	Hire	NCFL	2.325079e+05
256	2007	4	Hire	NCFL	3.984227e+03
257	2008	1	Hire	NCFL	3.435103e+03
258	2008	2	Hire	NCFL	5.630532e+05
259	2008	3	Hire	NCFL	6.296220e+05
260	2008	4	Hire	NCFL	2.797176e+04
261	2009	1	Hire	NCFL	1.553554e+04
262	2009	2	Hire	NCFL	2.437370e+05
263	2009	3	Hire	NCFL	2.048967e+05

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264	2009	4	Hire	NCFL	5.340980e+03
265	2010	1	Hire	NCFL	3.657414e+03
266	2010	2	Hire	NCFL	2.856638e+05
267	2010	3	Hire	NCFL	2.244790e+05
268	2010	4	Hire	NCFL	2.191794e+04
269	2011	1	Hire	NCFL	1.988053e+03
270	2011	2	Hire	NCFL	1.798123e+05
271	2011	3	Hire	NCFL	8.224854e+04
272	2011	4	Hire	NCFL	1.139008e+04
273	2012	1	Hire	NCFL	2.961105e+03
274	2012	2	Hire	NCFL	2.766761e+05
275	2012	3	Hire	NCFL	2.701667e+05
276	2012	4	Hire	NCFL	6.606708e+04
277	2013	1	Hire	NCFL	1.779384e+03
278	2013	2	Hire	NCFL	2.194358e+05
279	2013	3	Hire	NCFL	1.575236e+05
280	2013	4	Hire	NCFL	1.566620e+04
281	2014	1	Hire	NCFL	4.518215e+03
282	2014	2	Hire	NCFL	1.092633e+05
283	2014	3	Hire	NCFL	8.058665e+04
284	2014	4	Hire	NCFL	1.502980e+04
285	2015	1	Hire	NCFL	4.176974e+03
286	2015	2	Hire	NCFL	7.924758e+04
287	2015	3	Hire	NCFL	7.648718e+04
288	2015	4	Hire	NCFL	3.612587e+04
289	2016	1	Hire	NCFL	3.698497e+03
290	2016	2	Hire	NCFL	1.550273e+05
291	2016	3	Hire	NCFL	2.872233e+05
292	2016	4	Hire	NCFL	8.996649e+03
293	2017	1	Hire	NCFL	2.040101e+03
294	2017	2	Hire	NCFL	7.402020e+04
295	2017	3	Hire	NCFL	7.692057e+04
296	2017	4	Hire	NCFL	9.207619e+03
297	2018	1	Hire	NCFL	7.239204e+02
298	2018	2	Hire	NCFL	5.468464e+04
299	2018	3	Hire	NCFL	4.147245e+04
300	2018	4	Hire	NCFL	6.945761e+03
301	2019	1	Hire	NCFL	2.719131e+03
302	2019	2	Hire	NCFL	1.240299e+05
303	2019	3	Hire	NCFL	1.203616e+05
304	2019	4	Hire	NCFL	5.238812e+03

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305	2020	1	Hire	NCFL	4.188082e+02
306	2020	2	Hire	NCFL	1.058325e+05
307	2020	3	Hire	NCFL	8.464547e+04
308	2020	4	Hire	NCFL	3.438865e+04
309	2021	1	Hire	NCFL	7.634698e+02
310	2021	2	Hire	NCFL	5.397084e+04
311	2021	3	Hire	NCFL	7.160624e+04
312	2021	4	Hire	NCFL	1.839464e+04
313	2022	1	Hire	NCFL	3.239231e+03
314	2022	2	Hire	NCFL	4.272998e+04
315	2022	3	Hire	NCFL	4.157946e+04
316	2022	4	Hire	NCFL	5.359988e+03
317	2023	1	Hire	NCFL	3.092289e+02
318	2023	2	Hire	NCFL	3.614031e+04
319	2023	3	Hire	NCFL	7.173598e+04
320	2023	4	Hire	NCFL	1.029238e+04
321	2024	1	Hire	NCFL	4.297564e+02
322	2024	2	Hire	NCFL	2.305942e+04
323	2024	3	Hire	NCFL	4.334767e+04
324	2024	4	Hire	NCFL	1.839772e+04
325	2025	1	Hire	NCFL	5.841041e+02
326	2025	2	Hire	NCFL	NA
327	2025	3	Hire	NCFL	NA
328	2025	4	Hire	NCFL	NA
329	1985	1	Hire	NNC	0.000000e+00
330	1985	2	Hire	NNC	2.925707e+04
331	1985	3	Hire	NNC	2.925707e+04
332	1985	4	Hire	NNC	0.000000e+00
333	1986	1	Hire	NNC	0.000000e+00
334	1986	2	Hire	NNC	4.331248e+05
335	1986	3	Hire	NNC	3.259995e+05
336	1986	4	Hire	NNC	5.640033e+04
337	1987	1	Hire	NNC	1.667679e+03
338	1987	2	Hire	NNC	6.634651e+05
339	1987	3	Hire	NNC	8.440390e+04
340	1987	4	Hire	NNC	1.011253e+05
341	1988	1	Hire	NNC	0.000000e+00
342	1988	2	Hire	NNC	7.807843e+05
343	1988	3	Hire	NNC	2.294885e+05
344	1988	4	Hire	NNC	1.043543e+05
345	1989	1	Hire	NNC	0.000000e+00

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346	1989	2	Hire	NNC	1.331147e+06
347	1989	3	Hire	NNC	4.566275e+05
348	1989	4	Hire	NNC	1.819174e+05
349	1990	1	Hire	NNC	1.084238e+03
350	1990	2	Hire	NNC	1.298744e+06
351	1990	3	Hire	NNC	2.762370e+05
352	1990	4	Hire	NNC	1.747245e+05
353	1991	1	Hire	NNC	2.602455e+02
354	1991	2	Hire	NNC	1.111538e+06
355	1991	3	Hire	NNC	3.386228e+05
356	1991	4	Hire	NNC	2.565000e+05
357	1992	1	Hire	NNC	9.123982e+01
358	1992	2	Hire	NNC	1.074642e+06
359	1992	3	Hire	NNC	4.056591e+05
360	1992	4	Hire	NNC	1.393842e+05
361	1993	1	Hire	NNC	0.000000e+00
362	1993	2	Hire	NNC	1.026905e+06
363	1993	3	Hire	NNC	3.961182e+05
364	1993	4	Hire	NNC	1.327154e+05
365	1994	1	Hire	NNC	1.114007e+02
366	1994	2	Hire	NNC	9.896425e+05
367	1994	3	Hire	NNC	5.843752e+05
368	1994	4	Hire	NNC	1.351256e+05
369	1995	1	Hire	NNC	4.838705e+03
370	1995	2	Hire	NNC	1.163134e+06
371	1995	3	Hire	NNC	3.596215e+05
372	1995	4	Hire	NNC	1.501280e+05
373	1996	1	Hire	NNC	5.518140e+02
374	1996	2	Hire	NNC	6.272307e+05
375	1996	3	Hire	NNC	2.266328e+05
376	1996	4	Hire	NNC	6.914937e+04
377	1997	1	Hire	NNC	1.159216e+02
378	1997	2	Hire	NNC	1.305418e+06
379	1997	3	Hire	NNC	5.384247e+05
380	1997	4	Hire	NNC	2.838386e+05
381	1998	1	Hire	NNC	1.790000e+03
382	1998	2	Hire	NNC	1.286262e+06
383	1998	3	Hire	NNC	3.720073e+05
384	1998	4	Hire	NNC	2.627216e+05
385	1999	1	Hire	NNC	8.427480e+02
386	1999	2	Hire	NNC	1.408221e+06

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387	1999	3	Hire	NNC	4.320121e+05
388	1999	4	Hire	NNC	1.742143e+05
389	2000	1	Hire	NNC	3.049959e+02
390	2000	2	Hire	NNC	2.310664e+06
391	2000	3	Hire	NNC	5.314265e+05
392	2000	4	Hire	NNC	2.973395e+05
393	2001	1	Hire	NNC	1.255034e+02
394	2001	2	Hire	NNC	1.246703e+06
395	2001	3	Hire	NNC	1.912235e+05
396	2001	4	Hire	NNC	2.001049e+05
397	2002	1	Hire	NNC	6.761108e+02
398	2002	2	Hire	NNC	1.545059e+06
399	2002	3	Hire	NNC	1.004864e+06
400	2002	4	Hire	NNC	4.540662e+05
401	2003	1	Hire	NNC	1.183999e+03
402	2003	2	Hire	NNC	6.743990e+05
403	2003	3	Hire	NNC	2.130953e+05
404	2003	4	Hire	NNC	3.523793e+04
405	2004	1	Hire	NNC	6.161745e+03
406	2004	2	Hire	NNC	8.934973e+05
407	2004	3	Hire	NNC	1.489910e+05
408	2004	4	Hire	NNC	7.388486e+04
409	2005	1	Hire	NNC	0.000000e+00
410	2005	2	Hire	NNC	2.585572e+06
411	2005	3	Hire	NNC	3.369815e+05
412	2005	4	Hire	NNC	2.302774e+05
413	2006	1	Hire	NNC	6.495267e+02
414	2006	2	Hire	NNC	1.798733e+06
415	2006	3	Hire	NNC	4.041334e+05
416	2006	4	Hire	NNC	4.271605e+05
417	2007	1	Hire	NNC	2.637966e+04
418	2007	2	Hire	NNC	1.993879e+06
419	2007	3	Hire	NNC	4.980747e+05
420	2007	4	Hire	NNC	2.590388e+04
421	2008	1	Hire	NNC	4.065881e+02
422	2008	2	Hire	NNC	1.103121e+06
423	2008	3	Hire	NNC	4.078345e+05
424	2008	4	Hire	NNC	8.570458e+04
425	2009	1	Hire	NNC	2.420607e+04
426	2009	2	Hire	NNC	6.344073e+05
427	2009	3	Hire	NNC	1.952106e+05

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428	2009	4	Hire	NNC	9.611454e+04
429	2010	1	Hire	NNC	2.687381e+03
430	2010	2	Hire	NNC	7.870523e+05
431	2010	3	Hire	NNC	4.898075e+05
432	2010	4	Hire	NNC	1.070805e+05
433	2011	1	Hire	NNC	4.946504e+02
434	2011	2	Hire	NNC	1.081341e+06
435	2011	3	Hire	NNC	4.121178e+05
436	2011	4	Hire	NNC	8.270966e+04
437	2012	1	Hire	NNC	1.942786e+03
438	2012	2	Hire	NNC	6.529994e+05
439	2012	3	Hire	NNC	4.872517e+05
440	2012	4	Hire	NNC	9.417821e+04
441	2013	1	Hire	NNC	1.180400e+03
442	2013	2	Hire	NNC	3.829178e+05
443	2013	3	Hire	NNC	8.469482e+04
444	2013	4	Hire	NNC	1.719227e+05
445	2014	1	Hire	NNC	6.398102e+02
446	2014	2	Hire	NNC	4.865816e+05
447	2014	3	Hire	NNC	3.421213e+05
448	2014	4	Hire	NNC	8.761344e+04
449	2015	1	Hire	NNC	3.796263e+02
450	2015	2	Hire	NNC	8.873625e+05
451	2015	3	Hire	NNC	3.237039e+05
452	2015	4	Hire	NNC	1.537656e+05
453	2016	1	Hire	NNC	1.007698e+04
454	2016	2	Hire	NNC	6.548437e+05
455	2016	3	Hire	NNC	3.086638e+05
456	2016	4	Hire	NNC	2.719797e+05
457	2017	1	Hire	NNC	1.078562e+03
458	2017	2	Hire	NNC	4.489343e+05
459	2017	3	Hire	NNC	2.993657e+05
460	2017	4	Hire	NNC	8.379875e+04
461	2018	1	Hire	NNC	2.982117e+03
462	2018	2	Hire	NNC	7.496388e+05
463	2018	3	Hire	NNC	3.938811e+05
464	2018	4	Hire	NNC	3.213315e+05
465	2019	1	Hire	NNC	2.361142e+02
466	2019	2	Hire	NNC	5.174288e+05
467	2019	3	Hire	NNC	3.287088e+05
468	2019	4	Hire	NNC	1.493810e+05

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469	2020	1	Hire	NNC	2.583884e+03
470	2020	2	Hire	NNC	3.859802e+05
471	2020	3	Hire	NNC	1.904235e+05
472	2020	4	Hire	NNC	9.085066e+04
473	2021	1	Hire	NNC	2.474158e+02
474	2021	2	Hire	NNC	4.377578e+05
475	2021	3	Hire	NNC	1.903206e+05
476	2021	4	Hire	NNC	2.433248e+05
477	2022	1	Hire	NNC	3.577490e+02
478	2022	2	Hire	NNC	1.877862e+05
479	2022	3	Hire	NNC	1.499484e+05
480	2022	4	Hire	NNC	8.883334e+04
481	2023	1	Hire	NNC	1.385437e+03
482	2023	2	Hire	NNC	6.917578e+05
483	2023	3	Hire	NNC	4.855651e+05
484	2023	4	Hire	NNC	1.039317e+05
485	2024	1	Hire	NNC	0.000000e+00
486	2024	2	Hire	NNC	2.081499e+05
487	2024	3	Hire	NNC	1.385556e+05
488	2024	4	Hire	NNC	1.597430e+05
489	2025	1	Hire	NNC	1.094585e+02
490	2025	2	Hire	NNC	NA
491	2025	3	Hire	NNC	NA
492	2025	4	Hire	NNC	NA
493	1985	1	Hire	VBM	0.000000e+00
494	1985	2	Hire	VBM	8.280864e+03
495	1985	3	Hire	VBM	9.521840e+01
496	1985	4	Hire	VBM	5.045926e+02
497	1986	1	Hire	VBM	1.827329e-01
498	1986	2	Hire	VBM	6.103580e+04
499	1986	3	Hire	VBM	2.263707e+02
500	1986	4	Hire	VBM	3.958740e+02
501	1987	1	Hire	VBM	1.433616e-01
502	1987	2	Hire	VBM	7.039843e+03
503	1987	3	Hire	VBM	1.014313e+02
504	1987	4	Hire	VBM	1.782641e+04
505	1988	1	Hire	VBM	1.946561e-01
506	1988	2	Hire	VBM	9.129205e+03
507	1988	3	Hire	VBM	9.912037e+01
508	1988	4	Hire	VBM	8.544770e+04
509	1989	1	Hire	VBM	1.902211e-01

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510	1989	2	Hire	VBM	3.274145e+05
511	1989	3	Hire	VBM	4.332058e+03
512	1989	4	Hire	VBM	1.197307e+05
513	1990	1	Hire	VBM	5.062703e-01
514	1990	2	Hire	VBM	2.132708e+04
515	1990	3	Hire	VBM	3.644628e+02
516	1990	4	Hire	VBM	9.387198e+03
517	1991	1	Hire	VBM	6.994376e-01
518	1991	2	Hire	VBM	9.711731e+04
519	1991	3	Hire	VBM	6.804028e+02
520	1991	4	Hire	VBM	6.137384e+04
521	1992	1	Hire	VBM	1.305756e+00
522	1992	2	Hire	VBM	1.311151e+05
523	1992	3	Hire	VBM	2.967570e+02
524	1992	4	Hire	VBM	2.973731e+04
525	1993	1	Hire	VBM	5.695040e-01
526	1993	2	Hire	VBM	7.241465e+05
527	1993	3	Hire	VBM	5.604508e+02
528	1993	4	Hire	VBM	1.724731e+04
529	1994	1	Hire	VBM	1.075557e+00
530	1994	2	Hire	VBM	5.608290e+05
531	1994	3	Hire	VBM	4.745844e+03
532	1994	4	Hire	VBM	7.976476e+03
533	1995	1	Hire	VBM	4.056151e-01
534	1995	2	Hire	VBM	1.151956e+05
535	1995	3	Hire	VBM	1.756989e+03
536	1995	4	Hire	VBM	2.880463e+04
537	1996	1	Hire	VBM	4.132299e+01
538	1996	2	Hire	VBM	4.977416e+04
539	1996	3	Hire	VBM	9.135979e+02
540	1996	4	Hire	VBM	9.128843e+04
541	1997	1	Hire	VBM	2.450390e-01
542	1997	2	Hire	VBM	1.036612e+04
543	1997	3	Hire	VBM	1.425417e+03
544	1997	4	Hire	VBM	9.832395e+04
545	1998	1	Hire	VBM	0.000000e+00
546	1998	2	Hire	VBM	7.553346e+05
547	1998	3	Hire	VBM	1.270259e+03
548	1998	4	Hire	VBM	2.276001e+04
549	1999	1	Hire	VBM	0.000000e+00
550	1999	2	Hire	VBM	2.312582e+05

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551	1999	3	Hire	VBM	5.659894e+04
552	1999	4	Hire	VBM	6.873502e+02
553	2000	1	Hire	VBM	8.064427e+01
554	2000	2	Hire	VBM	1.590883e+03
555	2000	3	Hire	VBM	4.318811e+02
556	2000	4	Hire	VBM	9.839474e+02
557	2001	1	Hire	VBM	9.876608e+00
558	2001	2	Hire	VBM	1.849069e+04
559	2001	3	Hire	VBM	3.416471e+03
560	2001	4	Hire	VBM	1.241709e+04
561	2002	1	Hire	VBM	1.098993e+01
562	2002	2	Hire	VBM	8.680582e+04
563	2002	3	Hire	VBM	6.614244e+04
564	2002	4	Hire	VBM	1.539632e+04
565	2003	1	Hire	VBM	6.222483e+01
566	2003	2	Hire	VBM	1.443916e+04
567	2003	3	Hire	VBM	2.469372e+02
568	2003	4	Hire	VBM	4.100432e+04
569	2004	1	Hire	VBM	0.000000e+00
570	2004	2	Hire	VBM	1.152953e+05
571	2004	3	Hire	VBM	1.585921e+03
572	2004	4	Hire	VBM	3.854991e+04
573	2005	1	Hire	VBM	7.965220e+01
574	2005	2	Hire	VBM	2.865057e+04
575	2005	3	Hire	VBM	3.255974e+02
576	2005	4	Hire	VBM	9.044386e+03
577	2006	1	Hire	VBM	0.000000e+00
578	2006	2	Hire	VBM	6.907945e+04
579	2006	3	Hire	VBM	2.770928e+03
580	2006	4	Hire	VBM	6.486364e+04
581	2007	1	Hire	VBM	1.455036e+00
582	2007	2	Hire	VBM	2.436482e+04
583	2007	3	Hire	VBM	2.983204e+03
584	2007	4	Hire	VBM	5.754550e+04
585	2008	1	Hire	VBM	1.478184e+01
586	2008	2	Hire	VBM	9.670219e+04
587	2008	3	Hire	VBM	8.914606e+02
588	2008	4	Hire	VBM	1.599922e+04
589	2009	1	Hire	VBM	0.000000e+00
590	2009	2	Hire	VBM	8.152937e+04
591	2009	3	Hire	VBM	1.413382e+04

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592	2009	4	Hire	VBM	1.059973e+04
593	2010	1	Hire	VBM	0.000000e+00
594	2010	2	Hire	VBM	9.819018e+04
595	2010	3	Hire	VBM	1.138712e+03
596	2010	4	Hire	VBM	2.072151e+04
597	2011	1	Hire	VBM	0.000000e+00
598	2011	2	Hire	VBM	1.512896e+04
599	2011	3	Hire	VBM	1.250153e+03
600	2011	4	Hire	VBM	1.465611e+03
601	2012	1	Hire	VBM	0.000000e+00
602	2012	2	Hire	VBM	1.535951e+04
603	2012	3	Hire	VBM	2.296609e+03
604	2012	4	Hire	VBM	4.231196e+03
605	2013	1	Hire	VBM	0.000000e+00
606	2013	2	Hire	VBM	1.036735e+04
607	2013	3	Hire	VBM	8.714024e+02
608	2013	4	Hire	VBM	4.825515e+03
609	2014	1	Hire	VBM	4.930504e-01
610	2014	2	Hire	VBM	4.515808e+04
611	2014	3	Hire	VBM	3.636352e+03
612	2014	4	Hire	VBM	7.943520e+03
613	2015	1	Hire	VBM	2.281061e+01
614	2015	2	Hire	VBM	1.544282e+05
615	2015	3	Hire	VBM	7.493783e+03
616	2015	4	Hire	VBM	1.159326e+05
617	2016	1	Hire	VBM	1.339044e+01
618	2016	2	Hire	VBM	1.659762e+05
619	2016	3	Hire	VBM	1.505553e+04
620	2016	4	Hire	VBM	1.480608e+04
621	2017	1	Hire	VBM	0.000000e+00
622	2017	2	Hire	VBM	3.833937e+03
623	2017	3	Hire	VBM	2.005246e+03
624	2017	4	Hire	VBM	2.533937e+04
625	2018	1	Hire	VBM	6.507979e+00
626	2018	2	Hire	VBM	3.341788e+04
627	2018	3	Hire	VBM	2.440593e+03
628	2018	4	Hire	VBM	3.634945e+04
629	2019	1	Hire	VBM	3.866868e+00
630	2019	2	Hire	VBM	6.313726e+04
631	2019	3	Hire	VBM	1.885881e+04
632	2019	4	Hire	VBM	1.164199e+04

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633	2020	1	Hire	VBM	8.064427e+01
634	2020	2	Hire	VBM	1.516255e+04
635	2020	3	Hire	VBM	6.019786e+03
636	2020	4	Hire	VBM	3.042971e+04
637	2021	1	Hire	VBM	9.104998e+00
638	2021	2	Hire	VBM	4.958988e+04
639	2021	3	Hire	VBM	5.388107e+02
640	2021	4	Hire	VBM	5.086524e+04
641	2022	1	Hire	VBM	4.050165e+00
642	2022	2	Hire	VBM	1.172606e+04
643	2022	3	Hire	VBM	1.064174e+03
644	2022	4	Hire	VBM	5.723903e+03
645	2023	1	Hire	VBM	0.000000e+00
646	2023	2	Hire	VBM	3.165680e+04
647	2023	3	Hire	VBM	1.931828e+03
648	2023	4	Hire	VBM	3.913939e+04
649	2024	1	Hire	VBM	0.000000e+00
650	2024	2	Hire	VBM	8.709882e+03
651	2024	3	Hire	VBM	2.099045e+03
652	2024	4	Hire	VBM	7.943312e+03
653	2025	1	Hire	VBM	0.000000e+00
654	2025	2	Hire	VBM	NA
655	2025	3	Hire	VBM	NA
656	2025	4	Hire	VBM	NA
657	1985	1	Rec	FLK	4.684663e+05
658	1985	2	Rec	FLK	4.452078e+06
659	1985	3	Rec	FLK	1.537705e+06
660	1985	4	Rec	FLK	1.443498e+06
661	1986	1	Rec	FLK	4.020906e+05
662	1986	2	Rec	FLK	1.537010e+06
663	1986	3	Rec	FLK	2.075606e+06
664	1986	4	Rec	FLK	1.225359e+06
665	1987	1	Rec	FLK	1.556820e+06
666	1987	2	Rec	FLK	2.192920e+06
667	1987	3	Rec	FLK	2.961240e+06
668	1987	4	Rec	FLK	4.098999e+05
669	1988	1	Rec	FLK	7.202998e+05
670	1988	2	Rec	FLK	3.018472e+06
671	1988	3	Rec	FLK	2.253282e+06
672	1988	4	Rec	FLK	7.786410e+05
673	1989	1	Rec	FLK	3.945315e+05

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674	1989	2	Rec	FLK	7.785107e+06
675	1989	3	Rec	FLK	7.490678e+06
676	1989	4	Rec	FLK	1.972244e+06
677	1990	1	Rec	FLK	7.318250e+05
678	1990	2	Rec	FLK	1.211452e+07
679	1990	3	Rec	FLK	5.174374e+06
680	1990	4	Rec	FLK	2.081267e+06
681	1991	1	Rec	FLK	8.229340e+05
682	1991	2	Rec	FLK	1.043784e+07
683	1991	3	Rec	FLK	6.015731e+06
684	1991	4	Rec	FLK	5.694630e+06
685	1992	1	Rec	FLK	2.178847e+06
686	1992	2	Rec	FLK	3.838259e+06
687	1992	3	Rec	FLK	9.266735e+06
688	1992	4	Rec	FLK	7.945853e+05
689	1993	1	Rec	FLK	9.367454e+05
690	1993	2	Rec	FLK	3.404812e+06
691	1993	3	Rec	FLK	3.461382e+06
692	1993	4	Rec	FLK	2.073350e+06
693	1994	1	Rec	FLK	8.204078e+05
694	1994	2	Rec	FLK	3.825881e+06
695	1994	3	Rec	FLK	3.779174e+06
696	1994	4	Rec	FLK	1.223023e+06
697	1995	1	Rec	FLK	8.304530e+05
698	1995	2	Rec	FLK	7.321132e+06
699	1995	3	Rec	FLK	6.059479e+06
700	1995	4	Rec	FLK	9.971934e+05
701	1996	1	Rec	FLK	8.944467e+05
702	1996	2	Rec	FLK	4.717398e+06
703	1996	3	Rec	FLK	4.144946e+06
704	1996	4	Rec	FLK	8.143585e+05
705	1997	1	Rec	FLK	8.229003e+05
706	1997	2	Rec	FLK	1.366213e+07
707	1997	3	Rec	FLK	7.460292e+06
708	1997	4	Rec	FLK	6.803199e+05
709	1998	1	Rec	FLK	5.126049e+05
710	1998	2	Rec	FLK	4.808405e+06
711	1998	3	Rec	FLK	6.697606e+06
712	1998	4	Rec	FLK	4.808060e+05
713	1999	1	Rec	FLK	2.224750e+06
714	1999	2	Rec	FLK	6.375714e+06

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715	1999	3	Rec	FLK	4.880669e+06
716	1999	4	Rec	FLK	4.465400e+05
717	2000	1	Rec	FLK	3.883957e+05
718	2000	2	Rec	FLK	6.358022e+06
719	2000	3	Rec	FLK	4.543957e+06
720	2000	4	Rec	FLK	1.945322e+06
721	2001	1	Rec	FLK	7.347928e+05
722	2001	2	Rec	FLK	5.589528e+06
723	2001	3	Rec	FLK	4.428842e+06
724	2001	4	Rec	FLK	1.006040e+06
725	2002	1	Rec	FLK	5.576250e+05
726	2002	2	Rec	FLK	2.214085e+06
727	2002	3	Rec	FLK	2.925741e+06
728	2002	4	Rec	FLK	1.707999e+06
729	2003	1	Rec	FLK	1.456263e+06
730	2003	2	Rec	FLK	4.995939e+06
731	2003	3	Rec	FLK	3.649030e+06
732	2003	4	Rec	FLK	7.827342e+05
733	2004	1	Rec	FLK	9.672219e+05
734	2004	2	Rec	FLK	2.849332e+06
735	2004	3	Rec	FLK	2.248755e+06
736	2004	4	Rec	FLK	3.599090e+05
737	2005	1	Rec	FLK	5.237065e+05
738	2005	2	Rec	FLK	2.074780e+06
739	2005	3	Rec	FLK	1.481262e+06
740	2005	4	Rec	FLK	4.284965e+05
741	2006	1	Rec	FLK	7.714475e+05
742	2006	2	Rec	FLK	2.360906e+06
743	2006	3	Rec	FLK	4.707929e+06
744	2006	4	Rec	FLK	7.207076e+05
745	2007	1	Rec	FLK	1.687930e+06
746	2007	2	Rec	FLK	3.048185e+06
747	2007	3	Rec	FLK	2.545278e+06
748	2007	4	Rec	FLK	1.088889e+06
749	2008	1	Rec	FLK	4.599396e+05
750	2008	2	Rec	FLK	3.741868e+06
751	2008	3	Rec	FLK	2.419131e+06
752	2008	4	Rec	FLK	1.130458e+06
753	2009	1	Rec	FLK	1.060551e+06
754	2009	2	Rec	FLK	2.307627e+06
755	2009	3	Rec	FLK	2.391186e+06

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756	2009	4	Rec	FLK	9.739787e+05
757	2010	1	Rec	FLK	9.141962e+05
758	2010	2	Rec	FLK	1.487555e+06
759	2010	3	Rec	FLK	1.119818e+06
760	2010	4	Rec	FLK	5.835857e+05
761	2011	1	Rec	FLK	1.585451e+05
762	2011	2	Rec	FLK	2.840895e+06
763	2011	3	Rec	FLK	1.874902e+06
764	2011	4	Rec	FLK	1.027396e+06
765	2012	1	Rec	FLK	5.832363e+05
766	2012	2	Rec	FLK	1.913413e+06
767	2012	3	Rec	FLK	2.147562e+06
768	2012	4	Rec	FLK	1.229422e+06
769	2013	1	Rec	FLK	3.905219e+05
770	2013	2	Rec	FLK	5.642923e+06
771	2013	3	Rec	FLK	3.496798e+06
772	2013	4	Rec	FLK	6.223060e+05
773	2014	1	Rec	FLK	6.260843e+05
774	2014	2	Rec	FLK	3.609216e+06
775	2014	3	Rec	FLK	4.087638e+06
776	2014	4	Rec	FLK	1.674867e+06
777	2015	1	Rec	FLK	2.513409e+05
778	2015	2	Rec	FLK	6.725790e+06
779	2015	3	Rec	FLK	3.086404e+06
780	2015	4	Rec	FLK	1.961057e+06
781	2016	1	Rec	FLK	4.919460e+05
782	2016	2	Rec	FLK	2.370240e+06
783	2016	3	Rec	FLK	1.718750e+06
784	2016	4	Rec	FLK	6.215559e+05
785	2017	1	Rec	FLK	1.519864e+05
786	2017	2	Rec	FLK	3.691597e+06
787	2017	3	Rec	FLK	9.818841e+05
788	2017	4	Rec	FLK	6.070687e+05
789	2018	1	Rec	FLK	2.533863e+05
790	2018	2	Rec	FLK	3.234459e+06
791	2018	3	Rec	FLK	2.474352e+06
792	2018	4	Rec	FLK	1.831259e+06
793	2019	1	Rec	FLK	4.962654e+05
794	2019	2	Rec	FLK	2.593726e+06
795	2019	3	Rec	FLK	1.113341e+06
796	2019	4	Rec	FLK	9.086961e+04

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797	2020	1	Rec	FLK	5.812551e+04
798	2020	2	Rec	FLK	1.336144e+06
799	2020	3	Rec	FLK	1.105212e+06
800	2020	4	Rec	FLK	3.407217e+05
801	2021	1	Rec	FLK	3.157635e+05
802	2021	2	Rec	FLK	1.821803e+06
803	2021	3	Rec	FLK	7.596949e+05
804	2021	4	Rec	FLK	7.848983e+05
805	2022	1	Rec	FLK	1.633176e+05
806	2022	2	Rec	FLK	1.574723e+06
807	2022	3	Rec	FLK	1.470855e+06
808	2022	4	Rec	FLK	6.657709e+05
809	2023	1	Rec	FLK	1.138258e+05
810	2023	2	Rec	FLK	1.143757e+06
811	2023	3	Rec	FLK	6.930613e+05
812	2023	4	Rec	FLK	1.022943e+06
813	2024	1	Rec	FLK	1.741675e+05
814	2024	2	Rec	FLK	1.667158e+06
815	2024	3	Rec	FLK	9.281034e+05
816	2024	4	Rec	FLK	6.951332e+05
817	2025	1	Rec	FLK	3.138698e+04
818	2025	2	Rec	FLK	NA
819	2025	3	Rec	FLK	NA
820	2025	4	Rec	FLK	NA
821	1985	1	Rec	NCFL	0.000000e+00
822	1985	2	Rec	NCFL	6.036182e+05
823	1985	3	Rec	NCFL	5.590902e+05
824	1985	4	Rec	NCFL	0.000000e+00
825	1986	1	Rec	NCFL	0.000000e+00
826	1986	2	Rec	NCFL	3.338597e+05
827	1986	3	Rec	NCFL	3.017922e+05
828	1986	4	Rec	NCFL	8.768057e+04
829	1987	1	Rec	NCFL	1.353952e+04
830	1987	2	Rec	NCFL	1.077096e+05
831	1987	3	Rec	NCFL	1.505825e+04
832	1987	4	Rec	NCFL	3.885919e+04
833	1988	1	Rec	NCFL	0.000000e+00
834	1988	2	Rec	NCFL	2.502125e+05
835	1988	3	Rec	NCFL	2.737798e+04
836	1988	4	Rec	NCFL	4.343268e+04
837	1989	1	Rec	NCFL	1.399500e+05

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838	1989	2	Rec	NCFL	2.442929e+05
839	1989	3	Rec	NCFL	2.253837e+05
840	1989	4	Rec	NCFL	2.617131e+05
841	1990	1	Rec	NCFL	2.086118e+05
842	1990	2	Rec	NCFL	1.494802e+05
843	1990	3	Rec	NCFL	4.926664e+04
844	1990	4	Rec	NCFL	1.135865e+05
845	1991	1	Rec	NCFL	0.000000e+00
846	1991	2	Rec	NCFL	2.595592e+05
847	1991	3	Rec	NCFL	1.863238e+05
848	1991	4	Rec	NCFL	1.178413e+04
849	1992	1	Rec	NCFL	0.000000e+00
850	1992	2	Rec	NCFL	1.727708e+05
851	1992	3	Rec	NCFL	4.607402e+04
852	1992	4	Rec	NCFL	2.655088e+04
853	1993	1	Rec	NCFL	0.000000e+00
854	1993	2	Rec	NCFL	4.355714e+05
855	1993	3	Rec	NCFL	2.752532e+05
856	1993	4	Rec	NCFL	2.290328e+04
857	1994	1	Rec	NCFL	3.723780e+04
858	1994	2	Rec	NCFL	6.328460e+05
859	1994	3	Rec	NCFL	5.279053e+05
860	1994	4	Rec	NCFL	3.281162e+05
861	1995	1	Rec	NCFL	1.918508e+05
862	1995	2	Rec	NCFL	5.730691e+05
863	1995	3	Rec	NCFL	8.825788e+05
864	1995	4	Rec	NCFL	6.613425e+04
865	1996	1	Rec	NCFL	2.835385e+04
866	1996	2	Rec	NCFL	4.246640e+05
867	1996	3	Rec	NCFL	6.274736e+05
868	1996	4	Rec	NCFL	6.980021e+04
869	1997	1	Rec	NCFL	0.000000e+00
870	1997	2	Rec	NCFL	4.617975e+05
871	1997	3	Rec	NCFL	6.912068e+05
872	1997	4	Rec	NCFL	1.634984e+05
873	1998	1	Rec	NCFL	0.000000e+00
874	1998	2	Rec	NCFL	1.689554e+05
875	1998	3	Rec	NCFL	1.873414e+05
876	1998	4	Rec	NCFL	1.458389e+05
877	1999	1	Rec	NCFL	1.459837e+05
878	1999	2	Rec	NCFL	7.204270e+05

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879	1999	3	Rec	NCFL	6.613227e+05
880	1999	4	Rec	NCFL	1.147492e+05
881	2000	1	Rec	NCFL	2.342309e+04
882	2000	2	Rec	NCFL	1.292909e+06
883	2000	3	Rec	NCFL	1.223197e+06
884	2000	4	Rec	NCFL	2.444075e+05
885	2001	1	Rec	NCFL	3.043074e+05
886	2001	2	Rec	NCFL	1.526796e+06
887	2001	3	Rec	NCFL	1.626879e+06
888	2001	4	Rec	NCFL	1.400650e+05
889	2002	1	Rec	NCFL	4.023345e+04
890	2002	2	Rec	NCFL	9.589750e+05
891	2002	3	Rec	NCFL	7.704207e+05
892	2002	4	Rec	NCFL	5.947019e+05
893	2003	1	Rec	NCFL	1.996363e+05
894	2003	2	Rec	NCFL	1.196275e+06
895	2003	3	Rec	NCFL	5.198576e+05
896	2003	4	Rec	NCFL	5.640292e+04
897	2004	1	Rec	NCFL	5.587967e+04
898	2004	2	Rec	NCFL	6.996628e+05
899	2004	3	Rec	NCFL	5.903592e+05
900	2004	4	Rec	NCFL	3.085426e+04
901	2005	1	Rec	NCFL	2.566053e+04
902	2005	2	Rec	NCFL	7.310031e+05
903	2005	3	Rec	NCFL	1.137113e+06
904	2005	4	Rec	NCFL	2.547820e+04
905	2006	1	Rec	NCFL	5.351815e+03
906	2006	2	Rec	NCFL	8.487273e+05
907	2006	3	Rec	NCFL	7.853118e+05
908	2006	4	Rec	NCFL	7.268633e+04
909	2007	1	Rec	NCFL	1.582543e+05
910	2007	2	Rec	NCFL	1.269099e+06
911	2007	3	Rec	NCFL	9.717851e+05
912	2007	4	Rec	NCFL	2.940661e+04
913	2008	1	Rec	NCFL	2.669028e+04
914	2008	2	Rec	NCFL	5.067091e+05
915	2008	3	Rec	NCFL	1.070780e+06
916	2008	4	Rec	NCFL	2.970883e+04
917	2009	1	Rec	NCFL	2.401520e+04
918	2009	2	Rec	NCFL	1.689311e+06
919	2009	3	Rec	NCFL	2.046079e+06

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920	2009	4	Rec	NCFL	3.594953e+04
921	2010	1	Rec	NCFL	2.454607e+04
922	2010	2	Rec	NCFL	8.796385e+05
923	2010	3	Rec	NCFL	2.247155e+06
924	2010	4	Rec	NCFL	1.240658e+04
925	2011	1	Rec	NCFL	3.455564e+04
926	2011	2	Rec	NCFL	9.178266e+05
927	2011	3	Rec	NCFL	5.412456e+05
928	2011	4	Rec	NCFL	1.574686e+05
929	2012	1	Rec	NCFL	0.000000e+00
930	2012	2	Rec	NCFL	8.746241e+05
931	2012	3	Rec	NCFL	1.894418e+06
932	2012	4	Rec	NCFL	9.855716e+04
933	2013	1	Rec	NCFL	2.144051e+04
934	2013	2	Rec	NCFL	1.133952e+06
935	2013	3	Rec	NCFL	8.739791e+05
936	2013	4	Rec	NCFL	3.488479e+04
937	2014	1	Rec	NCFL	0.000000e+00
938	2014	2	Rec	NCFL	6.355006e+05
939	2014	3	Rec	NCFL	5.279255e+05
940	2014	4	Rec	NCFL	1.063765e+05
941	2015	1	Rec	NCFL	2.674703e+03
942	2015	2	Rec	NCFL	8.061179e+05
943	2015	3	Rec	NCFL	5.417106e+05
944	2015	4	Rec	NCFL	2.005999e+05
945	2016	1	Rec	NCFL	3.551940e+04
946	2016	2	Rec	NCFL	1.880964e+06
947	2016	3	Rec	NCFL	2.411574e+06
948	2016	4	Rec	NCFL	3.674186e+04
949	2017	1	Rec	NCFL	2.128542e+04
950	2017	2	Rec	NCFL	1.121239e+06
951	2017	3	Rec	NCFL	1.145773e+06
952	2017	4	Rec	NCFL	1.479721e+04
953	2018	1	Rec	NCFL	4.798984e+04
954	2018	2	Rec	NCFL	1.292812e+06
955	2018	3	Rec	NCFL	1.214296e+06
956	2018	4	Rec	NCFL	5.313994e+04
957	2019	1	Rec	NCFL	1.805194e+04
958	2019	2	Rec	NCFL	8.596577e+05
959	2019	3	Rec	NCFL	1.202456e+06
960	2019	4	Rec	NCFL	4.788760e+04

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961	2020	1	Rec	NCFL	4.941575e+03
962	2020	2	Rec	NCFL	8.708714e+05
963	2020	3	Rec	NCFL	1.093047e+06
964	2020	4	Rec	NCFL	7.473885e+04
965	2021	1	Rec	NCFL	1.356446e+04
966	2021	2	Rec	NCFL	4.814084e+05
967	2021	3	Rec	NCFL	6.071702e+05
968	2021	4	Rec	NCFL	2.963646e+05
969	2022	1	Rec	NCFL	4.356505e+04
970	2022	2	Rec	NCFL	1.398654e+06
971	2022	3	Rec	NCFL	1.267020e+06
972	2022	4	Rec	NCFL	3.594554e+04
973	2023	1	Rec	NCFL	2.716461e+04
974	2023	2	Rec	NCFL	1.207814e+06
975	2023	3	Rec	NCFL	1.299211e+06
976	2023	4	Rec	NCFL	1.052561e+05
977	2024	1	Rec	NCFL	0.000000e+00
978	2024	2	Rec	NCFL	6.938219e+05
979	2024	3	Rec	NCFL	5.986244e+05
980	2024	4	Rec	NCFL	3.303640e+04
981	2025	1	Rec	NCFL	3.303640e+04
982	2025	2	Rec	NCFL	NA
983	2025	3	Rec	NCFL	NA
984	2025	4	Rec	NCFL	NA
985	1985	1	Rec	NNC	0.000000e+00
986	1985	2	Rec	NNC	3.873855e+04
987	1985	3	Rec	NNC	3.873855e+04
988	1985	4	Rec	NNC	0.000000e+00
989	1986	1	Rec	NNC	0.000000e+00
990	1986	2	Rec	NNC	2.549586e+03
991	1986	3	Rec	NNC	0.000000e+00
992	1986	4	Rec	NNC	0.000000e+00
993	1987	1	Rec	NNC	0.000000e+00
994	1987	2	Rec	NNC	7.412236e+04
995	1987	3	Rec	NNC	0.000000e+00
996	1987	4	Rec	NNC	5.418005e+03
997	1988	1	Rec	NNC	0.000000e+00
998	1988	2	Rec	NNC	1.909134e+05
999	1988	3	Rec	NNC	1.185557e+04
1000	1988	4	Rec	NNC	0.000000e+00
1001	1989	1	Rec	NNC	0.000000e+00

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1002	1989	2	Rec	NNC	1.902856e+05
1003	1989	3	Rec	NNC	1.465598e+05
1004	1989	4	Rec	NNC	1.588620e+03
1005	1990	1	Rec	NNC	9.376058e+02
1006	1990	2	Rec	NNC	1.017529e+05
1007	1990	3	Rec	NNC	1.379817e+04
1008	1990	4	Rec	NNC	2.988903e+03
1009	1991	1	Rec	NNC	0.000000e+00
1010	1991	2	Rec	NNC	7.471727e+04
1011	1991	3	Rec	NNC	1.882011e+05
1012	1991	4	Rec	NNC	1.447126e+04
1013	1992	1	Rec	NNC	0.000000e+00
1014	1992	2	Rec	NNC	4.499024e+04
1015	1992	3	Rec	NNC	2.068819e+04
1016	1992	4	Rec	NNC	2.348243e+04
1017	1993	1	Rec	NNC	0.000000e+00
1018	1993	2	Rec	NNC	2.759011e+05
1019	1993	3	Rec	NNC	1.677273e+04
1020	1993	4	Rec	NNC	3.725324e+03
1021	1994	1	Rec	NNC	0.000000e+00
1022	1994	2	Rec	NNC	1.311286e+05
1023	1994	3	Rec	NNC	1.101410e+05
1024	1994	4	Rec	NNC	2.273649e+04
1025	1995	1	Rec	NNC	9.011241e+02
1026	1995	2	Rec	NNC	1.396278e+05
1027	1995	3	Rec	NNC	1.128793e+05
1028	1995	4	Rec	NNC	1.893502e+04
1029	1996	1	Rec	NNC	0.000000e+00
1030	1996	2	Rec	NNC	1.076565e+05
1031	1996	3	Rec	NNC	6.420212e+04
1032	1996	4	Rec	NNC	2.472091e+04
1033	1997	1	Rec	NNC	0.000000e+00
1034	1997	2	Rec	NNC	4.358563e+05
1035	1997	3	Rec	NNC	1.733239e+05
1036	1997	4	Rec	NNC	5.107759e+04
1037	1998	1	Rec	NNC	0.000000e+00
1038	1998	2	Rec	NNC	2.813807e+05
1039	1998	3	Rec	NNC	8.236200e+04
1040	1998	4	Rec	NNC	4.856163e+04
1041	1999	1	Rec	NNC	0.000000e+00
1042	1999	2	Rec	NNC	7.658788e+05

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1043	1999	3	Rec	NNC	1.105604e+05
1044	1999	4	Rec	NNC	4.383661e+05
1045	2000	1	Rec	NNC	0.000000e+00
1046	2000	2	Rec	NNC	4.822238e+05
1047	2000	3	Rec	NNC	3.086571e+05
1048	2000	4	Rec	NNC	3.609943e+04
1049	2001	1	Rec	NNC	0.000000e+00
1050	2001	2	Rec	NNC	1.306376e+06
1051	2001	3	Rec	NNC	4.407184e+05
1052	2001	4	Rec	NNC	1.362622e+04
1053	2002	1	Rec	NNC	0.000000e+00
1054	2002	2	Rec	NNC	5.714948e+05
1055	2002	3	Rec	NNC	2.149395e+05
1056	2002	4	Rec	NNC	8.549641e+04
1057	2003	1	Rec	NNC	0.000000e+00
1058	2003	2	Rec	NNC	6.097702e+05
1059	2003	3	Rec	NNC	2.393101e+05
1060	2003	4	Rec	NNC	1.911989e+04
1061	2004	1	Rec	NNC	0.000000e+00
1062	2004	2	Rec	NNC	1.130043e+05
1063	2004	3	Rec	NNC	4.759809e+05
1064	2004	4	Rec	NNC	2.136272e+03
1065	2005	1	Rec	NNC	2.136272e+03
1066	2005	2	Rec	NNC	5.906869e+05
1067	2005	3	Rec	NNC	2.595178e+05
1068	2005	4	Rec	NNC	5.415554e+04
1069	2006	1	Rec	NNC	0.000000e+00
1070	2006	2	Rec	NNC	6.079730e+05
1071	2006	3	Rec	NNC	1.069085e+05
1072	2006	4	Rec	NNC	2.871739e+04
1073	2007	1	Rec	NNC	5.942542e+03
1074	2007	2	Rec	NNC	7.446139e+05
1075	2007	3	Rec	NNC	4.911765e+05
1076	2007	4	Rec	NNC	1.638259e+05
1077	2008	1	Rec	NNC	8.401426e+04
1078	2008	2	Rec	NNC	1.520610e+05
1079	2008	3	Rec	NNC	7.150417e+04
1080	2008	4	Rec	NNC	0.000000e+00
1081	2009	1	Rec	NNC	0.000000e+00
1082	2009	2	Rec	NNC	1.264628e+06
1083	2009	3	Rec	NNC	1.087706e+06

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1084	2009	4	Rec	NNC	6.844584e+03
1085	2010	1	Rec	NNC	2.232589e+04
1086	2010	2	Rec	NNC	6.534632e+05
1087	2010	3	Rec	NNC	4.043579e+05
1088	2010	4	Rec	NNC	3.033275e+04
1089	2011	1	Rec	NNC	0.000000e+00
1090	2011	2	Rec	NNC	1.069911e+06
1091	2011	3	Rec	NNC	4.698892e+05
1092	2011	4	Rec	NNC	5.235156e+05
1093	2012	1	Rec	NNC	2.302064e+03
1094	2012	2	Rec	NNC	4.907943e+05
1095	2012	3	Rec	NNC	3.902222e+05
1096	2012	4	Rec	NNC	1.761655e+05
1097	2013	1	Rec	NNC	2.312303e+04
1098	2013	2	Rec	NNC	4.094121e+05
1099	2013	3	Rec	NNC	1.233905e+05
1100	2013	4	Rec	NNC	7.874372e+03
1101	2014	1	Rec	NNC	0.000000e+00
1102	2014	2	Rec	NNC	1.110407e+06
1103	2014	3	Rec	NNC	1.588618e+05
1104	2014	4	Rec	NNC	2.043554e+04
1105	2015	1	Rec	NNC	0.000000e+00
1106	2015	2	Rec	NNC	2.063566e+06
1107	2015	3	Rec	NNC	1.061391e+06
1108	2015	4	Rec	NNC	3.802375e+05
1109	2016	1	Rec	NNC	0.000000e+00
1110	2016	2	Rec	NNC	1.047769e+06
1111	2016	3	Rec	NNC	6.005003e+05
1112	2016	4	Rec	NNC	5.639231e+04
1113	2017	1	Rec	NNC	0.000000e+00
1114	2017	2	Rec	NNC	4.679636e+05
1115	2017	3	Rec	NNC	3.251755e+05
1116	2017	4	Rec	NNC	8.185031e+03
1117	2018	1	Rec	NNC	0.000000e+00
1118	2018	2	Rec	NNC	6.791719e+05
1119	2018	3	Rec	NNC	3.564936e+05
1120	2018	4	Rec	NNC	3.120586e+05
1121	2019	1	Rec	NNC	0.000000e+00
1122	2019	2	Rec	NNC	1.086972e+06
1123	2019	3	Rec	NNC	3.921515e+05
1124	2019	4	Rec	NNC	2.081160e+05

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1125	2020	1	Rec	NNC	0.000000e+00
1126	2020	2	Rec	NNC	6.442826e+05
1127	2020	3	Rec	NNC	4.355690e+05
1128	2020	4	Rec	NNC	9.393537e+03
1129	2021	1	Rec	NNC	0.000000e+00
1130	2021	2	Rec	NNC	3.354591e+05
1131	2021	3	Rec	NNC	2.612134e+05
1132	2021	4	Rec	NNC	0.000000e+00
1133	2022	1	Rec	NNC	0.000000e+00
1134	2022	2	Rec	NNC	1.095702e+05
1135	2022	3	Rec	NNC	3.506203e+04
1136	2022	4	Rec	NNC	6.676231e+04
1137	2023	1	Rec	NNC	0.000000e+00
1138	2023	2	Rec	NNC	1.497840e+05
1139	2023	3	Rec	NNC	2.540520e+04
1140	2023	4	Rec	NNC	0.000000e+00
1141	2024	1	Rec	NNC	0.000000e+00
1142	2024	2	Rec	NNC	8.349211e+04
1143	2024	3	Rec	NNC	2.281517e+04
1144	2024	4	Rec	NNC	0.000000e+00
1145	2025	1	Rec	NNC	0.000000e+00
1146	2025	2	Rec	NNC	NA
1147	2025	3	Rec	NNC	NA
1148	2025	4	Rec	NNC	NA
1149	1985	1	Rec	VBM	0.000000e+00
1150	1985	2	Rec	VBM	1.129443e+05
1151	1985	3	Rec	VBM	0.000000e+00
1152	1985	4	Rec	VBM	0.000000e+00
1153	1986	1	Rec	VBM	0.000000e+00
1154	1986	2	Rec	VBM	7.339022e+04
1155	1986	3	Rec	VBM	0.000000e+00
1156	1986	4	Rec	VBM	3.055492e+04
1157	1987	1	Rec	VBM	0.000000e+00
1158	1987	2	Rec	VBM	4.206392e+04
1159	1987	3	Rec	VBM	0.000000e+00
1160	1987	4	Rec	VBM	1.007585e+04
1161	1988	1	Rec	VBM	0.000000e+00
1162	1988	2	Rec	VBM	2.932505e+04
1163	1988	3	Rec	VBM	0.000000e+00
1164	1988	4	Rec	VBM	4.241116e+03
1165	1989	1	Rec	VBM	0.000000e+00

2 NOAA Fisheries recreational landings data

1166	1989	2	Rec	VBM	6.821138e+05
1167	1989	3	Rec	VBM	6.273420e+03
1168	1989	4	Rec	VBM	3.085581e+05
1169	1990	1	Rec	VBM	0.000000e+00
1170	1990	2	Rec	VBM	1.469085e+05
1171	1990	3	Rec	VBM	0.000000e+00
1172	1990	4	Rec	VBM	3.625165e+05
1173	1991	1	Rec	VBM	0.000000e+00
1174	1991	2	Rec	VBM	3.136398e+05
1175	1991	3	Rec	VBM	0.000000e+00
1176	1991	4	Rec	VBM	1.511470e+05
1177	1992	1	Rec	VBM	0.000000e+00
1178	1992	2	Rec	VBM	8.693982e+04
1179	1992	3	Rec	VBM	1.026340e+04
1180	1992	4	Rec	VBM	4.120940e+05
1181	1993	1	Rec	VBM	0.000000e+00
1182	1993	2	Rec	VBM	6.988338e+05
1183	1993	3	Rec	VBM	3.495753e+03
1184	1993	4	Rec	VBM	1.696190e+05
1185	1994	1	Rec	VBM	0.000000e+00
1186	1994	2	Rec	VBM	5.119812e+05
1187	1994	3	Rec	VBM	2.273618e+03
1188	1994	4	Rec	VBM	8.021262e+04
1189	1995	1	Rec	VBM	0.000000e+00
1190	1995	2	Rec	VBM	5.290627e+05
1191	1995	3	Rec	VBM	8.023987e+04
1192	1995	4	Rec	VBM	0.000000e+00
1193	1996	1	Rec	VBM	0.000000e+00
1194	1996	2	Rec	VBM	4.084946e+05
1195	1996	3	Rec	VBM	0.000000e+00
1196	1996	4	Rec	VBM	7.761164e+04
1197	1997	1	Rec	VBM	0.000000e+00
1198	1997	2	Rec	VBM	1.498605e+05
1199	1997	3	Rec	VBM	0.000000e+00
1200	1997	4	Rec	VBM	4.801143e+04
1201	1998	1	Rec	VBM	0.000000e+00
1202	1998	2	Rec	VBM	4.708166e+05
1203	1998	3	Rec	VBM	0.000000e+00
1204	1998	4	Rec	VBM	1.359527e+05
1205	1999	1	Rec	VBM	0.000000e+00
1206	1999	2	Rec	VBM	2.411140e+05

2 NOAA Fisheries recreational landings data

1207	1999	3	Rec	VBM	2.572690e+04
1208	1999	4	Rec	VBM	2.567828e+05
1209	2000	1	Rec	VBM	0.000000e+00
1210	2000	2	Rec	VBM	1.440373e+06
1211	2000	3	Rec	VBM	0.000000e+00
1212	2000	4	Rec	VBM	1.681063e+05
1213	2001	1	Rec	VBM	0.000000e+00
1214	2001	2	Rec	VBM	2.879419e+05
1215	2001	3	Rec	VBM	5.977503e+04
1216	2001	4	Rec	VBM	2.490633e+04
1217	2002	1	Rec	VBM	0.000000e+00
1218	2002	2	Rec	VBM	6.060759e+05
1219	2002	3	Rec	VBM	1.599474e+05
1220	2002	4	Rec	VBM	9.663327e+05
1221	2003	1	Rec	VBM	0.000000e+00
1222	2003	2	Rec	VBM	1.223911e+05
1223	2003	3	Rec	VBM	0.000000e+00
1224	2003	4	Rec	VBM	7.823895e+05
1225	2004	1	Rec	VBM	0.000000e+00
1226	2004	2	Rec	VBM	1.283020e+05
1227	2004	3	Rec	VBM	0.000000e+00
1228	2004	4	Rec	VBM	5.823660e+04
1229	2005	1	Rec	VBM	0.000000e+00
1230	2005	2	Rec	VBM	2.756822e+05
1231	2005	3	Rec	VBM	0.000000e+00
1232	2005	4	Rec	VBM	2.135543e+05
1233	2006	1	Rec	VBM	0.000000e+00
1234	2006	2	Rec	VBM	6.831538e+05
1235	2006	3	Rec	VBM	1.111672e+05
1236	2006	4	Rec	VBM	4.201761e+05
1237	2007	1	Rec	VBM	0.000000e+00
1238	2007	2	Rec	VBM	2.089240e+05
1239	2007	3	Rec	VBM	0.000000e+00
1240	2007	4	Rec	VBM	2.784509e+04
1241	2008	1	Rec	VBM	0.000000e+00
1242	2008	2	Rec	VBM	6.659730e+04
1243	2008	3	Rec	VBM	2.248205e+04
1244	2008	4	Rec	VBM	7.739115e+03
1245	2009	1	Rec	VBM	0.000000e+00
1246	2009	2	Rec	VBM	1.573958e+06
1247	2009	3	Rec	VBM	3.543890e+03

2 NOAA Fisheries recreational landings data

1248	2009	4	Rec	VBM	3.041695e+04
1249	2010	1	Rec	VBM	0.000000e+00
1250	2010	2	Rec	VBM	5.114340e+05
1251	2010	3	Rec	VBM	5.630438e+03
1252	2010	4	Rec	VBM	2.003465e+05
1253	2011	1	Rec	VBM	0.000000e+00
1254	2011	2	Rec	VBM	2.626410e+06
1255	2011	3	Rec	VBM	0.000000e+00
1256	2011	4	Rec	VBM	0.000000e+00
1257	2012	1	Rec	VBM	0.000000e+00
1258	2012	2	Rec	VBM	1.041694e+06
1259	2012	3	Rec	VBM	0.000000e+00
1260	2012	4	Rec	VBM	0.000000e+00
1261	2013	1	Rec	VBM	0.000000e+00
1262	2013	2	Rec	VBM	1.374911e+05
1263	2013	3	Rec	VBM	0.000000e+00
1264	2013	4	Rec	VBM	2.270012e+05
1265	2014	1	Rec	VBM	0.000000e+00
1266	2014	2	Rec	VBM	1.746051e+06
1267	2014	3	Rec	VBM	1.453196e+05
1268	2014	4	Rec	VBM	1.489590e+05
1269	2015	1	Rec	VBM	0.000000e+00
1270	2015	2	Rec	VBM	2.279425e+06
1271	2015	3	Rec	VBM	2.682508e+05
1272	2015	4	Rec	VBM	2.098120e+06
1273	2016	1	Rec	VBM	0.000000e+00
1274	2016	2	Rec	VBM	1.667758e+06
1275	2016	3	Rec	VBM	2.730686e+05
1276	2016	4	Rec	VBM	2.453000e+05
1277	2017	1	Rec	VBM	0.000000e+00
1278	2017	2	Rec	VBM	1.287459e+06
1279	2017	3	Rec	VBM	2.537905e+04
1280	2017	4	Rec	VBM	5.139850e+05
1281	2018	1	Rec	VBM	0.000000e+00
1282	2018	2	Rec	VBM	8.945892e+05
1283	2018	3	Rec	VBM	5.947017e+03
1284	2018	4	Rec	VBM	1.970588e+06
1285	2019	1	Rec	VBM	0.000000e+00
1286	2019	2	Rec	VBM	8.636894e+05
1287	2019	3	Rec	VBM	1.059053e+05
1288	2019	4	Rec	VBM	4.708263e+05

2 NOAA Fisheries recreational landings data

1289	2020	1	Rec	VBM	0.000000e+00
1290	2020	2	Rec	VBM	4.699253e+05
1291	2020	3	Rec	VBM	7.429021e+03
1292	2020	4	Rec	VBM	2.117452e+05
1293	2021	1	Rec	VBM	0.000000e+00
1294	2021	2	Rec	VBM	6.502247e+05
1295	2021	3	Rec	VBM	1.280529e+04
1296	2021	4	Rec	VBM	4.192489e+05
1297	2022	1	Rec	VBM	9.898678e+02
1298	2022	2	Rec	VBM	6.221143e+05
1299	2022	3	Rec	VBM	2.531032e+04
1300	2022	4	Rec	VBM	5.588537e+05
1301	2023	1	Rec	VBM	0.000000e+00
1302	2023	2	Rec	VBM	4.341456e+05
1303	2023	3	Rec	VBM	3.106470e+03
1304	2023	4	Rec	VBM	4.352388e+05
1305	2024	1	Rec	VBM	0.000000e+00
1306	2024	2	Rec	VBM	9.457560e+05
1307	2024	3	Rec	VBM	7.382052e+02
1308	2024	4	Rec	VBM	4.001127e+05
1309	2025	1	Rec	VBM	0.000000e+00
1310	2025	2	Rec	VBM	NA
1311	2025	3	Rec	VBM	NA
1312	2025	4	Rec	VBM	NA

```
names(findat1) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
```

```
head(findat1)
```

	Year	Quarter	Fleet	Area	Catch_lbs
1	1985	1	Hire	FLK	4462.09
2	1985	2	Hire	FLK	253487.38
3	1985	3	Hire	FLK	380811.37
4	1985	4	Hire	FLK	41469.99
5	1986	1	Hire	FLK	40545.41
6	1986	2	Hire	FLK	501184.61

```
#plot(findat1$Catch_lbs, type = "l")
#plot(findat1$Catch_lbs ~ factor(findat1$Area))
#plot(findat1$Catch_lbs ~ factor(findat1$Quarter))
```

2 NOAA Fisheries recreational landings data

```
#plot(findat1$Catch_lbs ~ factor(findat1$Year))

findat1 <- findat1[which(findat1$Year <= 2022), ]
findat1 <- findat1[which(findat1$Year >= 1986), ]
#apply(findat1, 2, table)

findat <- findat[which(findat$year >=1986 & findat$year <= 2022), ]
#check that the numbers are exactly the same
tapply(findat1$Catch_lbs, list(findat1$Area, findat1$Fleet), sum, na.rm = T)
```

	Hire	Rec
FLK	54012811	357091651
NCFL	16532764	65643193
NNC	56097190	32197622
VBM	5736592	37749487

```
colSums(findat[3:10], na.rm = T)
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
54012811	16532764	56097190	5736592	357091651	65643193	32197622	37749487

```
write.csv(findat1, file = "data/FINAL_files/rec_catch_TomFormat.csv", row.names = FALSE)
```

3 Analysis of international landings in the high seas

Here we will explore commercial and recreational landings data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported landings. For each country, project scientists assemble available data on landings and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total landings from all fishing sectors and for all species globally.

3.1 Data access and upload

The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are downloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024 (the data files can be found on github in the SEFSC-dolphin-analyses/data /SAU/ folder). This was still the most recent version of data available as of June 2025.

3.2 Read in the high seas landings

First we read in the high seas landings for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

```
# clear workspace
rm(list = ls())

if(!require("pals")) install.packages("pals")
library(pals)
```


3 Analysis of international landings in the high seas

```
# find data files
hslis <- dir("data/SAU/") [grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
hslis[1]; hslis[3]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
#unique(h1$area_name)
#unique(h3$area_name)

hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the landings. Most of the landings are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name) [grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

```
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]

# look at the categories within each column of data
apply(hs[1:13], 2, table)
```

3 Analysis of international landings in the high seas

\$area_name

Atlantic, Northwest Atlantic, Western Central	113	160
---	-----	-----

\$area_type

high_seas
273

\$year

1985	1986	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
1	1	1	1	1	1	1	6	4	1	5	10	5	8	10	12
2013	2014	2015	2016	2017	2018	2019									
24	22	30	33	33	29	34									

\$scientific_name

Coryphaena hippurus
273

\$common_name

Common dolphinfish
273

\$functional_group

Large pelagics (≥ 90 cm)
273

\$commercial_group

Perch-likes
273

\$fishing_entity

Barbados	Belize
6	3

3 Analysis of international landings in the high seas

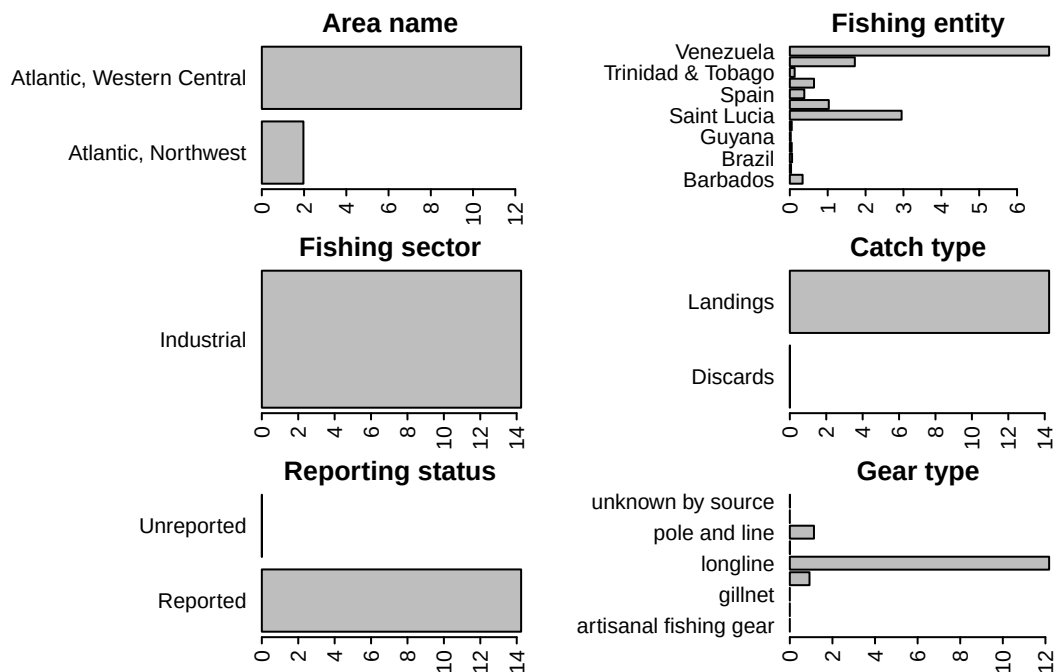
	Brazil	Canada
	19	18
	Guyana	Panama
	1	1
	Saint Lucia	Saint Vincent & the Grenadines
	5	28
	Spain	Suriname
	80	1
	Trinidad & Tobago	USA
	14	74
	Venezuela	
	23	
\$fishing_sector		
Industrial		
	273	
\$catch_type		
Discards Landings		
	5	268
\$reporting_status		
	Reported	Unreported
	253	20
\$gear_type		
artisanal fishing gear		
	7	bottom trawl 9 gillnet 11
	hand lines	longline 187 mixed gear 1
	17	187
	pole and line	pots or traps 9 unknown by source 13
	19	9
\$end_use_type		
	Direct human consumption	Fishmeal and fish oil
	5	194 37

3 Analysis of international landings in the high seas

Other
37

```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
par(mar = c(3, 20, 3, 1), mfrow = c(3, 2), mex = 0.5)
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, xlab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of pound
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$fishing_sector, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, xlab = "millions of
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(hs$pounds/10^6, hs$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```



```
round(tapply(hs$pounds, hs$catch_type, sum, na.rm = T) / sum(hs$pounds) * 100, 2) # << 1%
```

Discards Landings
0.03 99.97

3 Analysis of international landings in the high seas

```
round(tapply(hs$pounds, hs$reporting_status, sum, na.rm = T) / sum(hs$pounds) * 100, 2) #
```

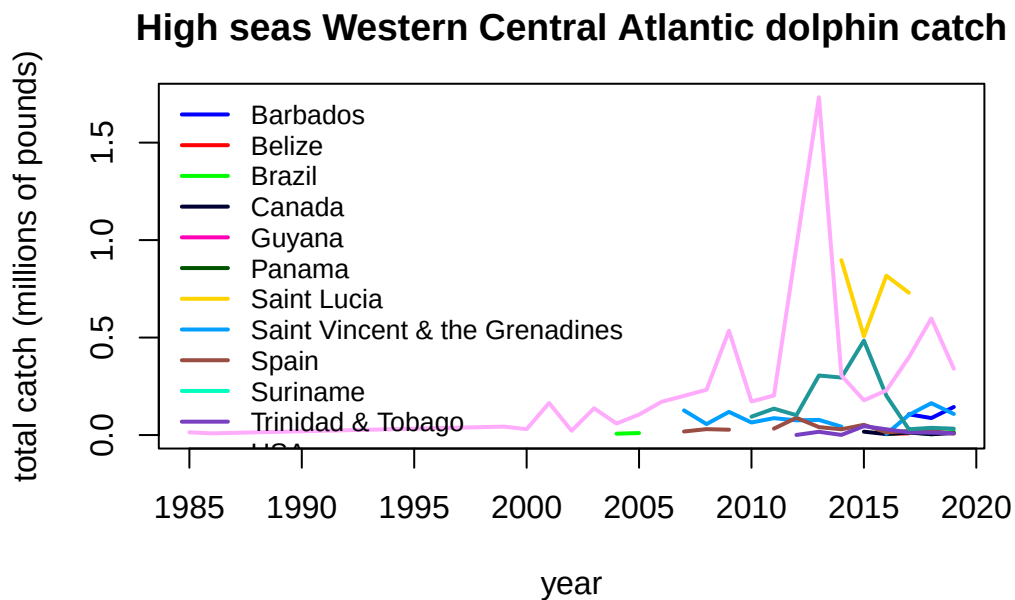
Reported	Unreported
99.97	0.03

3.3 Summarize the high seas landings

We plot the annual catches by fishing entity to look at trends over time.

```
hs$fishing_entity <- as.factor(hs$fishing_entity)
hs$fishing_entity <- droplevels(hs$fishing_entity)

par(mar = c(5, 5, 3, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```



3 Analysis of international landings in the high seas

Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA landings in our national reporting and are aiming to only summarize international landings here, we filter out the USA landings from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]

hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tail(tab, 40)
```

	Atlantic, Northwest	Atlantic, Western Central
1980	0.000	0.000
1981	0.000	0.000
1982	0.000	0.000
1983	0.000	0.000
1984	0.000	0.000
1985	0.000	14037.129
1986	0.000	8748.399
1987	0.000	0.000
1988	0.000	0.000
1989	0.000	0.000
1990	0.000	0.000
1991	0.000	0.000
1992	0.000	0.000
1993	0.000	0.000
1994	0.000	0.000
1995	0.000	0.000
1996	0.000	0.000
1997	0.000	0.000
1998	0.000	0.000
1999	0.000	43168.296
2000	0.000	30284.627
2001	0.000	164272.223
2002	0.000	22147.477
2003	0.000	137103.181
2004	0.000	66058.258
2005	0.000	115171.863
2006	0.000	170522.101

3 Analysis of international landings in the high seas

2007	14687.527	330838.620
2008	28552.636	340863.719
2009	18719.121	662720.661
2010	4510.085	232093.837
2011	21589.424	300498.224
2012	72398.812	1059210.392
2013	34741.604	2471791.396
2014	27922.953	1247462.983
2015	90313.638	710059.125
2016	15319.433	1094893.891
2017	34806.933	1374259.739
2018	29328.358	885814.754
2019	21870.621	632747.157

```
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

3.4 Distribute the high seas landings across quarters of the year

The international landings are only reported as annual catches and do not have any month or season associated with them. However, the MSE operating model requires seasonal catches. The majority of the international landings are caught by longline, so we will assume that the fleet distribution of the U.S. pelagic longline is representative of the seasonality of the international high seas fleets. Now we parse the annual catches according to the seasonality of the U.S. fleet.

At the time of analysis, the SAU database only contained catches to 2019, so we interpolated the 2019 values to cover years 2020 - 2022.

```
# read in the percentage of the catch by area and year-quarter combinations from the U.S. P
load("data/outputs/per_PLLcatch_by_area_yearquarter.RData")

tab <- tab[which(rownames(tab) >= 1997), ] # PLL data are only available from 1997 on
#tab[nrow(tab), ]
tab <- rbind(tab, tab[nrow(tab), ], tab[nrow(tab), ], tab[nrow(tab), ]) # interpolate 2019
tail(tab)
```

Atlantic, Northwest Atlantic, Western Central

3 Analysis of international landings in the high seas

2017	34806.93	1374259.7
2018	29328.36	885814.8
2019	21870.62	632747.2
	21870.62	632747.2
	21870.62	632747.2
	21870.62	632747.2

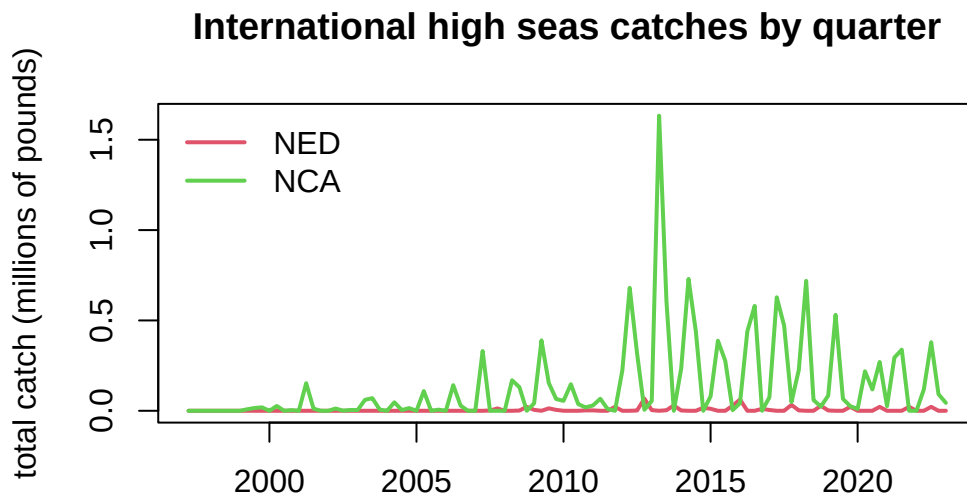
```
NED <- percatch[, , 1] # create matrices for final catch data
NCA <- percatch[, , 1]

for (i in 1:ncol(percatch)) {
  NED[, i] <- percatch[, i, 7] * tab[i, 1] # these are the percentages for the NED (Atlant
  NCA[, i] <- percatch[, i, 1] * tab[i, 2] # these are the percentages for the NCA (Weste
}

# plot the parsed out data by year-quarter
yrs <- sort(rep(as.numeric(colnames(NCA)), 4))
qrt <- rep(1:4, length(yrs)/4)

matplot(yrs + qrt/4, cbind(matrix(NED), matrix(NCA))/10^6, lty = 1, lwd = 2,
        col = 2:3, type = "l", main = "International high seas catches by quarter",
        xlab = "", ylab = "total catch (millions of pounds)")
legend("topleft", c("NED", "NCA"), lwd = 2, col = 2:3, bty = "n")
```


3 Analysis of international landings in the high seas



```
# save NED data in this format because it needs to be combined with Canada EEZ catches
save(NED, file = "data/outputs/NED_year_quarter.RData")
```

Finally, we format the data in the format for input into the operating model.

```
# format for operating model
flt <- rep("Intl", length(yrs))
are <- rep("NED", length(yrs))
fin1 <- data.frame(cbind(yrs, qrt, flt, are, matrix(NED)))
are <- rep("NCA", length(yrs))
fin2 <- data.frame(cbind(yrs, qrt, flt, are, matrix(NCA)))

findat <- data.frame(rbind(fin1, fin2), stringsAsFactors = FALSE)
names(findat) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
findat$Catch_lbs <- as.numeric(findat$Catch_lbs)
head(findat)
```

	Year	Quarter	Fleet	Area	Catch_lbs
1	1997	1	Intl	NED	0
2	1997	2	Intl	NED	0

3 Analysis of international landings in the high seas

3	1997	3	Intl	NED	0
4	1997	4	Intl	NED	0
5	1998	1	Intl	NED	0
6	1998	2	Intl	NED	0

```
#plot(findat$Catch_lbs, type = "l")
#plot(findat$Catch_lbs ~ factor(findat$Area))
#plot(findat$Catch_lbs ~ factor(findat$Quarter))
#plot(findat$Catch_lbs ~ factor(findat$Year))
#apply(findat, 2, table)
#barplot(tapply(findat$Catch_lbs, list(findat$Quarter, findat$Area), sum, na.rm = T), beside = TRUE)
write.csv(findat, file = "data/FINAL_files/Intl_highseas_TomFormat.csv", row.names = FALSE)
```

4 Analysis of international landings in territorial waters

4.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded from the Sea Around Us site (Pauly, Zeller, and Palomares, 2020.). In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (*Coryphaena hippurus*). After clicking "View graphs" one can access the full data set using the "Download Data" button.

4.2 Read in the landings within exclusive economic zones (EEZs)

Now we read in the landings reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin landings for EEZs worldwide, we need to separate out the landings for the Atlantic Ocean basin.

```
# load libraries
library(pals)

rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",") # v. 50.1 - Jun
head(d)
```

	area_name	area_type	year	scientific_name	common_name
1	Australia	eez	1950	<i>Coryphaena hippurus</i>	Common dolphinfish
2	Australia	eez	1950	<i>Coryphaena hippurus</i>	Common dolphinfish
3	Australia	eez	1950	<i>Coryphaena hippurus</i>	Common dolphinfish
4	Bahamas	eez	1950	<i>Coryphaena hippurus</i>	Common dolphinfish
5	Barbados	eez	1950	<i>Coryphaena hippurus</i>	Common dolphinfish

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```

6 Bermuda (UK)          eez 1950 Coryphaena hippurus Common dolphinfish
      functional_group commercial_group fishing_entity fishing_sector
1 Large pelagics (>=90 cm)      Perch-likes      Australia      Industrial
2 Large pelagics (>=90 cm)      Perch-likes      Australia      Industrial
3 Large pelagics (>=90 cm)      Perch-likes      Australia      Industrial
4 Large pelagics (>=90 cm)      Perch-likes      Bahamas        Recreational
5 Large pelagics (>=90 cm)      Perch-likes      Barbados        Recreational
6 Large pelagics (>=90 cm)      Perch-likes      Bermuda (UK)     Artisanal
      catch_type reporting_status      gear_type
1 Landings      Reported      longline
2 Landings      Reported      longline
3 Landings      Reported      longline
4 Landings      Unreported recreational fishing gear
5 Landings      Unreported recreational fishing gear
6 Landings      Reported      small scale lines
      end_use_type      tonnes landed_value
1 Fishmeal and fish oil 0.003373202 3.438170e-05
2 Direct human consumption 6.739657104 1.040254e+04
3      Other 0.003373202 3.438170e-05
4 Direct human consumption 48.665105143 8.262728e+00
5 Direct human consumption 0.009158559 4.846886e-04
6 Direct human consumption 1.511085514 3.199403e+03

```

```

#dim(d)
#apply(d[1:13], 2, table)

# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462

```

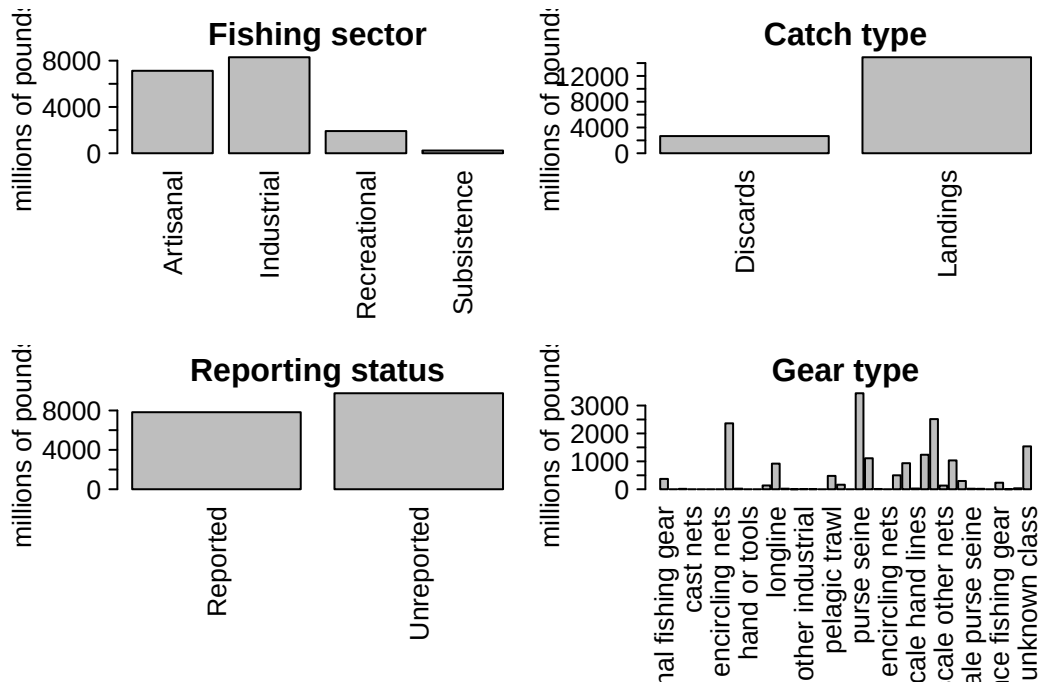
We look at the characteristics of the landings within EEZs. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```

# look at characteristics of data
par(mfrow = c(2, 2), mar = c(12, 7, 3, 1), mex = 0.5, mgp = c(5, 1, 0))
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")

```

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```
round(tapply(d$lbs, d$catch_type, sum, na.rm = T) / sum(d$lbs) * 100, 2) # 15% are discards
```

```
Discards Landings
15.18    84.82
```

```
round(tapply(d$lbs, d$reporting_status, sum, na.rm = T) / sum(d$lbs) * 100, 2) # 55% are unreported
```

```
Reported Unreported
44.52    55.48
```

4.3 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into "other oceans." This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

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```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type) # this should be all "eez"
```

```
eez
63056
```

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Grenada",
        "Aruba (Netherlands)", "Haiti", "Guadeloupe (France)", "Jamaica", "Martinique (France)",
        "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Nevis",
        "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin Islands",
        "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire",
        "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras (Caribbean)",
        "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
        "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba",
        "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
        "Panama (Caribbean)", "Bermuda (UK)", "Curaçao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
        "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <- c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <- c("USA (East Coast)", "USA (Gulf of Mexico)")

# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwest Africa)",
        "Spain (mainland, Med and Gulf of Cadiz)",
        "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
        "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
```

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```
"Côte d'Ivoire", "CÃ´te d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria"
"Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)"
"Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
        "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")

# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)", "Sardinia (Italy)",
        "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
        "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
        "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)",
        "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
        "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central Atlantic EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast Atlantic EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central Atlantic EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast Atlantic EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest Atlantic EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

```
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

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othlis

[1] "Australia"	"China"
[3] "Taiwan"	"Mayotte (France)"
[5] "French Polynesia"	"Hong Kong (China)"
[7] "Andaman & Nicobar Isl. (India)"	"Indonesia (Eastern)"
[9] "Japan (main islands)"	"Japan (Daito Islands)"
[11] "Korea (South)"	"Hawaii Northwest Islands (USA)"
[13] "Nauru"	"Niue (New Zealand)"
[15] "Northern Marianas (USA)"	"Micronesia (Federated States of)"
[17] "Marshall Isl."	"Palau"
[19] "Pakistan"	"Papua New Guinea"
[21] "Philippines"	"Seychelles"
[23] "Viet Nam"	"Tokelau (New Zealand)"
[25] "Tonga"	"Hawaii Main Islands (USA)"
[27] "USA (West Coast)"	"Wake Isl. (USA)"
[29] "Wallis & Futuna Isl. (France)"	"Samoa"
[31] "Guatemala (Pacific)"	"Kiribati (Gilbert Islands)"
[33] "Mexico (Pacific)"	"Japan (Ogasawara Islands)"
[35] "Korea (North, Yellow Sea)"	"Korea (North, Sea of Japan)"
[37] "Mauritius"	"Solomon Isl."
[39] "Christmas Isl. (Australia)"	"Johnston Atoll (USA)"
[41] "Timor Leste"	"Palmyra Atoll & Kingman Reef (USA)"
[43] "Jarvis Isl. (USA)"	"Howland & Baker Isl. (USA)"
[45] "Indonesia (Central)"	"Indonesia (Indian Ocean)"
[47] "Kiribati (Line Islands)"	"Kiribati (Phoenix Islands)"
[49] "New Caledonia (France)"	"Vanuatu"
[51] "Lord Howe Isl. (Australia)"	"Myanmar"
[53] "Ecuador (mainland)"	"El Salvador"
[55] "Fiji"	"Malaysia (Peninsula West)"
[57] "Norfolk Isl. (Australia)"	"Peru"
[59] "Pitcairn (UK)"	"Colombia (Pacific)"
[61] "Costa Rica (Pacific)"	"Nicaragua (Pacific)"
[63] "Panama (Pacific)"	"Thailand (Andaman Sea)"
[65] "Thailand (Gulf of Thailand)"	"Cook Islands"
[67] "Kermadec Isl. (New Zealand)"	"Tuvalu"
[69] "Honduras (Pacific)"	"American Samoa"
[71] "Galapagos Isl. (Ecuador)"	"New Zealand"
[73] "Brunei Darussalam"	"Malaysia (Sabah)"
[75] "Malaysia (Sarawak)"	"Guam (USA)"

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[77] "Clipperton Isl. (France)"	"South Africa (Indian Ocean Coast)"
[79] "Singapore"	"Easter Isl. (Chile)"
[81] "Madagascar"	"Réunion (France)"
[83] "Chagos Archipelago (UK)"	"Comoros Isl."
[85] "Mozambique Channel Isl. (France)"	"Kenya"
[87] "Mozambique"	"Somalia"
[89] "Tanzania"	"Glorieuse Islands (France)"
[91] "Yemen (Socotra)"	"Oman"
[93] "St Paul & Amsterdam Isl. (France)"	"Iran (Sea of Oman)"
[95] "Desventuradas Isl. (Chile)"	"India (mainland)"
[97] "Maldives"	"Tromelin Isl. (France)"
[99] "Cambodia"	"Sri Lanka"
[101] "Malaysia (Peninsula East)"	"Bangladesh"

```
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("USA", othlis)]
```

[1] "Hawaii Northwest Islands (USA)"	"Northern Marianas (USA)"
[3] "Hawaii Main Islands (USA)"	"USA (West Coast)"
[5] "Wake Isl. (USA)"	"Johnston Atoll (USA)"
[7] "Palmyra Atoll & Kingman Reef (USA)"	"Jarvis Isl. (USA)"
[9] "Howland & Baker Isl. (USA)"	"Guam (USA)"

```
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("France", othlis)]
```

[1] "Mayotte (France)"	"Wallis & Futuna Isl. (France)"
[3] "New Caledonia (France)"	"Clipperton Isl. (France)"
[5] "Réunion (France)"	"Mozambique Channel Isl. (France)"
[7] "Glorieuse Islands (France)"	"St Paul & Amsterdam Isl. (France)"
[9] "Tromelin Isl. (France)"	

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```
unique(d$area_name[which(d$reg == "Western Central Atlantic EEZs")]) # Western Central EE
```

```
[1] "Bahamas"
[2] "Barbados"
[3] "Bermuda (UK)"
[4] "Cayman Isl. (UK)"
[5] "Dominica"
[6] "Dominican Republic"
[7] "Grenada"
[8] "Guadeloupe (France)"
[9] "Haiti"
[10] "Jamaica"
[11] "Martinique (France)"
[12] "Montserrat (UK)"
[13] "Puerto Rico (USA)"
[14] "Saint Kitts & Nevis"
[15] "Saint Lucia"
[16] "Saint Vincent & the Grenadines"
[17] "Trinidad & Tobago"
[18] "US Virgin Isl."
[19] "St Barthelemy (France)"
[20] "St Martin (France)"
[21] "Curaçao (Netherlands)"
[22] "Bonaire (Netherlands)"
[23] "Saba and Sint Eustatius (Netherlands)"
[24] "Sint Maarten (Netherlands)"
[25] "Honduras (Caribbean)"
[26] "Colombia (Caribbean)"
[27] "Mexico (Atlantic)"
[28] "Turks & Caicos Isl. (UK)"
[29] "Costa Rica (Caribbean)"
[30] "Panama (Caribbean)"
[31] "Antigua & Barbuda"
[32] "Belize"
[33] "British Virgin Isl. (UK)"
[34] "Cuba"
[35] "French Guiana"
[36] "Guyana"
[37] "Anguilla (UK)"
[38] "Suriname"
```

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```
[39] "Venezuela"
[40] "Guatemala (Caribbean)"
[41] "Aruba (Netherlands)"
[42] "Nicaragua (Caribbean)"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")]) # Brazil EEZs:
```

```
[1] "Brazil (mainland)"
[2] "Fernando de Noronha (Brazil)"
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")]) # Canadian EEZs:
```

```
[1] "Canada (East Coast)" "Saint Pierre & Miquelon (France)"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")]) # United States EEZs:
```

```
[1] "USA (East Coast)" "USA (Gulf of Mexico)"
```

```
unique(d$area_name[which(d$reg == "Northeast Atlantic EEZs")]) # Northeast Atlantic EEZs:
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

```
unique(d$area_name[which(d$reg == "Eastern Central Atlantic EEZs")]) # Eastern Central Atlantic EEZs:
```

```
[1] "Cape Verde" "Equatorial Guinea"
[3] "Sao Tome & Principe" "Madeira Isl. (Portugal)"
[5] "Togo" "Ghana"
[7] "Guinea" "Liberia"
[9] "Guinea-Bissau" "Senegal"
[11] "Sierra Leone" "Morocco (Central)"
```

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```
[13] "Morocco (South)"      "Côte d'Ivoire"
[15] "Gabon"                "Benin"
[17] "Gambia"               "Mauritania"
[19] "Nigeria"             "Cameroon"
[21] "Congo, R. of"         "Congo (ex-Zaire)"
```

```
unique(d$area_name[which(d$reg == "Southeast Atlantic EEZs")]) # Southeast Atlantic EEZs:
```

```
[1] "Ascension Isl. (UK)"      "South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"       "Angola"
[5] "Namibia"                 "Tristan da Cunha Isl. (UK)"
```

```
unique(d$area_name[which(d$reg == "Southwest Atlantic EEZs")]) # Southwest Atlantic EEZs:
```

```
[1] "Trindade & Martim Vaz Isl. (Brazil)" "Uruguay"
```

```
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")]) # Mediterranean Sea EEZs:
```

```
[1] "Italy (mainland)"      "Libya"
[3] "Malta"                 "Syria"
[5] "Sicily (Italy)"        "Sardinia (Italy)"
[7] "Balearic Islands (Spain)" "Tunisia"
[9] "Algeria"               "Morocco (Mediterranean)"
[11] "France (Mediterranean)" "Corsica (France)"
[13] "Croatia"               "Lebanon"
[15] "Albania"               "Greece (without Crete)"
[17] "Crete (Greece)"        "Cyprus (South)"
```

4.4 Summarize catch by regional EEZs

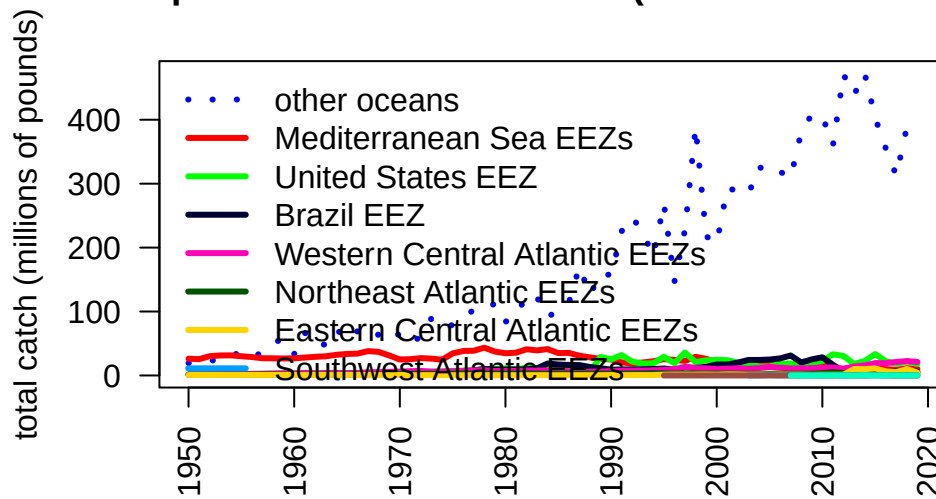
Now we will summarize the catch by region by grouping the catch across the EEZs by region and plot the results. The U.S. landings should be very close to what is reported in Section 2 because the majority of U.S. catch is reported through the various data collection systems that are compiled within Sea Around Us.

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```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
#head(tab)
tab2 <- t(tab)
tab3 <- tab2[order(rowSums(tab2, na.rm = T), decreasing = T), ]
tab_glob <- t(tab3)
#head(tab_glob)
yrs <- as.numeric(rownames(tab))

par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_glob/10^6, las = 2, type = "l", lty = c(3, rep(1, 9)), lwd = 3, col = glasbey(ncol(tab_glob)),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Global dolphin catches within EEZs (commercial + recreational)")
legend("topleft", colnames(tab_glob), col = glasbey(ncol(tab_glob)),
        lty = c(3, rep(1, 9)), lwd = 3, bty = "n")
```

Global dolphin catches within EEZs (commercial + recreation)



We can see that the EEZ catches globally are dominated by areas outside of the Atlantic ("other oceans") so we will replot the data after moving this category. Here we can see that most of the catches historically were coming from the Mediterranean Sea, though these catches have been declining drastically since the 1970s. Catches in the United States have showed an opposite pattern, with low historical catches that increased in

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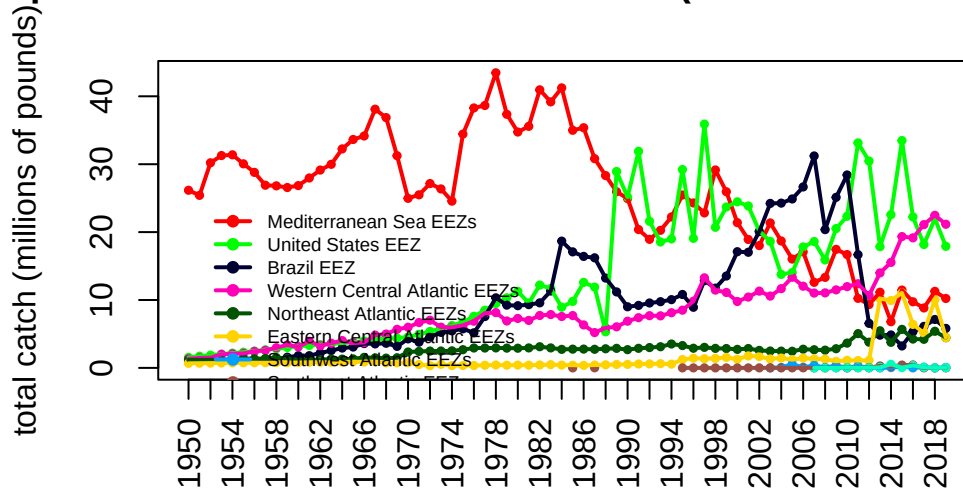
the 1980s (largely due to the introduction of recreational monitoring). Catches in the Western Central EEZs have increased gradually over time.

We save these data for later use, and move on to analyzing only the Western Central and Northwest Atlantic jurisdictions.

```
# remove other oceans
#head(tab_glob)
tab <- tab_glob[, -which(colnames(tab_glob) == "other oceans")]
#head(tab)

matplot(yrs, tab/10^6, las = 2, type = "b", pch = 19, cex = 0.5, lty = 1, col = glasbey(ncol(tab)-1),
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Dolphin catches from the Atlantic EEZs (commercial + recreational)")
matplot(yrs, tab/10^6, las = 2, type = "l", pch = 19, lty = 1, lwd = 2, add = T, col = glasbey(ncol(tab)-1))
axis(1, at = yrs[seq(1, 90, 2)], las = 2); axis(2); box()
legend(1952, 25, colnames(tab), col = glasbey(ncol(tab)+1)[-1], pch = 19, lty = 1, lwd = 2,
```

Dolphin catches from the Atlantic EEZs (commercial + recreational)



```
write.csv(data.frame(tab), file = "data/outputs/intl_catches_allAtlantic.csv", row.names = FALSE)
dwest <- d[which(d$reg == "Canadian EEZ" | d$reg == "Western Central Atlantic EEZs"), ]
```

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```
save(dwest, file = "data/outputs/SAU_EEZs_WCA.RData")
```

```
tab <- tapply(dwest$lbs, list(dwest$year, dwest$area_name), sum, na.rm = T)
sort(colSums(tab, na.rm = T), decreasing = T)[1:15] # eight countries are responsible for
```

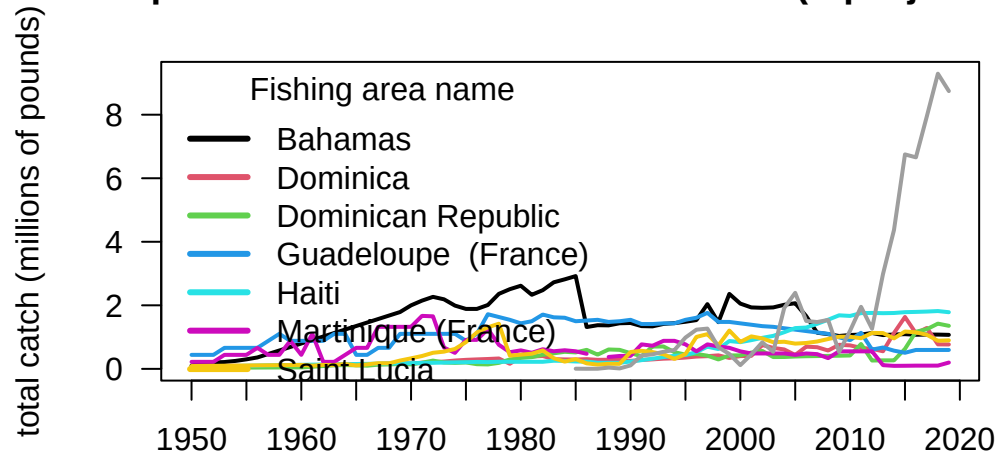
Bahamas	Guadeloupe (France)	Venezuela
102439744	77112296	69319813
Martinique (France)	Haiti	Saint Lucia
41322610	41295995	39294031
Dominica	Dominican Republic	Mexico (Atlantic)
26516078	25336178	16150763
Grenada	Trinidad & Tobago	Aruba (Netherlands)
15619929	14306678	13517868
Jamaica	Puerto Rico (USA)	Curaçao (Netherlands)
10236038	9784950	9667059

```
lis <- names(sort(colSums(tab, na.rm = T), decreasing = T)[1:8])
```

```
tab1 <- tab[, colnames(tab) %in% lis]
```

```
matplot(rownames(tab1), tab1/10^6, las = 2, type = "l", lty = 1, lwd = 2, pch = 19, col = 1:8,
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in Caribbean EEZs (top 8 jurisdictions)")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab1), col = 1:8, lty = 1, lwd = 3, bty = "n", title = "Fishing")
```

Total dolphinfish catches in Caribbean EEZs (top 8 jurisdictions)



Note the rapid eight-fold increase in catches listed within the Venezuelan EEZ which occurs from 2013 - 2019. This increase is responsible for a rapid increase in estimated international catches within the Western Atlantic seen in the prior plot.

4.5 Look at Venezuela catches in detail

The figures above indicate that the database shows a rapid increase in international catches within the Western Atlantic, driven by a sudden increase in catches reported within Sea Around Us database within the Venezuelan EEZ. We investigate these catches in detail, as it seems odd that a relatively small fleet from a single country would be able to ramp up effort so rapidly. Data can be downloaded on a country-specific basis, the catches by taxon in the waters of Venezuela are accessed here.

```
rm(list = ls()) # clear the workspace

# download the data for waters of Venezuela
ven <- read.csv("data/SAU/SAU EEZ 862 v50-1.csv")

# subset dolphinfish from the data
```

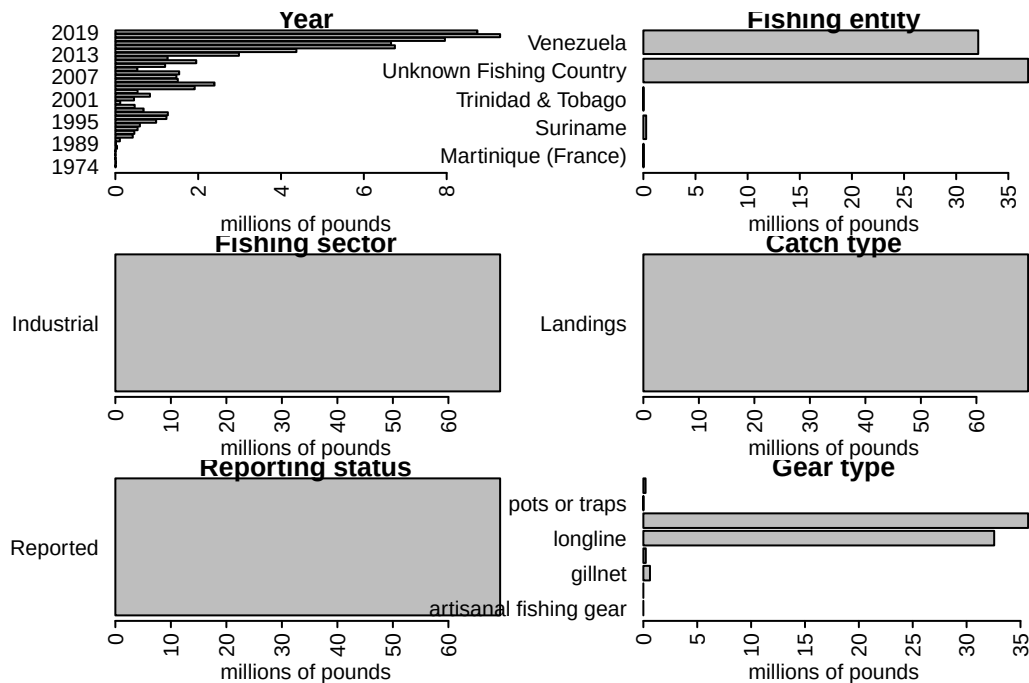

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```
v <- ven[which(ven$common_name == "Common dolphinfish"), ]

#apply(v[1:13], 2, table)

#convert to pounds
v$pounds <- v$tonnes * 1000 * 2.20462

# look at the characteristics of the data
par(mar = c(5, 10, 1, 1), mfrow = c(3, 2), mex = 0.5)
#barplot(tapply(v$pounds/10^6, v$area_name, sum, na.rm = T), las = 2, xlab = "millions of p
#barplot(tapply(v$pounds/10^6, v$area_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$year, sum, na.rm = T), las = 2, xlab = "millions of pounds"
barplot(tapply(v$pounds/10^6, v$fishing_entity, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$fishing_sector, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$catch_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$reporting_status, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v$pounds/10^6, v$gear_type, sum, na.rm = T), las = 2, xlab = "millions of po
```



Note that a large portion of the catch in the database is listed as coming from an “Unknown Fishing Country.” We extract these catches and further investigate the characteristics.

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```
# extract the Unknown Fishing Country catch
v1 <- v[which(v$fishing_entity == "Unknown Fishing Country"), ]
apply(v1[1:15], 2, table)
```

\$area_name

Venezuela
39

\$area_type

eez
39

\$data_layer

Assigned Tuna RFMO catch
39

\$uncertainty_score
< table of extent 0 >

\$year

2013	2014	2015	2016	2017	2018	2019
5	5	5	6	7	5	6

\$scientific_name

Coryphaena hippurus
39

\$common_name

Common dolphinfish
39

\$functional_group

Large pelagics (>=90 cm)

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39

\$commercial_group

Perch-likes

39

\$fishing_entity

Unknown Fishing Country

39

\$fishing_sector

Industrial

39

\$catch_type

Landings

39

\$reporting_status

Reported

39

\$gear_type

artisanal fishing gear

1

bottom trawl

5

gillnet

7

hand lines

3

longline

4

pole and line

7

pots or traps

5

unknown by source

7

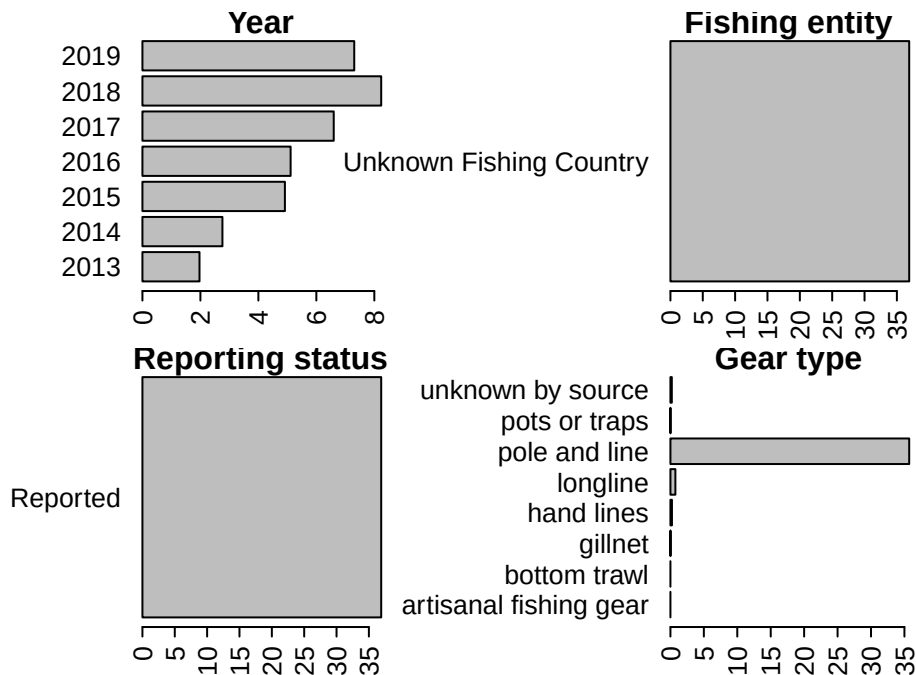
\$end_use_type

Direct human consumption

39

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```
# look at characteristics
par(mar = c(3, 10, 1, 5), mfrow = c(2, 2), mex = 0.6)
barplot(tapply(v1$pounds/10^6, v1$year, sum, na.rm = T), las = 2, xlab = "millions of pound
barplot(tapply(v1$pounds/10^6, v1$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v1$pounds/10^6, v1$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(v1$pounds/10^6, v1$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```



Here is another look at the catches within the venezuelan EEZ, separated by fishing entity.

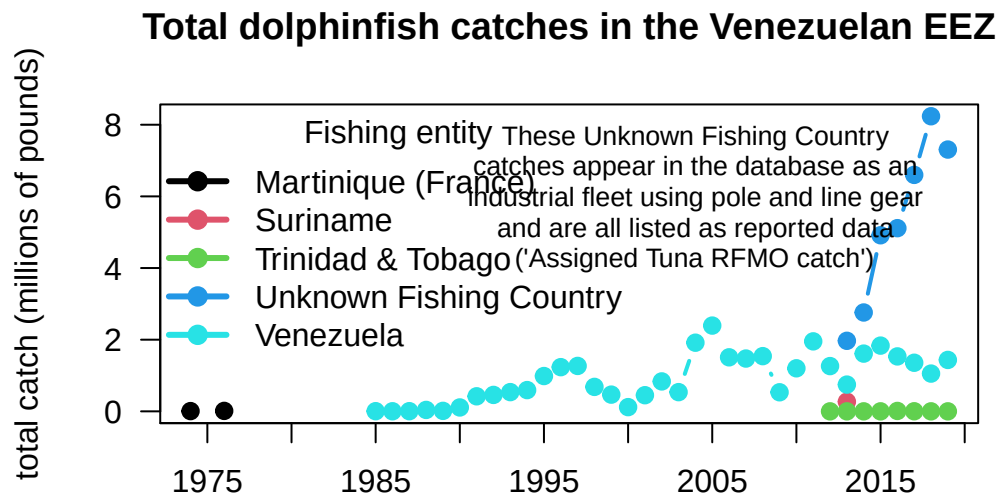
```
# summarize Venezuela EEZ catches by fishing entity
tab <- tapply(v$pounds, list(v$year, v$fishing_entity), sum, na.rm = T)
head(tab)
```

	Martinique (France)	Suriname	Trinidad & Tobago	Unknown Fishing Country
1974	7320.429	NA	NA	NA
1976	12814.641	NA	NA	NA
1985	NA	NA	NA	NA
1986	NA	NA	NA	NA
1987	NA	NA	NA	NA

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Year	NA	NA	NA	NA
1988				
	Venezuela			
1974	NA			
1976	NA			
1985	4699.563			
1986	2229.484			
1987	4585.610			
1988	39960.149			

```
matplot(rownames(tab), tab/10^6, las = 2, type = "b", lty = 1, lwd = 2, pch = 19, col = 1:5,
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in the Venezuelan EEZ")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab), col = 1:5, pch = 19, lty = 1, lwd = 3, bty = "n", title =
text(2004, 6, cex = 0.8,
     "These Unknown Fishing Country\ncatches appear in the database as an\nindustrial fleet
```



The very substantial increase in international Western Atlantic dolphin catches can be attributed solely to this Unknown Fishing Country catches which appear in the database beginning in 2013. The documentation of the catch reconstruction can be

4 Analysis of international landings in territorial waters

found in the Sea Around Us documentation files (see Mendoza et al. 2015). We spoke to Jeremy Mendoza and other experts who were involved in the reconstruction of domestic catches in Venezuela, and they noted that the “Assigned tuna RFMO catch” actually emerge from spatially reshuffled landing reports to ICCAT, according to the methods of Coulter et al. (2020). In this work, the Sea Around Us team developed a global spatial reallocation of landings of tunas and associated species that were reported to ICCAT. In short, this work uses the fraction of reported spatially disaggregated data to reallocate all data reported to ICCAT. This step is necessary because there are not many countries that report spatial data to ICCAT in the region, and this method may lead to a significant reallocation of landings from the Caribbean to the Venezuelan EEZ, where the local pole and line fishery is significant (J. Mendoza, personal communication). Thus, it is likely that landings from other EEZs, that were included in the reconstructed catch data layer, could have been reassigned to the Venezuelan EEZ.

Because there may be some overlap between the reconstructed data used in the national EEZ estimates, and the ICCAT reporting that was reassigned, there could potentially be a “double counting” of these catch estimates in the Sea Around Us database. We verified with Sea Around Us database managers that their method led to an error resulting in an overestimate of the catch, and they advised us to remove these Unknown Fishing Country catches as they are difficult to trace back to the data source.

4.6 Compile catch in CAR and NED jurisdictional waters by discards and reporting type

Now that we have separated the EEZ catch by region, we will summarize it according to the areas used in the dolphin management strategy evaluation operating model. We will separate the catch by discards versus landings and reported versus unreported data so that we can use these data in various sensitivity analyses.

```
rm(list = ls()) # clear the workspace

# load data
load("data/outputs/SAU_EEZs_WCA.RData") # SAU EEZ data for NED and NCA saved from earlier
load("data/outputs/erroneous_catch_to_be_removed.RData") # double-counted catch values to

d <- dwest

# summarize the catch by area, discards/landings and reported/unreported
tapply(d$lbs, list(d$catch_type, d$reporting_status, d$reg), sum, na.rm = T)
```

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, , Canadian EEZ

	Reported	Unreported
Discards	NA	NA
Landings	1220146	NA

, , Western Central Atlantic EEZs

	Reported	Unreported
Discards	NA	2179444
Landings	393279187	176215080

The catch from the Canadian EEZs is 100% landings, with no unreported data and no discards. In the WCA there are no reported discards, only unreported discards. Thus, we only need to separate out unreported discards and landings for the WCA.

```
# summarize the catch by year, area, discards/landings and reported/unreported

d$year <- factor(d$year, levels = 1950:2019)
dc <- d[which(d$reg == "Canadian EEZ"), ]
dw <- d[which(d$reg == "Western Central Atlantic EEZs"), ]

# verify proper identification of the erroneous Venezuelan Unknown Fishing Country catches
bad <- dw[which(dw$area_name == "Venezuela" & dw$fishing_entity == "Unknown Fishing Country"), ]
#dim(bad)
bad_totals <- tapply(bad$lbs, bad$year, sum, na.rm = T)

cbind(tail(bad_totals, 10), tail(removecatch, 10)) # the numbers match exactly
```

	[,1]	[,2]
2010	NA	NA
2011	NA	NA
2012	NA	NA
2013	1968566	1968566
2014	2759603	2759603
2015	4915156	4915156
2016	5112450	5112450
2017	6599737	6599737
2018	8236857	8236857
2019	7307245	7307245

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```
# now remove those bad catches from the database prior to summarizing
#dim(dw)
dw <- dw[-which(dw$area_name == "Venezuela" & dw$fishing_entity == "Unknown Fishing Country", ) , ]
#dim(dw)

CAN <- tapply(dc$lbs, list(dc$year, dc$catch_type, dc$reporting_status), sum, na.rm = T)
tail(CAN)
```

, , Reported

	Landings
2014	531022.55
2015	38945.76
2016	354370.65
2017	83776.88
2018	12588.88
2019	36112.19

```
tab_dw <- tapply(dw$lbs, list(dw$year, dw$catch_type, dw$reporting_status), sum, na.rm = T)
tail(tab_dw)
```

, , Reported

	Discards	Landings
2014	NA	9923818
2015	NA	11590469
2016	NA	11225710
2017	NA	11702729
2018	NA	11460758
2019	NA	11066090

, , Unreported

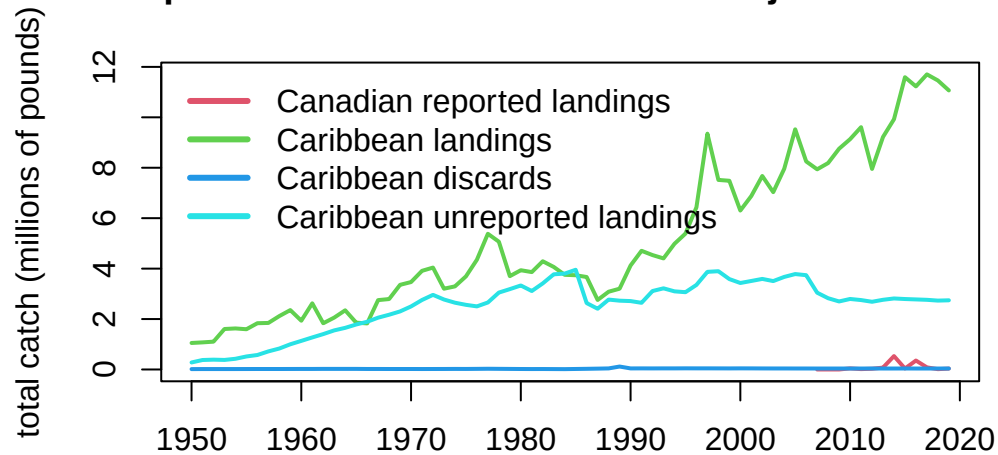
	Discards	Landings
2014	38062.63	2816498
2015	39134.61	2794662
2016	38973.17	2778493
2017	38728.81	2762926
2018	38769.93	2732351
2019	38655.51	2743182

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```
# combine CAR reported landings, unreported discards, unreported landings
CAR <- data.frame(cbind(tab_dw[, 2, 1], tab_dw[, , 2]))
names(CAR) <- c("landings", "discards", "unreported")

matplot(rownames(CAR), cbind(CAN, CAR)/10^6, type = "l", lwd = 2, lty = 1, col = 2:5,
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Total dolphinfish catches in NED and NCA jurisdictional waters")
legend("topleft", col = 2:5, c("Canadian reported landings", "Caribbean landings",
                               "Caribbean discards", "Caribbean unreported landings"), lty = 1, lwd =
```

Total dolphinfish catches in NED and NCA jurisdictional wa



```
round(sum(CAR$discards, na.rm = T) / (sum(CAR$discards, na.rm = T) + sum(CAR$landings, na.rm = T)), 2)
```

```
[1] 0.61
```

Note that the Caribbean unreported discards in the database are negligible (< 1%) and we exclude them from further analysis.

4.7 Distribute the territorial international catches across quarters of the year

The international landings are only reported as annual catches and do not have any month or season associated with them. However, the MSE operating model requires seasonal catches. Lacking other information on the seasonality of these fleets, we will assume that the fleet distribution of the U.S. pelagic longline is representative of the seasonality of the jurisdictional fleets. Now we parse the annual catches according to the seasonality of the U.S. fleet.

The pelagic longline logbook data is only available going back to 1997, but there are Caribbean catches extending back to the 1950s. To interpolate the seasonality for 1986 - 1996, we take the average seasonality of the oldest 10 years of logbook data (1997 - 2006) and use those averages for the unknown years.

At the time of analysis, the SAU database only contained catches to 2019, so we interpolated the 2019 values to cover years 2020 - 2022.

```
#par(mar = c(5, 10, 3, 1))
#barplot(tapply(dw$lbs, dw$gear_type, sum, na.rm = T), horiz = T, las = 2)
#barplot(tapply(dc$lbs, dc$gear_type, sum, na.rm = T), horiz = T, las = 2)

# read in the percentage of the catch by area and year-quarter combinations from the U.S. P
load("data/outputs/per_PLLcatch_by_area_yearquarter.RData")

percatch[1, 1, ] # matrix 2 == Caribbean, matrix 7 == NED (Atlantic Northwest)
```

	NCA	CAR	FLK	NCFL	NNC	VBM
0.755794702	0.547263682	0.049739219	0.001825132	0.000000000	0.000000000	
NED						
0.000000000						

```
CAN <- CAN[which(rownames(CAN) >= 1997)] # PLL data are only available from 1997 on
CAN <- c(CAN, CAN[length(CAN)], CAN[length(CAN)], CAN[length(CAN)]) # interpolate 2019 cat

CANqrt <- percatch[, , 7] # create matrix for final catch data
for (i in 1:ncol(percatch)) {
  CANqrt[, i] <- percatch[, i, 7] * CAN[i] }

percatch_car <- percatch[, , 2]
```

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```
int <- rowMeans(percatch_car[, 1:10])
percatch_car <- cbind(matrix(rep(int, length(1986:1996)), nrow = 4), percatch_car)
colnames(percatch_car) <- 1986:2022 # full data frame for Caribbean seasonality 1986 - 2022

CAR <- CAR[which(rownames(CAR) >= 1986), ] # only need Caribbean data from 1986 on
CAR <- rbind(CAR, CAR[nrow(CAR), ], CAR[nrow(CAR), ], CAR[nrow(CAR), ]) # interpolate 2019

#dim(CAR)
#dim(percatch_car)

CARrep <- percatch_car # create matrices for final catch data
CARunr <- percatch_car

for (i in 1:ncol(percatch_car)) {
  CARrep[, i] <- percatch_car[, i] * CAR[i, 1]
  CARunr[, i] <- percatch_car[, i] * CAR[i, 3]
}

# ensure sums are equal after parsing
sum(CARrep); sum(CAR$landings)
```

```
[1] 286220280
```

```
[1] 286220280
```

```
sum(CARunr); sum(CAR$unreported)
```

```
[1] 112754888
```

```
[1] 112754888
```

```
sum(CAN, na.rm = T); sum(CANqrt, na.rm = T)
```

```
[1] 1328482
```

```
[1] 1328482
```

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The Canada jurisdictional data need to be combined with the NED high seas data calculated in the previous section, because both of these are a component of the international catches in the NED area.

```
load("data/outputs/NED_year_quarter.RData")
```

```
NEDtot <- NED + CANqrt
```

```
sum(NED, na.rm = T) + sum(CANqrt, na.rm = T); sum(NEDtot, na.rm = T)
```

```
[1] 1808855
```

```
[1] 1808855
```

```
NEDtot <- NEDtot[, !is.na(colSums(NEDtot))] # remove NA columns from CAN data
```

```
# plot the parsed out data by year-quarter
```

```
yrs <- sort(rep(as.numeric(colnames(CARrep)), 4))
```

```
qrt <- rep(1:4, length(yrs)/4)
```

```
yrs1 <- sort(rep(as.numeric(colnames(NEDtot)), 4))
```

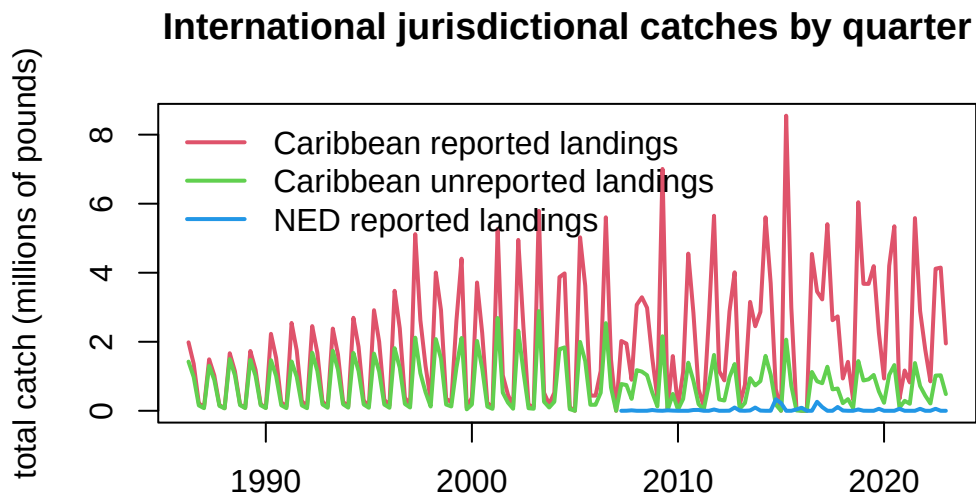
```
qrt1 <- rep(1:4, length(yrs1)/4)
```

```
matplot(yrs + qrt/4, cbind(matrix(CARrep), matrix(CARunr))/10^6, lty = 1, lwd = 2,  
      col = 2:3, type = "l", main = "International jurisdictional catches by quarter",  
      xlab = "", ylab = "total catch (millions of pounds)")
```

```
lines(yrs1 + qrt1/4, as.numeric(matrix(NEDtot))/10^6, lwd = 2, col = 4)
```

```
legend("topleft", c("Caribbean reported landings", "Caribbean unreported landings", "NED re  
      lwd = 2, col = 2:4, bty = "n")
```

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Finally, we format the data in the format for input into the operating model.

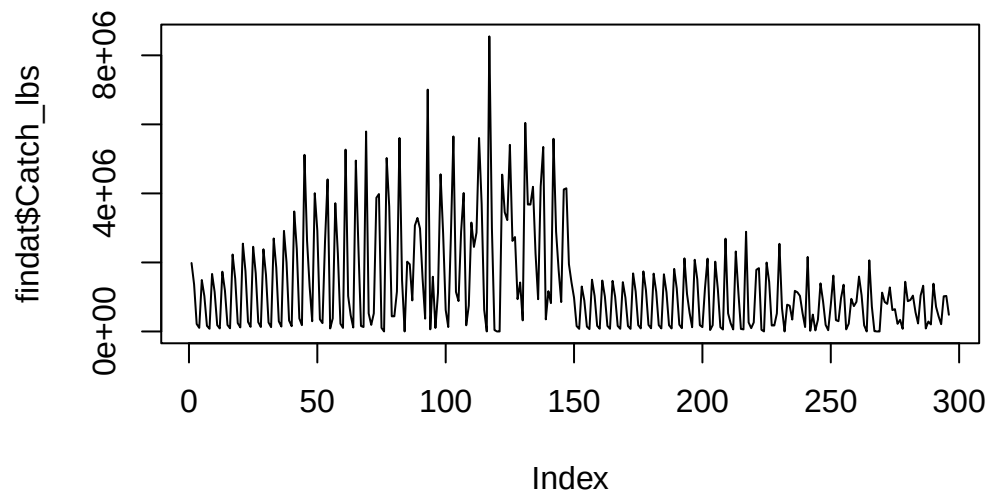
```
# format for operating model - Caribbean
flt <- rep("Intl", length(yrs))
are <- rep("CAR", length(yrs))
fin1 <- data.frame(cbind(yrs, qrt, flt, are, matrix(CARrep)))
flt <- rep("UnRep", length(yrs))
fin2 <- data.frame(cbind(yrs, qrt, flt, are, matrix(CARunr)))

findat <- data.frame(rbind(fin1, fin2), stringsAsFactors = FALSE)
names(findat) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
findat$Catch_lbs <- as.numeric(findat$Catch_lbs)
head(findat)
```

	Year	Quarter	Fleet	Area	Catch_lbs
1	1986	1	Intl	CAR	1983610.3
2	1986	2	Intl	CAR	1363432.3
3	1986	3	Intl	CAR	216194.7
4	1986	4	Intl	CAR	104564.4
5	1987	1	Intl	CAR	1491251.1
6	1987	2	Intl	CAR	1025009.7

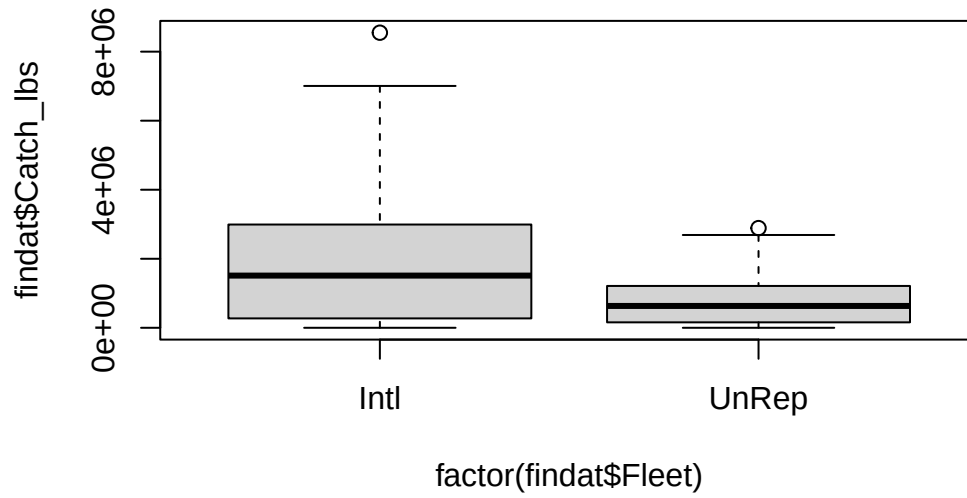
4 Analysis of international landings in territorial waters

```
plot(findat$Catch_lbs, type = "l")
```

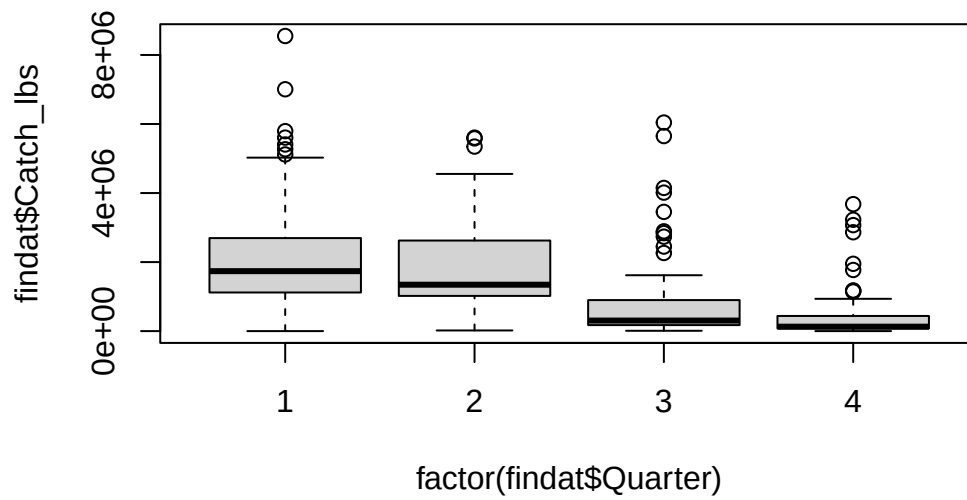


```
plot(findat$Catch_lbs ~ factor(findat$Fleet))
```

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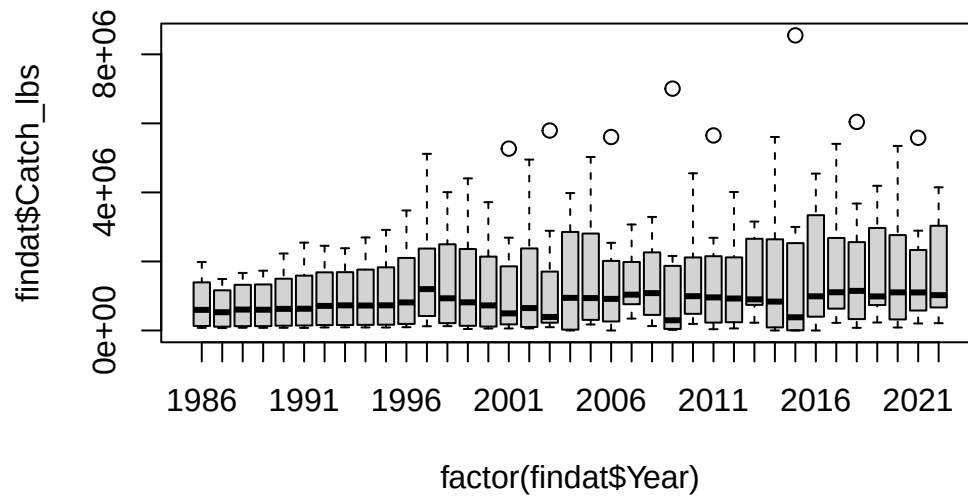


```
plot(findat$Catch_lbs ~ factor(findat$Quarter))
```



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```
plot(findat$Catch_lbs ~ factor(findat$Year))
```



```
apply(findat, 2, table)
```

\$Year

1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
2018	2019	2020	2021	2022											
8	8	8	8	8											

\$Quarter

1	2	3	4
74	74	74	74

\$Fleet

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Intl UnRep
148 148

\$Area

CAR
296

\$Catch_lbs

0.000	9788.658	18795.593	31325.988	36496.425	40597.091
10	1	1	1	1	1
42107.302	47942.517	56401.113	58679.944	60096.556	60355.398
1	1	1	1	1	1
60985.763	68676.820	73889.208	75072.838	75366.606	76555.925
1	1	1	1	1	1
77324.643	77796.214	78610.095	79013.468	86809.568	87372.358
1	1	1	1	1	1
87909.943	87912.183	88337.800	88700.132	91332.089	91760.040
1	1	1	1	1	1
95367.294	95587.693	101642.938	103649.945	103959.370	104564.413
1	1	1	1	1	1
114975.069	117566.580	122091.821	122788.214	125568.197	125577.532
1	1	1	1	1	1
127378.801	127939.292	128377.939	129341.470	134181.182	141994.444
1	1	1	1	1	1
142082.599	153611.205	155218.688	155826.075	157841.728	159874.464
1	1	1	1	1	1
160849.471	162532.231	163366.235	174374.503	177603.665	178729.645
1	1	1	2	1	1
180648.863	181760.360	181800.152	181826.986	182644.985	183296.695
1	1	1	1	1	1
183394.133	188835.902	189720.723	190653.498	191294.330	197634.680
1	1	1	1	1	1
203382.417	211165.836	216194.717	221034.105	224371.646	224865.912
1	1	1	1	1	1
231889.538	238171.536	242198.267	243077.667	259621.606	261005.227
1	1	1	1	1	2
267423.130	277429.596	289365.228	293766.363	296609.364	297403.409
1	1	1	1	1	1
317602.615	321111.411	328467.823	338237.998	342580.505	344719.080

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1	1	1	1	1	1
350192.731	350633.439	370608.639	378979.578	379620.789	438906.151
1	1	1	1	1	2
439008.388	466663.514	483510.559	488685.417	523276.184	523542.377
1	1	1	1	1	2
524667.731	532547.301	541712.711	560990.412	619518.125	622719.273
1	1	1	1	1	1
625779.400	645979.955	675898.404	716799.006	723381.857	734121.066
1	1	1	1	1	1
746891.341	750312.866	769344.634	775617.931	798676.095	820451.542
1	1	1	1	1	1
851850.118	854920.890	858458.688	868863.599	876913.328	880696.803
1	1	1	1	1	1
895488.156	898560.296	911356.265	935450.233	936218.338	945759.571
1	1	1	1	1	1
972684.090	978886.865	982717.351	1008248.523	1014397.405	1020305.956
1	1	1	1	1	1
1025009.732	1028012.380	1028200.006	1029911.297	1030269.383	1038946.142
1	1	1	1	1	1
1041714.819	1047861.626	1088240.645	1118948.693	1124895.908	1135461.213
1	1	1	1	1	1
1139262.299	1140356.772	1146271.954	1146409.210	1151850.846	1154659.868
1	1	1	1	1	1
1156575.350	1167309.105	1177790.191	1190893.874	1196473.997	1228880.724
1	1	1	1	1	1
1246383.379	1250459.272	1276394.128	1302814.665	1308570.723	1325292.742
1	1	1	1	1	1
1354934.942	1363432.275	1383992.545	1394153.707	1418728.597	1424148.555
1	1	1	1	1	1
1429721.395	1438589.646	1440643.324	1466865.812	1475811.608	1491251.114
1	1	1	1	1	1
1491435.663	1498903.198	1512675.756	1532969.639	1542658.460	1585629.832
1	1	1	1	1	1
1591032.836	1618981.402	1637304.009	1657473.212	1667671.316	1675787.852
1	1	1	1	1	1
1682661.369	1686504.331	1732590.200	1740708.527	1749610.120	1770974.671
1	1	1	1	1	1
1786489.583	1813319.956	1835946.074	1852637.965	1946880.641	1950497.933
1	1	1	1	1	1
1983610.336	1996588.054	2002961.321	2021760.666	2022477.562	2061491.464
1	1	1	1	1	1

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2077364.264	2110469.016	2118698.602	2159404.791	2230264.367	2258349.389
1	1	1	1	1	1
2263054.276	2317658.148	2382056.825	2390035.223	2436024.009	2449550.826
1	1	1	1	1	1
2453636.667	2540069.809	2545447.091	2610725.425	2624048.582	2628773.185
1	1	1	1	1	1
2685145.127	2688922.121	2695338.730	2736131.059	2837913.353	2864429.705
1	1	1	1	1	1
2880396.604	2888123.223	2891591.411	2914038.968	2917460.171	2983399.544
1	1	1	1	1	1
3000124.991	3070081.011	3155727.641	3226823.465	3289151.671	3454064.554
1	1	1	1	1	1
3477179.365	3620975.746	3676441.965	3678185.252	3692098.460	3716765.085
1	1	1	1	1	1
3873854.414	3981096.712	4006560.875	4012351.024	4115948.700	4147793.564
1	1	1	1	1	1
4191143.840	4202312.777	4406265.156	4544821.781	4553634.683	4950968.882
1	1	1	1	1	1
5025475.429	5117965.494	5268563.446	5346275.700	5406331.249	5583072.690
1	1	1	1	1	1
5604911.442	5605940.459	5650523.616	5793197.766	6042732.914	7006586.524
1	1	1	1	1	1
8549747.268					
1					

```
write.csv(findat, file = "data/FINAL_files/CAR_rep_unrep_TomFormat.csv", row.names = FALSE)

# format for operating model - NED
flt <- rep("Intl", length(yrs1))
are <- rep("NED", length(yrs1))
fin <- data.frame(cbind(yrs1, qrt1, flt, are, matrix(NEDtot)), stringsAsFactors = FALSE)
names(fin) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
fin$Catch_lbs <- as.numeric(fin$Catch_lbs)
head(fin)
```

	Year	Quarter	Fleet	Area	Catch_lbs
1	2007	1	Intl	NED	0.000
2	2007	2	Intl	NED	1450.366
3	2007	3	Intl	NED	13279.912
4	2007	4	Intl	NED	0.000

4 Analysis of international landings in territorial waters

```
5 2008      1 Intl  NED      0.000
6 2008      2 Intl  NED  1312.750
```

```
apply(fin, 2, table)
```

\$Year

```
2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022
      4     4     4     4     4     4     4     4     4     4     4     4     4     4     4     4
```

\$Quarter

```
  1  2  3  4
16 16 16 16
```

\$Fleet

```
Intl
  64
```

\$Area

```
NED
  64
```

\$Catch_lbs

```
      0.0000      172.2071      1312.7496      1450.3658      2242.1859      2661.4120
      33          1          1          1          1          1
5080.1088      5124.9962      5250.9982      5461.1683      7529.1307      9708.7436
      1          1          1          1          1          1
13279.9117      13518.2557      22170.8813      23742.5934      24859.8919      37049.2036
      1          1          1          1          1          1
39255.8271      39772.1232      57982.8061      89487.2772      93932.0946      94171.8058
      1          1          4          1          1          1
108201.9757      111054.6782      224735.2547      261488.1078      334210.2501
      1          1          1          1          1
```

4 Analysis of international landings in territorial waters

```
write.csv(fin, file = "data/FINAL_files/NED_Intl_TomFormat.csv", row.names = FALSE)

#barplot(tapply(fin$Catch_lbs, list(fin$Quarter), sum, na.rm = T), col = rainbow(4))
```

4.8 References

Pauly D., Zeller D., Palomares M.L.D. (Editors), 2020. Sea Around Us Concepts, Design and Data (searoundus.org).

5 Data compilation for all fleets in the operating model

In this last step we will compile all of the output data files from the different data sources (NMFS commercial trip tickets, NMFS recreational statistics, Sea Around Us) and check the files for accuracy.

Data upload

First we access all of the final data files from the previous steps.

```
# clear workspace
rm(list = ls())

# view data files in directory
dir("data/FINAL_files/")
```

```
[1] "CAR_rep_unrep_TomFormat.csv"      "commercial_TomFormat.csv"
[3] "complete_catch_dataset_09052025.csv" "Intl_highseas_TomFormat.csv"
[5] "NED_Intl_TomFormat.csv"           "rec_catch_TomFormat.csv"
```

```
intc <- read.csv("data/FINAL_files/CAR_rep_unrep_TomFormat.csv")
inth <- read.csv("data/FINAL_files/Intl_highseas_TomFormat.csv")
intn <- read.csv("data/FINAL_files/NED_Intl_TomFormat.csv")
com <- read.csv("data/FINAL_files/commercial_TomFormat.csv")
rec <- read.csv("data/FINAL_files/rec_catch_TomFormat.csv")
```

```
#head(intc)
#head(inth)
#head(intn)
#head(com)
#head(rec)
```

```
d <- rbind(intc, inth, intn, com, rec)
```

5 Data compilation for all fleets in the operating model

```
# check for zeros and NAs
#hist(d$Catch_lbs)
table(d$Catch_lbs == 0)
```

```
FALSE TRUE
2078   710
```

```
table(is.na(d$Catch_lbs))
```

```
FALSE
2788
```

```
apply(d, 2, function (x) table(is.na(x)))
```

Year	Quarter	Fleet	Area	Catch_lbs
2788	2788	2788	2788	2788

```
# resort levels geographically
d$Area <- factor(d$Area, levels = c("NCA", "CAR", "FLK", "NCFL", "NNC", "VBM", "NED"))
d$Fleet <- factor(d$Fleet, levels = c("Rec", "Hire", "UScom", "Intl", "UnRep"))
```

Recall that there are seven defined areas in the operating model, over which the catches are summed. Four of these areas are within the U.S. EEZ and the other three areas are international waters (high seas and jurisdictional waters of other countries).

5 Data compilation for all fleets in the operating model

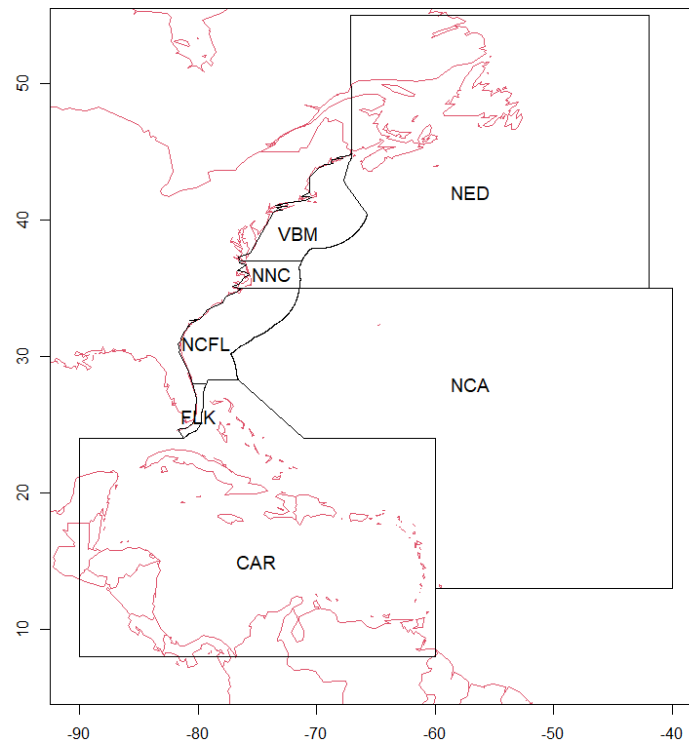


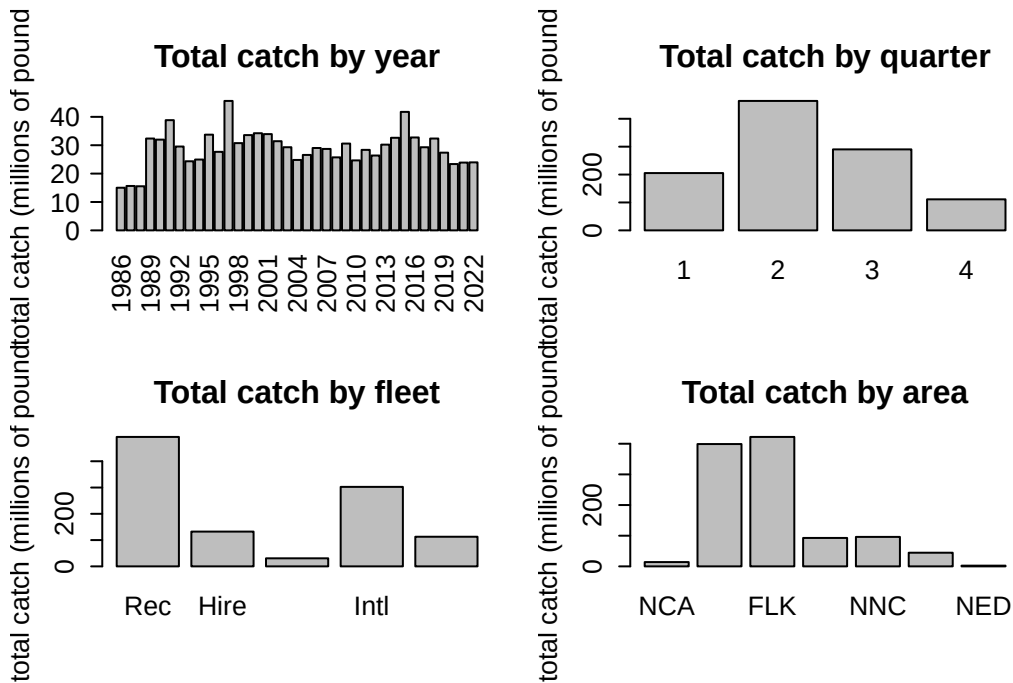
Figure 5.1: Areas defined in the operating model.

5.1 Visualizing the final data set

We plot the final data set in a number of different ways to detect any errors that may have occurred during processing.

```
# plot the total catches by each factor
par(mfrow = c(2, 2), mex = 0.7)
barplot(tapply(d$Catch_lbs, d$Year, sum, na.rm = T)/10^6, las = 2,
        main = "Total catch by year", ylab = "total catch (millions of pounds)")
barplot(tapply(d$Catch_lbs, d$Quarter, sum, na.rm = T)/10^6,
        ylab = "total catch (millions of pounds)", main = "Total catch by quarter")
barplot(tapply(d$Catch_lbs, d$Fleet, sum, na.rm = T)/10^6,
        main = "Total catch by fleet", ylab = "total catch (millions of pounds)")
barplot(tapply(d$Catch_lbs, d$Area, sum, na.rm = T)/10^6,
        main = "Total catch by area", ylab = "total catch (millions of pounds)")
```

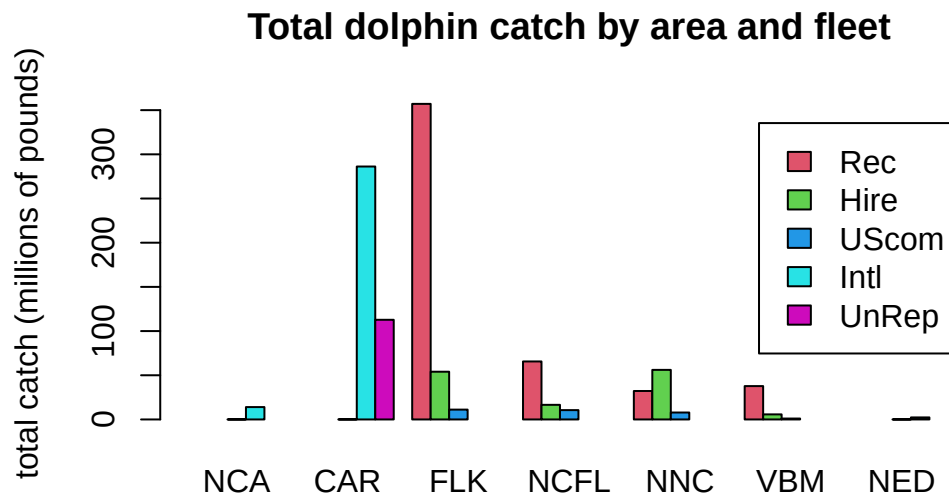

5 Data compilation for all fleets in the operating model



The total catch in the regions analyzed is highly variable but relatively stable over time with a slight decrease in recent years. Most of the catch is caught in quarter 2 (March - May). The largest sector is the U.S. private recreational sector, followed by international fleets (all types), the for-hire fleet and then the U.S. commercial fleet. Most of the fishing activity takes place in the Florida Keys and the Greater Caribbean, with other regions contributing much less.

```
par(mfrow = c(1, 1))
tab <- tapply(d$Catch_lbs, list(d$Fleet, d$Area), sum, na.rm = T)/10^6
barplot(tab, beside = T, col = 2:6, legend = rownames(tab),
        main = "Total dolphin catch by area and fleet",
        ylab = "total catch (millions of pounds)")
```

5 Data compilation for all fleets in the operating model



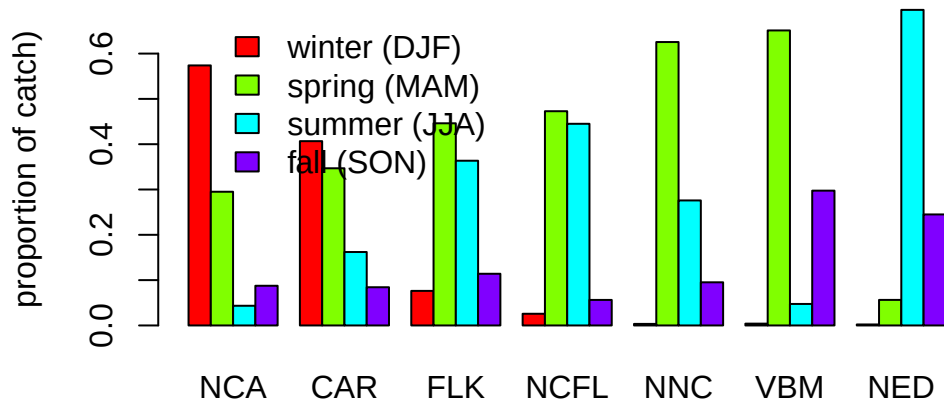
As expected, U.S. recreational fishing (private and for-hire) only takes place in the four U.S. EEZ regions (FLK, NCFL, NNC and VBM). The U.S. commercial fishery operates largely in those four U.S. EEZ regions, with small amounts of catch in the high seas of NCA, CAR and NED regions. International catches occur largely in the Caribbean, with lesser amounts in the NCA and NED high seas regions.

```
tab <- tapply(d$Catch_lbs, list(d$Quarter, d$Area), sum, na.rm = T)/10^6
tabp <- apply(tab, 2, function(x) x / sum(x, na.rm = T))

barplot(tabp, beside = T, col = rainbow(4),
        legend = c("winter (DJF)", "spring (MAM)", "summer (JJA)", "fall (SON)"),
        args.legend = list(x = 15, y = 0.7, bty = "n"),
        main = "Total dolphin catch by area and season",
        ylab = "proportion of catch")
```

5 Data compilation for all fleets in the operating model

Total dolphin catch by area and season

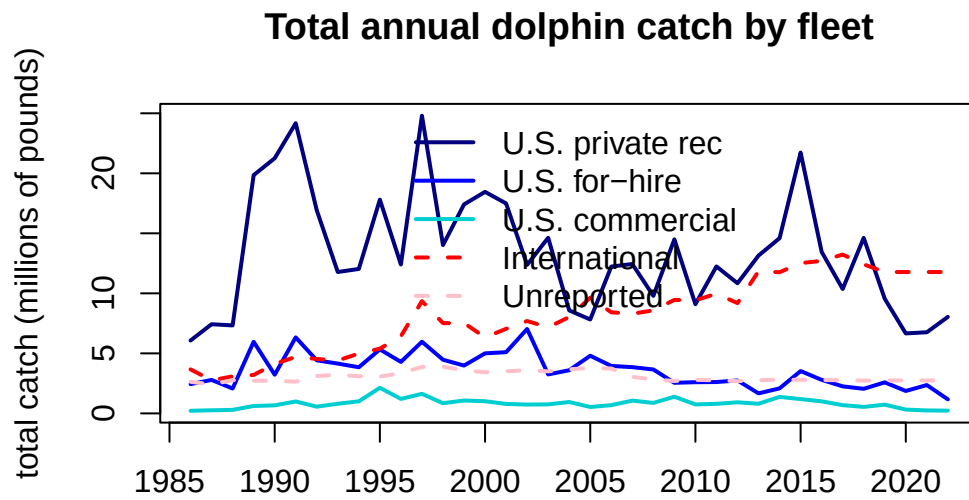


This plot shows the seasonal movements of dolphin across the different regions, as expressed through the proportion of catch that occurs in each quarter within each area. In the NCA and CAR regions, most of the catch occurs in winter, whereas in the U.S. South Atlantic regions, most of the catch occurs in spring and summer. In the U.S. Mid-Atlantic the catch also occurs largely in spring, and also fall. In the NED region most of the catch occurs in summer.

```
tab <- tapply(d$Catch_lbs, list(d$Year, d$Fleet), sum, na.rm = T)/10^6

matplot(rownames(tab), tab,
        col = c("navy", "blue", "darkturquoise", "red", "pink"),
        type = "l", lwd = 2, lty = c(rep(1, 3), 2, 2),
        main = "Total annual dolphin catch by fleet", xlab = "",
        ylab = "total catch (millions of pounds)")
legend("top", lwd = 2, lty = c(rep(1, 3), 2, 2), bty = "n",
       c("U.S. private rec", "U.S. for-hire", "U.S. commercial",
         "International", "Unreported"),
       col = c("navy", "blue", "darkturquoise", "red", "pink"))
```

5 Data compilation for all fleets in the operating model



This plot shows the time series of total annual dolphin catches by fleet. We can compare these figures with the plots from the previous chapters to ensure the data were processed correctly. The magnitude and variability of the landings match the expectations.

```
write.csv(d, file = "data/FINAL_files/complete_catch_dataset_09052025.csv", row.names = FALSE)
```

6 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

6.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to the present. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded as a .csv file under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94,984	55,676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0

6 NOAA Fisheries landings data

4	1980	South Atlantic	76,042	45,852	Commercial	34
5	1981	Gulf	73,075	46,951	Commercial	33
6	1981	Gulf	1,559,544		Recreational	707

```
# take commas out of file -----
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94984	55676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0
4	1980	South Atlantic	76042	45852	Commercial	34
5	1981	Gulf	73075	46951	Commercial	33
6	1981	Gulf	1559544	NA	Recreational	707

6.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions -----
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "com_NE")
dat
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200
1981	1559544	73075	8421898	59037	0	0	0	0

6 NOAA Fisheries landings data

1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370397	13164377	740255	241212	79832	19542	44707
2008	3937153	384954	10417536	627939	264017	57764	0	16328
2009	3860527	496701	13251102	1007355	1538112	55271	0	92010
2010	1216198	111790	8543542	572031	813738	35869	0	14939
2011	3023914	309324	10774812	543811	2173136	15376	187289	26587
2012	3229090	146256	9983276	704240	755558	44918	26892	46171
2013	5425636	187966	8483569	583556	302421	41936	226988	18564
2014	3074373	201943	11925345	1041856	2129222	58868	183673	61505
2015	6286577	154133	14821772	920454	4501793	36190	1198628	37808
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421
2022	1664771	15906	6845573	182798	708494	17360	466897	9131

6 NOAA Fisheries landings data

2023 1427518 13305 6630236 125302 734866 28467 122550 8310

6.3 Plot the summary

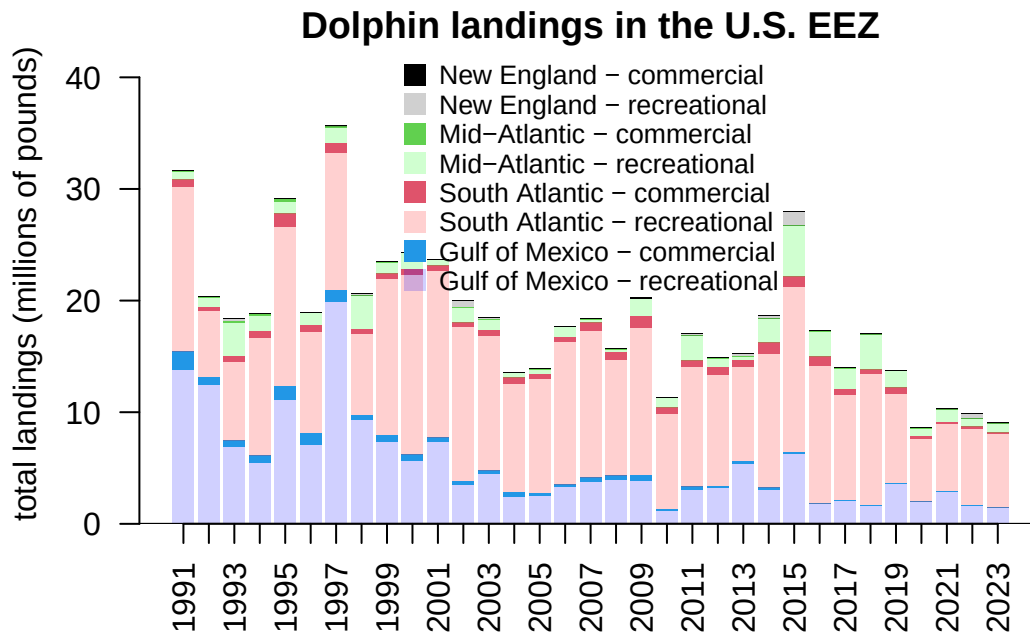
Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```
# select years -----
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot -----
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
               "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
               "South Atlantic - commercial", "South Atlantic - recreational",
               "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
       col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```

6.4 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep = ",")
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

Mode TRUE

6 NOAA Fisheries landings data

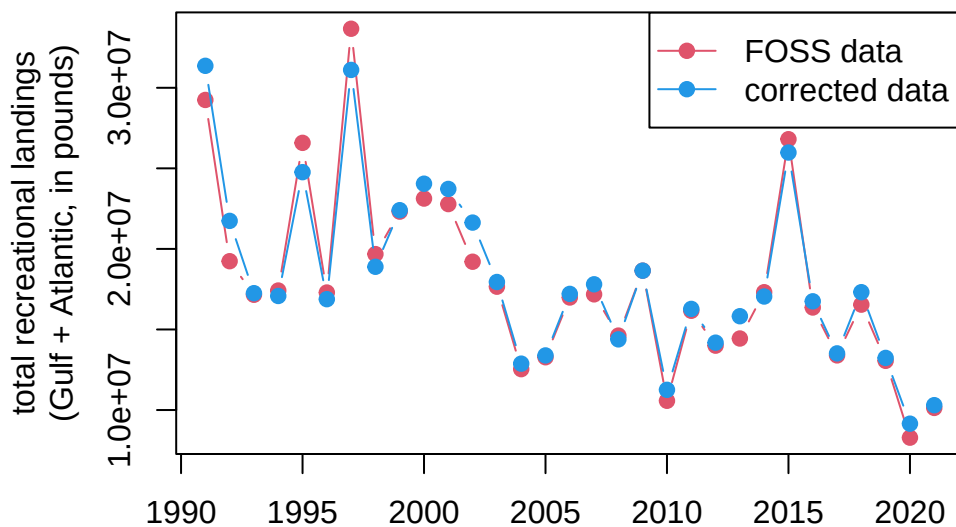
logical 31

```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with data corrected for Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```

Comparison of FOSS data with data corrected for Monroe



```
# average % difference between FOSS and updated data
mean((old - new)/old*100)
```

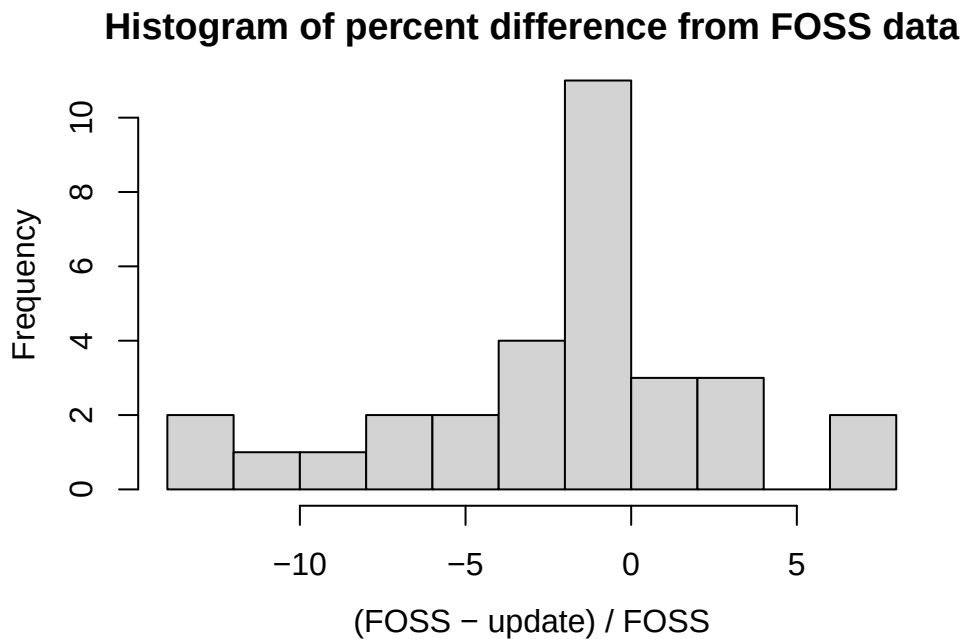
[1] -2.023886

6 NOAA Fisheries landings data

```
# min and max % difference between FOSS and updated data
range((old - new)/old*100)
```

```
[1] -13.008552  7.587789
```

```
hist((old - new)/old*100, main = "Histogram of percent difference from FOSS data", breaks = 10,
      xlab = "(FOSS - update) / FOSS")
```



```
# replace FOSS data with update data
datnew <- dat
datnew$rec_Gulf <- ml$GOM
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)

# old data set
tail(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969

6 NOAA Fisheries landings data

2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421

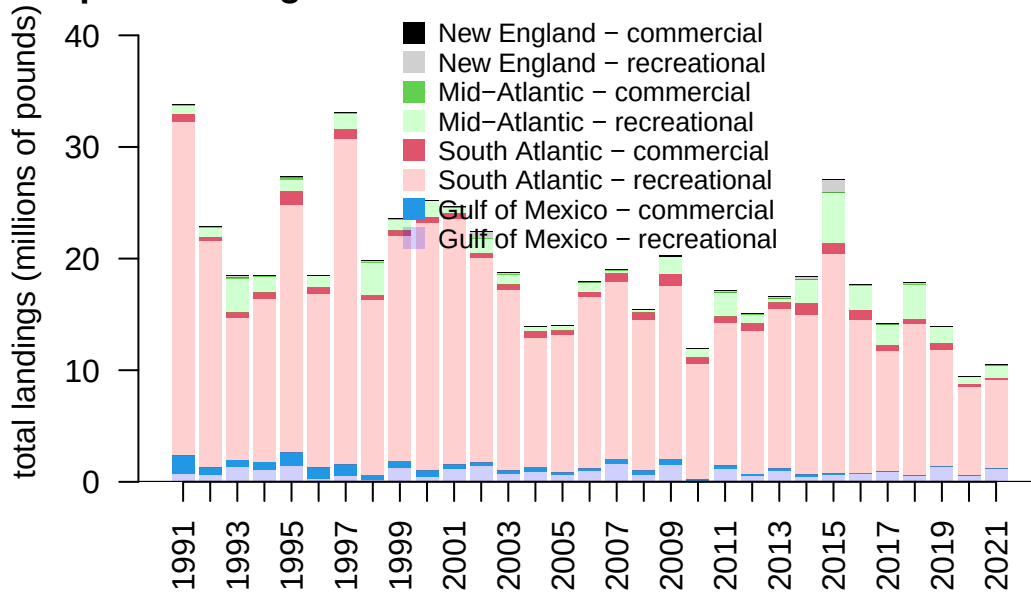
```
# updated data set  
tail(datnew)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	751499	37411	13766851	828133	2212513	61336	16499	15747
2017	868489	65141	10777104	532973	1820541	16179	48607	30969
2018	508570	56188	13603875	435059	3104018	20514	97107	12667
2019	1304865	83956	10410599	563132	1518629	48224	66	16918
2020	555695	26332	7916891	260311	679556	14040	0	1880
2021	1174412	26570	7950716	178822	1051785	25318	123038	11421

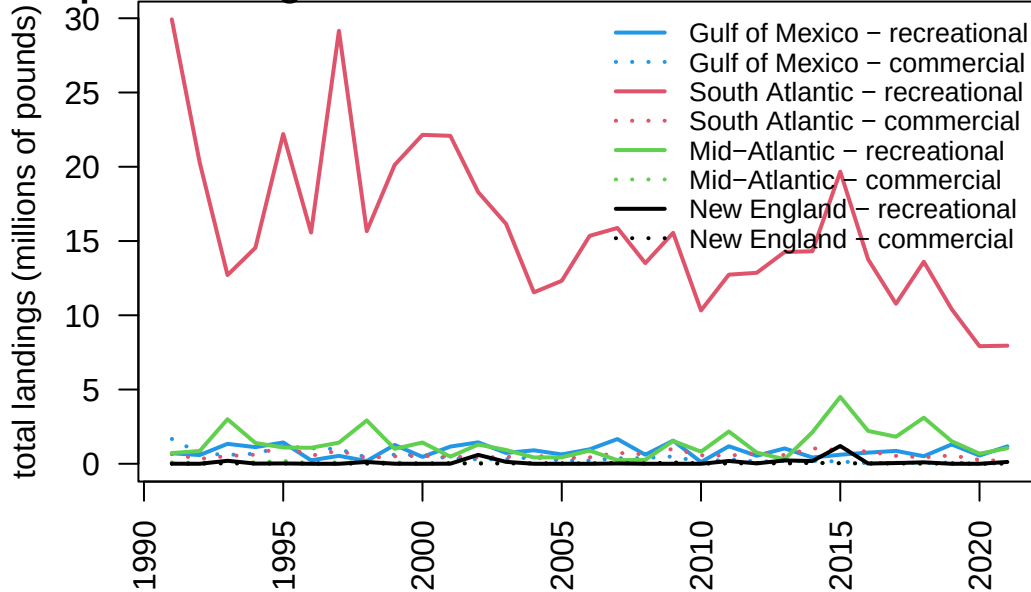
6.5 Plots with updated data

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and re-plot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.

Dolphin landings in the U.S. EEZ -- correct Monroe Coun



Dolphin landings in the U.S. EEZ -- correct Monroe Coun



7 SST

8 Analysis of ocean temperature trends

Coming soon